



IEEE Standard for Ethernet

Amendment 2: Physical Layers and Management Parameters for Increased-Reach Point-to-Multipoint Ethernet Optical Subscriber Access (Super-PON)

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.3cs™-2022
(Amendment to IEEE Std 802.3™-2022
as revised by IEEE Std 802.3dd™-2022)

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Approved 21 September 2022
IEEE SA Standards Board

Abstract: Super-PON optical subscriber access networks, which are in the family of Ethernet passive optical networks (EPONs), are defined in this amendment to IEEE Std 802.3-2022. Super-PON has a reach of up to 50 km and up to 1024 ONUs over a point-to-multipoint passive optical distribution network (ODN) through wavelength division multiplexing (WDM). A Super-PON ODN contains a passive wavelength router that determines the channels used by the ODN. This standard specifies the Super-PON Reconciliation Sublayer (RS), Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer at a MAC data rate of 10 Gb/s in the downstream direction and of 10 Gb/s or 2.5 Gb/s in the upstream direction.

Keywords: amendment, IEEE 802.3™, IEEE 802.3cs™, Multipoint MAC Control (MPMC), Optical Distribution Network (ODN), Passive Optical Network (PON), Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA), Physical Medium Dependent (PMD), Point-to-Multipoint (P2MP), Reconciliation Sublayer (RS), Super-PON

The Institute of Electrical and Electronics Engineers, Inc.
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PDF: ISBN 978-1-5044-9168-6 STD25765
Print: ISBN 978-1-5044-9169-3 STDPD25765

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Introduction

This introduction is not part of IEEE Std 802.3cs-2022, IEEE Standard for Ethernet—Amendment 2: Physical Layers and Management Parameters for Increased-Reach Point-to-Multipoint Ethernet Optical Subscriber Access (Super-PON).

IEEE Std 802.3™ was first published in 1985. Since the initial publication, many projects have added functionality or provided maintenance updates to the specifications and text included in the standard. Each IEEE 802.3 project/amendment is identified with a suffix (e.g., IEEE Std 802.3ba™-2010).

The half duplex Media Access Control (MAC) protocol specified in IEEE Std 802.3-1985 is Carrier Sense Multiple Access with Collision Detection (CSMA/CD). This MAC protocol was key to the experimental Ethernet developed at Xerox Palo Alto Research Center, which had a 2.94 Mb/s data rate. Ethernet at 10 Mb/s was jointly released as a public specification by Digital Equipment Corporation (DEC), Intel and Xerox in 1980. “Local Area Networks: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications” was approved as an IEEE standard by the IEEE Standards Board in 1983 and subsequently published in 1985 as IEEE Std 802.3-1985. Since 1985, new media options, new speeds of operation, and new capabilities have been added to IEEE Std 802.3. A full duplex MAC protocol and the ability to use an EtherType to specify the MAC client protocol were added in 1997. The title was changed to Standard for Ethernet with the 2012 Revision.

Some of the major additions to IEEE Std 802.3 are identified in the marketplace with their project number. This is most common for projects adding higher speeds of operation or new protocols. For example, IEEE Std 802.3u™ added 100 Mb/s operation (also called Fast Ethernet), IEEE Std 802.3z™ added 1000 Mb/s operation (also called Gigabit Ethernet), IEEE Std 802.3ae™ added 10 Gb/s operation (also called 10 Gigabit Ethernet), IEEE Std 802.3ah™ specified access network Ethernet (also called Ethernet in the First Mile) and IEEE Std 802.3ba added 40 Gb/s operation (also called 40 Gigabit Ethernet) and 100 Gb/s operation (also called 100 Gigabit Ethernet). These major additions are all now included in and are superseded by IEEE Std 802.3-2022 and are not maintained as separate documents.

At the date of this IEEE Std 802.3dd-2022 publication, IEEE Std 802.3 was composed of the following documents:

IEEE Std 802.3-2022

Section One—Includes Clause 1 through Clause 20 and Annex A through Annex K and Annex 4A. Section One includes the specifications for 10 Mb/s operation and the MAC, frame formats and service interfaces used for all speeds of operation.

Section Two—Includes Clause 21 through Clause 33 and Annex 22A through Annex 33A. Section Two includes management attributes for multiple protocols and speed of operation as well as specifications for providing power over twisted pair cabling for multiple operational speeds. It also includes general information on 100 Mb/s operation as well as most of the 100 Mb/s Physical Layer specifications.

Section Three—Includes Clause 34 through Clause 43 and Annex 36A through Annex 43C. Section Three includes general information on 1000 Mb/s operation as well as most of the 1000 Mb/s Physical Layer specifications.

Section Four—Includes Clause 44 through Clause 55 and Annex 44A through Annex 55B. Section Four includes general information on 10 Gb/s operation as well as most of the 10 Gb/s Physical Layer specifications.

Section Five—Includes Clause 56 through Clause 77 and Annex 57A through Annex 76A. Clause 56 through Clause 67 and Clause 75 through Clause 77, as well as associated annexes, specify subscriber access and other Physical Layers and sublayers for operation from 512 kb/s to 10 Gb/s, and defines services and protocol elements that enable the exchange of IEEE Std 802.3 format frames between stations in a subscriber access network. Clause 68 specifies a 10 Gb/s Physical Layer specification. Clause 69 through Clause 74 and associated annexes specify Ethernet operation over electrical backplanes at speeds of 1000 Mb/s and 10 Gb/s.

Section Six—Includes Clause 78 through Clause 95 and Annex 83A through Annex 93C. Clause 78 specifies Energy-Efficient Ethernet. Clause 79 specifies IEEE 802.3 Organizationally Specific Link Layer Discovery Protocol (LLDP) type, length, and value (TLV) information elements. Clause 80 through Clause 95 and associated annexes include general information on 40 Gb/s and 100 Gb/s operation as well as 40 Gb/s and 100 Gb/s Physical Layer specifications. Clause 90 specifies Ethernet support for time synchronization protocols.

Section Seven—Includes Clause 96 through Clause 115 and Annex 97A through Annex 115A. Clause 96 through Clause 98, Clause 104, and associated annexes, specify Physical Layers and optional features for 100 Mb/s and 1000 Mb/s operation over a single twisted pair. Clause 100 through Clause 103, as well as associated annexes, specify Physical Layers for the operation of the EPON protocol over coaxial distribution networks. Clause 105 through Clause 114 and associated annexes include general information on 25 Gb/s operation as well as 25 Gb/s Physical Layer specifications. Clause 99 specifies a MAC merge sublayer for the interspersing of express traffic. Clause 115 and its associated annex specify a Physical Layer for 1000 Mb/s operation over plastic optical fiber.

Section Eight—Includes Clause 116 through Clause 140 and Annex 119A through Annex 136D. Clause 116 through Clause 124 and associated annexes include general information on 200 Gb/s and 400 Gb/s operation as well as 200 Gb/s and 400 Gb/s Physical Layer specifications. Clause 125 includes general information on 2.5 Gb/s and 5 Gb/s operation. Clause 126 through Clause 130 and associated annexes include 2.5 Gb/s and 5 Gb/s Physical Layer specifications. Clause 131 provides general information on 50 Gb/s operation. Clause 132 through Clause 140 and associated annexes include 50 Gb/s Physical Layer specifications and additional 100 Gb/s, 200 Gb/s, and 400 Gb/s Physical Layer specifications.

Section Nine—Includes Clause 141 through Clause 160 and Annex 142A through Annex 154A. Clause 141 through Clause 144 and associated annexes specify symmetric and asymmetric operation of Ethernet passive optical networks over multiple 25 Gb/s channels. Clause 145 and associated annexes specify increased power delivery using all four pairs in the structured wiring plant. Clause 146 through Clause 149 and associated annexes specify Physical Layers for 10 Mb/s, 2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation over a single balanced pair of conductors. Clause 150 and Clause 151 include additional 400 Gb/s Physical Layer specifications. Clause 153 and Clause 154 specify 100 Gb/s operation over DWDM channels. Clause 157 through Clause 160 include 10 Gb/s, 25 Gb/s, and 50 Gb/s bidirectional Physical Layer specifications.

IEEE Std 802.3dd™-2022

Amendment 1—This amendment includes editorial and technical corrections, refinements, and clarifications to Clause 104, Power over Data Lines of Single Pair Ethernet, and related portions of the standard.

IEEE Std 802.3cs™-2022

Amendment 2—This amendment to IEEE Std 802.3-2022 defines Super-PON optical subscriber access networks, in the family of Ethernet passive optical networks (EPONs). Super-PON has a reach of up to 50 km and up to 1024 ONUs over a point-to-multipoint passive optical distribution network (ODN)

through wavelength division multiplexing (WDM). A Super-PON ODN contains a passive wavelength router that determines the channels used by the ODN. This standard specifies the Super-PON Reconciliation Sublayer (RS), Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer at a MAC data rate of 10 Gb/s in the downstream direction and of 10 Gb/s or 2.5 Gb/s in the upstream direction.

Two companion documents exist, IEEE Std 802.3.1 and IEEE Std 802.3.2. IEEE Std 802.3.1 describes Ethernet management information base (MIB) modules for use with the Simple Network Management Protocol (SNMP). IEEE Std 802.3.2 describes YANG data models for Ethernet. IEEE Std 802.3.1 and IEEE Std 802.3.2 are updated to add management capability for enhancements to IEEE Std 802.3 after approval of those enhancements.

IEEE Std 802.3 will continue to evolve. New Ethernet capabilities are anticipated to be added within the next few years as amendments to this standard.

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IEEE Standard for Ethernet

Amendment 2: Physical Layers and Management Parameters for Increased-Reach Point-to-Multipoint Ethernet Optical Subscriber Access (Super-PON)

(This amendment is based on IEEE Std 802.3™-2022 as amended by IEEE Std 802.3dd™-2022.)

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strike through~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

Cross references that refer to clauses, tables, equations, or figures not covered by this amendment are highlighted in green.⁶

⁶Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

1. Introduction

1.3 Normative references

Insert the following references in alphanumerical order:

ITU-T Recommendation G.989.2, 2019—40-Gigabit-capable passive optical networks 2 (NG-PON2): Physical media dependent (PMD) layer specification.

1.4 Definitions

Insert the following new definition after 1.4.197 arbitration:

1.4.197a Asymmetric Super-PON PMDs: Family of Super-PON PMDs supporting the upstream MAC data rate of 2.5 Gb/s. See Clause 164.

Change 1.4.216, to read as follows:

1.4.216 black link approach: The specification of the input, output, and transfer characteristics of the unidirectional transmission path ~~from TP2 to TP3 for a given DWDM channel within a DWDM black link, without specifying how the transmission path is implemented. (See, for example, IEEE Std 802.3, Clause 154, Figure 154-3)~~

- a) From TP2 to TP3 for a given DWDM channel within a DWDM black link (see, for example, IEEE Std 802.3, Clause 154, Figure 154-3).
- b) From MDI to MDI for a given Super-PON channel within a Super-PON black link (see, for example, IEEE Std 802.3, Clause 164, Figure 164-3).

Insert the following new definition after 1.4.289 differential skew:

1.4.289a dispersion compensation module (DCM): A device having a negative dispersion value that reverses the optical fiber chromatic dispersion in an optical network.

Change 1.4.313 as follows:

1.4.313 EQT: The unit of measurement of time for time-related parameters specified in IEEE Std 802.3, Clause 144 and Clause 164 for Nx25G EPON Multipoint MAC Control. Each EQT is equal to the time required to transmit one envelope quantum EQ between the MCRC and the PCS. When an EQ is transmitted across 25GMII (Clause 144), and the EQT is equal to 2.56 ns. When an EQ is transmitted across XGMII (Clause 164), the EQT is equal to 6.4 ns.

Insert the following new definition after 1.4.344 frame loss ratio:

1.4.344a free spectral range (FSR): The spacing in frequency or wavelength between successive repetitions of the properties of an optical filter.

Insert the following new definitions after 1.4.543 stream:

1.4.543a Super-PON PMDs: Family of PMDs for point-to-multipoint (P2MP) networks, operating at a MAC data rate of 10 Gb/s in the downstream direction and at a MAC data rate of 10 Gb/s or 2.5 Gb/s in the upstream direction. See Clause 164.

1.4.543b Symmetric Super-PON PMDs: Family of Super-PON PMDs supporting the upstream MAC data rate of 10 Gb/s. See Clause 164.

1.5 Abbreviations

Insert the following new abbreviations into the list, in alphanumeric order:

CD	chromatic dispersion
DCM	dispersion compensation module
FSR	free spectral range
MPMC	multipoint mac control
PON	passive optical network

30. Management

30.3 Layer management for DTEs

30.3.2 PHY device managed object class

30.3.2.1 PHY device attributes

30.3.2.1.2 aPhyType

Insert new entries in the APPROPRIATE SYNTAX section of 30.3.2.1.2 after the entry for 10/1GBASE-PRX as follows:

10/2.5GBASE-SP	164.3 10 Gb/s downstream and 2.5 Gb/s upstream Super-PON 256B/257B
----------------	---

Insert new entries in the APPROPRIATE SYNTAX section of 30.3.2.1.2 after the entry for 10GBASE-R as follows:

10GBASE-SP	164.3 10 Gb/s Super-PON 256B/257B
------------	-----------------------------------

30.3.2.1.3 aPhyTypeList

Insert new entries in the APPROPRIATE SYNTAX section of 30.3.2.1.3 after the entry for 10/1GBASE-PRX as follows:

10/2.5GBASE-SP	164.3 10 Gb/s downstream and 2.5 Gb/s upstream Super-PON 256B/257B
----------------	---

Insert new entries in the APPROPRIATE SYNTAX section of 30.3.2.1.3 after the entry for 10GBASE-R as follows:

10GBASE-SP	164.3 10 Gb/s Super-PON 256B/257B
------------	-----------------------------------

30.3.5 MPCP managed object class

30.3.5.1 MPCP Attributes

30.3.5.1.2 aMPCPAdminState

Change the text of the BEHAVIOUR DEFINED AS section of 30.3.5.1.2 as follows:

BEHAVIOUR DEFINED AS:

A read-only value that identifies the operational state of the Multipoint MAC Control sublayer. An interface that can provide the Multipoint MAC Control sublayer functions specified in [Clause 64](#), [Clause 77](#), ~~or Clause 103~~, ~~or 164.5~~ is enabled to do so when this attribute has the enumeration “enabled”. When this attribute has the enumeration “disabled”, the interface acts as it would if it had no Multipoint MAC Control sublayer. The operational state of the Multipoint MAC Control sublayer can be changed using the acMPCPAdminControl action.;

30.3.5.1.3 aMPCPMode

Change the text of the BEHAVIOUR DEFINED AS section of 30.3.5.1.3 as follows:

BEHAVIOUR DEFINED AS:

A read-only value that identifies the operational mode of the Multipoint MAC Control sublayer.
An interface that can provide the Multipoint MAC Control sublayer functions specified in [Clause 64](#), [Clause 77](#), ~~or Clause 103~~, or [164.5](#). When this attribute has the enumeration “OLT”, the interface acts as an OLT. When this attribute has the enumeration “ONU”, the interface acts as an ONU. When this attribute has the enumeration “CLT”, the interface acts as a CLT. When this attribute has the enumeration “CNU”, the interface acts as a CNU.;

30.5 Layer management for medium attachment units (MAUs)

30.5.1 MAU managed object class

30.5.1.1 MAU attributes

30.5.1.1.2 aMAUType

Insert new entries in the APPROPRIATE SYNTAX section of 30.5.1.1.2 after the entry for 10/1GBASE-PRX-U4 as follows:

10/2.5GBASE-SP1-Dx	One single-mode fiber 10.3125 GBd continuous downstream / 2.578125 GBd burst mode upstream OLT PHY supporting channel x of Super-PON FSR set 1, as specified in 164.2
10/2.5GBASE-SP1-Uxy	One single-mode fiber 10.3125 GBd continuous downstream / 2.578125 GBd burst mode upstream ONU PHY supporting channels x to y of Super-PON FSR set 1, as specified in 164.2

Insert new entries in the APPROPRIATE SYNTAX section of 30.5.1.1.2 after the entry for 10GBASE-R as follows:

10GBASE-SP1-Dx	One single-mode fiber 10.3125 GBd continuous downstream / 10.3125 GBd burst mode upstream OLT PHY supporting channel x of Super-PON FSR set 1, as specified in 164.2
10GBASE-SP1-Uxy	One single-mode fiber 10.3125 GBd continuous downstream / 10.3125 GBd burst mode upstream ONU PHY supporting channels x to y of Super-PON FSR set 1, as specified in 164.2

45. Management Data Input/Output (MDIO) Interface

45.2 MDIO Interface registers

45.2.1 PMA/PMD registers

Change rows in Table 45–3 as shown below (unchanged rows not shown):

Table 45–3—PMA/PMD registers

Register address	Register name	Subclause
...		
1.1003	Super-PON PMA/PMD extended ability	45.2.1.159a
1.1004 through 1.1099	Reserved	
...		

Change 45.2.1.27 as follows:

45.2.1.27 PMA/PMD control 3 register (Register 1.29)

The assignment of bits in the PMA/PMD control 3 register is shown in Table 45–30.

45.2.1.27.1 Downstream differential encoding (1.29.15)

Downstream differential encoding is selected using bit 1.29.15. This bit is read/write in the OLT and read only in the ONU with the default value of 0 (indicating that the downstream differential encoding is not enabled).

In the OLT, this bit controls whether downstream differential encoding is selected for the transmit PMA output.

In the ONU, this bit indicates whether the downstream differential decoding is active~~enabled~~ in the ONU receive PMA.

45.2.1.27.1a Super-PON PMA/PMD transmit channel (1.29.14:11)

The Super-PON PMA/PMD operating transmit channel (see 164.2.2) is selected using bits 14 to 11. The Super-PON operating transmit number is a four-bit number, 0 to 15, represented by bits 14 to 11, as an unsigned integer with bit 14 the most significant bit. The channel interval supported by the PMA/PMD is advertised in the Super-PON PMA/PMD extended ability register (register 1.1003, see 45.2.1.159a).

45.2.1.27.2 PMA/PMD type selection (1.29.5:0)

The PMA/PMD type of the PMA/PMD is selected using bits 5 to 0. The PMA/PMD type abilities of the PMA/PMD for Nx25G-EPON are advertised in the Nx25G-EPON PMA/PMD extended ability registers (registers 1.1000 through 1.1002, see 45.2.1.159). The PMA/PMD type abilities of the PMA/PMD for Super-PON are advertised in the Super-PON PMA/PMD extended ability register (register 1.1003, see 45.2.1.159a). A PMA/PMD shall ignore writes to the PMA/PMD type selection bits that select PMA/PMD types ~~it has~~ not advertised. It is the responsibility of the STA entity to ensure that mutually acceptable MMD types are applied consistently across all the MMDs on a particular PHY.

Table 45–30—PMA/PMD control 3 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.29.15	Downstream differential encoding	1 = Downstream differential encoding enabled/ <u>active</u> 0 = Downstream differential encoding disabled/ <u>inactive</u>	R/W in OLT RO in ONU
<u>1.29.14:1</u> <u>1</u>	<u>Super-PON PMA/PMD</u> <u>transmit channel</u>	<u>Super-PON PMA/PMD operating transmit channel</u>	<u>R/W</u>
<u>1.29.14:1</u> <u>0:6</u>	Reserved	Value always 0	RO
1.29.5:0	PMA/PMD type selection	5 4 3 2 1 0 1 1 x x x x = Reserved 1 0 1 x x x = Reserved 1 0 1 1 x x = Reserved 1 0 1 0 1 1 = 10GBASE-SP1-D 1 0 1 0 1 0 = 10GBASE-SP1-U 1 0 1 0 0 1 = 10/2.5GBASE-SP1-D 1 0 1 0 0 0 = 10/2.5GBASE-SP1-U 1 0 0 1 1 1 = 50GBASE-PQX-U3 1 0 0 1 1 0 = 50GBASE-PQX-U2 1 0 0 1 0 1 = 50GBASE-PQX-D3 1 0 0 1 0 0 = 50GBASE-PQX-D2 1 0 0 0 1 1 = 50GBASE-PQG-U3 1 0 0 0 1 0 = 50GBASE-PQG-U2 1 0 0 0 0 1 = 50GBASE-PQG-D3 1 0 0 0 0 0 = 50GBASE-PQG-D2 0 1 1 1 1 1 = 50/25GBASE-PQX-U3 0 1 1 1 1 0 = 50/25GBASE-PQX-U2 0 1 1 1 0 1 = 50/25GBASE-PQX-D3 0 1 1 1 0 0 = 50/25GBASE-PQX-D2 0 1 1 0 1 1 = 50/25GBASE-PQG-U3 0 1 1 0 1 0 = 50/25GBASE-PQG-U2 0 1 1 0 0 1 = 50/25GBASE-PQG-D3 0 1 1 0 0 0 = 50/25GBASE-PQG-D2 0 1 0 1 1 1 = 50/10GBASE-PQX-U3 0 1 0 1 1 0 = 50/10GBASE-PQX-U2 0 1 0 1 0 1 = 50/10GBASE-PQX-D3 0 1 0 1 0 0 = 50/10GBASE-PQX-D2 0 1 0 0 1 1 = 50/10GBASE-PQG-U3 0 1 0 0 1 0 = 50/10GBASE-PQG-U2 0 1 0 0 0 1 = 50/10GBASE-PQG-D3 0 1 0 0 0 0 = 50/10GBASE-PQG-D2 0 0 1 1 1 1 = 25GBASE-PQX-U3 0 0 1 1 1 0 = 25GBASE-PQX-U2 0 0 1 1 0 1 = 25GBASE-PQX-D3 0 0 1 1 0 0 = 25GBASE-PQX-D2 0 0 1 0 1 1 = 25GBASE-PQG-U3 0 0 1 0 1 0 = 25GBASE-PQG-U2 0 0 1 0 0 1 = 25GBASE-PQG-D3 0 0 1 0 0 0 = 25GBASE-PQG-D2 0 0 0 1 1 1 = 25/10GBASE-PQX-U3 0 0 0 1 1 0 = 25/10GBASE-PQX-U2 0 0 0 1 0 1 = 25/10GBASE-PQX-D3 0 0 0 1 0 0 = 25/10GBASE-PQX-D2 0 0 0 0 1 1 = 25/10GBASE-PQG-U3 0 0 0 0 1 0 = 25/10GBASE-PQG-U2 0 0 0 0 0 1 = 25/10GBASE-PQG-D3 0 0 0 0 0 0 = 25/10GBASE-PQG-D2	R/W

^aR/W = Read/Write, RO = Read only

Insert 45.2.1.159a as follows:

45.2.1.159a Super-PON PMA/PMD extended ability register (Register 1.1003)

The assignment of bits in the Super-PON PMA/PMD extended ability register is shown in Table 45–128a.

Table 45–128a—Super-PON PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1003.15	10GBASE-SP1-D	1 = PMA/PMD is able to perform 10GBASE-SP1-Dxy 0 = PMA/PMD is not able to perform 10GBASE-SP1-Dxy	RO
1.1003.14	10GBASE-SP1-U	1 = PMA/PMD is able to perform 10GBASE-SP1-Uxy 0 = PMA/PMD is not able to perform 10GBASE-SP1-Uxy	RO
1.1003.13	10/2.5GBASE-SP1-D	1 = PMA/PMD is able to perform 10/2.5GBASE-SP1-Dxy 0 = PMA/PMD is not able to perform 10/2.5GBASE-SP1-Dxy	RO
1.1003.12	10/2.5GBASE-SP1-U	1 = PMA/PMD is able to perform 10G/2.5BASE-SP1-Uxy 0 = PMA/PMD is not able to perform 10/2.5GBASE-SP1-Uxy	RO
1.1003.11:8	Reserved	Value always 0	RO
1.1003.7:4	Super-PON channel y	PMA/PMD last transmit channel supported	RO
1.1003.3:0	Super-PON channel x	PMA/PMD first transmit channel supported	RO

^aRO = Read only

45.2.1.159a.1 10GBASE-SP1-D (1.1003.15)

When read as a one, bit 1.1003.15 indicates that the PMA/PMD is able to support a 10GBASE-SP1-Dxy PMA/PMD type, able to operate on any of the Super-PON FSR set 1 channels x (1.1003.3:0) to y (1.1003.7:4). When read as a zero, bit 1.1003.15 indicates that the PMA/PMD is not able to support a 10GBASE-SP1-Dxy PMA/PMD type.

45.2.1.159a.2 10GBASE-SP1-U (1.1003.14)

When read as a one, bit 1.1003.14 indicates that the PMA/PMD is able to support a 10GBASE-SP1-Uxy PMA/PMD type, able to operate on any of the Super-PON FSR set 1 channels x (1.1003.3:0) to y (1.1003.7:4). When read as a zero, bit 1.1003.14 indicates that the PMA/PMD is not able to support a 10GBASE-SP1-Uxy PMA/PMD type.

45.2.1.159a.3 10/2.5GBASE-SP1-D (1.1003.13)

When read as a one, bit 1.1003.13 indicates that the PMA/PMD is able to support a 10/2.5GBASE-SP1-Dxy PMA/PMD type, able to operate on any of the Super-PON FSR set 1 channels x (1.1003.3:0) to y (1.1003.7:4). When read as a zero, bit 1.1003.13 indicates that the PMA/PMD is not able to support a 10/2.5GBASE-SP1-Dxy PMA/PMD type.

45.2.1.159a.4 10/2.5GBASE-SP1-U (1.1003.12)

When read as a one, bit 1.1003.12 indicates that the PMA/PMD is able to support a 10/2.5GBASE-SP1-Uxy PMA/PMD type, able to operate on any of the Super-PON FSR set 1 channels x (1.1003.3:0) to y (1.1003.7:4). When read as a zero, bit 1.1003.12 indicates that the PMA/PMD is not able to support a 10/2.5GBASE-SP1-Uxy PMA/PMD type.

45.2.1.159a.5 Super-PON channel y (1.1003.7:4)

This 4-bit field indicates the highest numbered Super-PON transmit channel supported by the PMA/PMD (see 164.2.2).

45.2.1.159a.6 Super-PON channel x (1.1003.3:0)

This 4-bit field indicates the lowest numbered Super-PON transmit channel supported by the PMA/PMD (see 164.2.2).

45.2.3 PCS registers

Change the identified rows in Table 45–233 (unchanged rows not shown):

Table 45–233—PCS registers

Register address	Register name	Subclause
...	...	...
3.76, 3.77	10G-EPON, and Nx25G-EPON, <u>and Super-PON</u> corrected FEC codewords counter	45.2.3.41
3.78, 3.79	10G-EPON, and Nx25G-EPON, <u>and Super-PON</u> uncorrected FEC codewords counter	45.2.3.42
3.80	10GBASE-PR and , 10/1GBASE-PRX, and Nx25G-EPON, <u>and Super-PON</u> BER monitor interval control	45.2.3.43
3.81	10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON, <u>and Super-PON</u> BER monitor status	45.2.3.44
3.82	10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON, <u>and Super-PON</u> BER monitor threshold control	45.2.3.45
3.83 through 3.134	Nx25G-EPON <u>and Super-PON</u> synchronization pattern	45.2.3.46
...	...	...

45.2.3.1 PCS control 1 register (Register 3.0)

Change the identified row in Table 45–234 as follows (unchanged rows not shown):

Table 45–234—PCS control 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
...	...	...	...
3.0.5:2	Speed selection	5 4 3 2 1 1 x x = Reserved 1 1 1 x = Reserved 1 1 0 1 = Reserved 1 1 0 0 = 10/2.5 Gb/s 1 0 1 1 = 25/10 Gb/s 1 0 1 0 = 400 Gb/s 1 0 0 1 = 200 Gb/s 1 0 0 0 = 5 Gb/s 0 1 1 1 = 2.5 Gb/s 0 1 1 0 = 50 Gb/s 0 1 0 1 = 25 Gb/s 0 1 0 0 = 100 Gb/s 0 0 1 1 = 40 Gb/s 0 0 1 0 = 10/1 Gb/s 0 0 0 1 = 10PASS-TS/2BASE-TL 0 0 0 0 = 10 Gb/s	R/W
...	...	...	...

^aRO = Read only, R/W = Read/Write, SC = Self-clearing

45.2.3.6 PCS control 2 register (Register 3.7)

Change the identified rows in Table 45–237 as follows (unchanged rows not shown):

Table 45–237—PCS control 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
3.7.15:5	Reserved	Value always 0	RO
3.7.4:0	PCS type selection	4 3 2 1 0 1 1 x x x = reserved 1 0 1 x x = reserved 1 0 1 1 x = reserved 1 0 1 0 1 = Select 10GBASE-SP PCS type 1 0 1 0 0 = Select 10/2.5GBASE-SP PCS type 1 0 0 1 1 = Select 25GBASE-PQ PCS type 1 0 0 1 0 = Select 25/10GBASE-PQ PCS type 1 0 0 0 1 = Select 25GBASE-PQ PCS type, Tx only 1 0 0 0 0 = Select 25GBASE-PQ PCS type, Rx only 0 1 1 1 1 = Select 5GBASE-R PCS type 0 1 1 1 0 = Select 2.5GBASE-X PCS type 0 1 1 0 1 = Select 400GBASE-R PCS type 0 1 1 0 0 = Select 200GBASE-R PCS type 0 1 0 1 1 = Select 5GBASE-T PCS type 0 1 0 1 0 = Select 2.5GBASE-T PCS type 0 1 0 0 1 = Select 25GBASE-T PCS type 0 1 0 0 0 = Select 50GBASE-R PCS type 0 0 1 1 1 = Select 25GBASE-R PCS type 0 0 1 1 0 = Select 40GBASE-T PCS type 0 0 1 0 1 = Select 100GBASE-R PCS type 0 0 1 0 0 = Select 40GBASE-R PCS type 0 0 0 1 1 = Select 10GBASE-T PCS type 0 0 0 1 0 = Select 10GBASE-W PCS type 0 0 0 0 1 = Select 10GBASE-X PCS type 0 0 0 0 0 = Select 10GBASE-R PCS type	R/W

^aRO = Read only, R/W = Read/Write

Change content of 45.2.3.6.1 as follows:

45.2.3.6.1 PCS type selection (3.7.4:0)

The PCS type shall be selected using bits 4 through 0. The PCS type abilities of the PCS are advertised in bits 3.8.13:12, 3.8.9:0 and 3.9.97:0. A PCS shall ignore writes to the PCS type selection bits that select PCS types it has not advertised in the PCS status 2 register or the PCS status 3 register. It is the responsibility of the STA entity to ensure that mutually acceptable MMD types are applied consistently across all the MMDs on a particular PHY. The PCS type selection defaults to a supported ability.

Table 45–237—PCS control 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
3.7.15:5	Reserved	Value always 0	RO
3.7.4:0	PCS type selection	4 3 2 1 0 1 1 x x x = reserved 1 0 1 x x = reserved 1 0 1 1 x = reserved 1 0 1 0 1 = Select 10GBASE-SP PCS type 1 0 1 0 0 = Select 10/2.5GBASE-SP PCS type 1 0 0 1 1 = Select 25GBASE-PQ PCS type 1 0 0 1 0 = Select 25/10GBASE-PQ PCS type 1 0 0 0 1 = Select 25GBASE-PQ PCS type, Tx only 1 0 0 0 0 = Select 25GBASE-PQ PCS type, Rx only 0 1 1 1 1 = Select 5GBASE-R PCS type 0 1 1 1 0 = Select 2.5GBASE-X PCS type 0 1 1 0 1 = Select 400GBASE-R PCS type 0 1 1 0 0 = Select 200GBASE-R PCS type 0 1 0 1 1 = Select 5GBASE-T PCS type 0 1 0 1 0 = Select 2.5GBASE-T PCS type 0 1 0 0 1 = Select 25GBASE-T PCS type 0 1 0 0 0 = Select 50GBASE-R PCS type 0 0 1 1 1 = Select 25GBASE-R PCS type 0 0 1 1 0 = Select 40GBASE-T PCS type 0 0 1 0 1 = Select 100GBASE-R PCS type 0 0 1 0 0 = Select 40GBASE-R PCS type 0 0 0 1 1 = Select 10GBASE-T PCS type 0 0 0 1 0 = Select 10GBASE-W PCS type 0 0 0 0 1 = Select 10GBASE-X PCS type 0 0 0 0 0 = Select 10GBASE-R PCS type	R/W

^aRO = Read only, R/W = Read/Write

45.2.3.8 PCS status 3 register (Register 3.9)

Change the row for bits 3.9.15:4 in Table 45-182 as follows (unchanged rows not shown):

Insert 45.2.3.8.a through 45.2.3.8.b (before 45.2.3.8.1) as follows:

45.2.3.8.a 10GBASE-SP capable (3.9.9)

When read as a one, bit 3.9.9 indicates that the PCS is able to support the 10GBASE-SP PCS type. When read as a zero, bit 3.9.9 indicates that the PCS is not able to support the 10GBASE-SP PCS type.

Table 45–239—PCS status 3 register bit definitions

Bit(s)	Name	Description	R/W ^a
3.9.15:10 8	Reserved	Value always 0	RO
3.9.9	<u>10GBASE-SP capable</u>	<u>1 = PCS is able to support 10GBASE-SP PCS type</u> <u>0 = PCS is not able to support 10GBASE-SP PCS type</u>	<u>RO</u>
3.9.8	<u>10/2.5GBASE-SP capable</u>	<u>1 = PCS is able to support 10/2.5GBASE-SP PCS type</u> <u>0 = PCS is not able to support 10/2.5GBASE-SP PCS type</u>	<u>RO</u>
...			

^aRO = Read only

45.2.3.8.b 10/2.5GBASE-SP capable (3.9.8)

When read as a one, bit 3.9.8 indicates that the PCS is able to support the 10/2.5GBASE-SP PCS type. When read as a zero, bit 3.9.8 indicates that the PCS is not able to support the 10/2.5GBASE-SP PCS type.

Change the title, text and table captions (unchanged table content is not shown) for 45.2.3.41 through 45.2.3.45 as follows:

45.2.3.41 10G-EPON, and Nx25G-EPON, and Super-PON corrected FEC codewords counter (Register 3.76, 3.77)

The assignment of bits in the 10G-EPON, and Nx25G-EPON, and Super-PON corrected FEC codewords counter register is shown in Table 45–270. See 76.3.3.3.2 for a definition of the 10G-EPON counters and 142.3.4 for the definition of the Nx25-EPON and Super-PON counters. These bits shall be reset to all zeros when the register is read by the management function or upon PCS reset. These bits shall be held at all ones in the case of overflow.

Table 45–270—10G-EPON, and Nx25G-EPON, and Super-PON corrected FEC codewords counter register bit definitions

Bit(s)	Name	Description	R/W ^a
...			

^aRO = Read only, MW = Multi-word, NR = Non Roll-over

45.2.3.42 10G-EPON, and Nx25G-EPON, and Super-PON uncorrected FEC codewords counter (Register 3.78, 3.79)

The assignment of bits in the 10G-EPON, and Nx25G-EPON, and Super-PON uncorrected FEC codewords counter register is shown in Table 45–271. See 76.3.3.3.2 for a definition of the 10G-EPON counters and 142.3.4 for the definition of the Nx25-EPON and Super-PON counters. These bits shall be reset to all zeros

when the register is read by the management function or upon PCS reset. These bits shall be held at all ones in the case of overflow.

Table 45–271—10G-EPON, ~~and Nx25G-EPON, and Super-PON~~ uncorrected FEC codewords counter register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
3.79.15:0	uncorrected FEC codewords lowerupper	uncorrected_FEC_codewords_counter[321:16]	RO, MW, NR

^aRO = Read only, MW = Multi-word, NR = Non Roll-over

45.2.3.43 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON, and Super-PON~~ BER monitor interval control register (Register 3.80)

The assignment of bits in the 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON, and Super-PON~~ BER monitor interval ~~timer~~ control register is shown in Table 45–272. This register is defined only when 10GBASE-PR, 10/1GBASE-PRX, ~~or Nx25G EPON, or Super-PON~~ ONU capability is supported. The 10G-EPON BER monitor is described in 76.3.3.4. The Nx25G EPON ~~and Super-PON~~ QC-LDPC BER monitor is described in 142.3.5.6.

Table 45–272—10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON, and Super-PON~~ BER monitor interval control register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
3.80.7:0	EPON BER monitor timer interval	For 10GBASE-PR and 10/1GBASE-PRX: Duration (in units of 5 micro-seconds) of the timer used by the BER monitor function. Default value is 25 (i.e., 125 micro seconds). A value of zero indicates that the BER monitor function is disabled. For Nx25G EPON and Super-PON : QC-LDPC codeword count (in units of 16 codewords) of the monitoring interval used by the BER monitor function. Default value is 12 (i.e., 192 codewords).	R/W

^aRO = Read only, R/W = Read/Write

45.2.3.44 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON~~, and Super-PON BER monitor status (register 3.81)

The assignment of bits in the 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON~~, and Super-PON BER monitor status register is shown in Table 45–273. This register is defined only when 10GBASE-PR ~~or~~, 10/1GBASE-PRX, ~~or~~ Nx25G EPON, or Super-PON ONU capability is supported.

Table 45–273—10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON~~, and Super-PON BER monitor status register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
3.81.1	Latched high BER	10GBASE-PR, 10/1GBASE-PRX, or Nx25G EPON, <u>or Super-PON</u> PCS: 1 = reported a high BER. 0 = did not report a high BER.	RO, LH
3.81.0	high BER	10GBASE-PR, 10/1GBASE-PRX, or Nx25G EPON, <u>or Super-PON</u> PCS: 1 = reporting a high BER. 0 = not reporting a high BER.	RO

^aRO Read only, LH = Latching high

45.2.3.44.1 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON~~, and Super-PON PCS high BER (3.81.0)

In the 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON~~, and Super-PON PCS, when read as a one, bit 3.81.0 indicates that the receiver is detecting a BER greater than the configurable threshold (high BER state). When read as a zero, bit 3.81.0 indicates that the receiver is detecting a BER lower than the configurable threshold (low BER state). This bit mirrors the state of the hi_ber variable, defined in 76.3.3.4 for 10GBASE-PR and, 10/1GBASE-PRX, and the HiBer variable in 142.3.5.2 for Nx25G EPON and Super-PON.

45.2.3.44.2 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON~~, and Super-PON PCS latched high BER (3.81.1)

In the 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON~~, and Super-PON PCS, when read as a one, bit 3.81.1 indicates that the receiver detected a BER greater than the configurable threshold (high BER state). When read as a zero, bit 3.81.1 indicates that the receiver detected BER lower than the configurable threshold (low BER state).

This bit is a latching high version of the 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON~~, and Super-PON high BER status bit (3.81.0).

45.2.3.45 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON~~, and Super-PON BER monitor threshold control (register 3.82)

The assignment of bits in the 10GBASE-PR, 10/1GBASE-PRX, ~~and Nx25G EPON~~, and Super-PON BER monitor threshold control register is shown in Table 45–274. This register is defined only when 10GBASE-PR ~~or~~, 10/1GBASE-PRX, ~~or~~ Nx25G EPON, or Super-PON ONU capability is supported. The 10G-EPON

BER monitor is described in 76.3.3.4. The Nx25G EPON and Super-PON QC-LDPC BER monitor is described in 142.3.5.6.

**Table 45–274—10GBASE-PR, 10/1GBASE-PRX, and Nx25G EPON, and Super-PON
BER monitor threshold control register bit definitions**

Bit(s)	Name	Description	R/W ^a
3.82.15:0	EPON BER monitor threshold	For 10G-EPON: number of sync header errors within a timer interval that triggers a high BER condition for the BER monitor function. Default value is 1600. A value of zero indicates that the BER monitor function is disabled. For Nx25G EPON and Super-PON: number of invalid QC-LDPC code-word parity checks within a BER monitor interval that triggers a high BER condition for the BER monitor function. Default value is 18. A value of zero indicates that the BER monitor function is disabled.	R/W

^aR/W = Read/Write

Change the subclause and table title and text for 45.2.3.46 and subclauses as follows:

45.2.3.46 Nx25G-EPON and Super-PON synchronization pattern registers (Registers 3.83 through 3.134)

The assignment of bits in registers 3.83 through 3.134 is shown in Table 45–275. The Nx25G-EPON synchronization pattern (see 142.1.3 and 144.3.6.7) is used in the upstream data transmissions to facilitate the OLT in locking to the incoming data burst.

**Table 45–275—Nx25G-EPON and Super-PON synchronization pattern registers
bit definitions**

45.2.3.46.1 SP3 bit 257 (3.83.5)

In the Nx25G-EPON and Super-PON PCS, bit 3.83.5 indicates the value to be used for the 257th bit of SP3. See 142.1.3 and 144.3.6.7 for additional details.

45.2.3.46.2 SP3 balanced (3.83.4)

In the Nx25G-EPON and Super-PON PCS, bit 3.83.4 indicates that repeating SP3 synchronization patterns are to have a balanced number of one and zero bits transmitted. When this bit is set to a zero then SP3 is to remain unbalanced, i.e., SP3 is always transmitted using the values from 3.83.5 and 3.118.0 through 3.133.15. When this bit is set to a one SP3 is to be balanced, i.e., each 257-bit block of SP3 (starting with the second block) is an inversion of the preceding block. See 142.1.3 and 144.3.6.7 for additional details.

45.2.3.46.3 SP2 bit 257 (3.83.3)

In the Nx25G-EPON and Super-PON PCS, bit 3.83.3 indicates the value to be used for the 257th bit of SP2. See 142.1.3 and 144.3.6.7 for additional details.

45.2.3.46.4 SP2 balanced (3.83.2)

In the Nx25G-EPON and Super-PON PCS, bit 3.83.2 indicates that repeating SP2 synchronization patterns are to have a balanced number of one and zero bits transmitted. When this bit is set to a zero then SP2 is to remain unbalanced, i.e., SP2 is always transmitted using the values from 3.83.3 and 3.101.0 through

3.116.15. When this bit is set to a one SP2 is to be balanced, i.e., each 257-bit block of SP2 (starting with the second block) is an inversion of the preceding block. See [142.1.3](#) and [144.3.6.7](#) for additional details.

45.2.3.46.5 SP1 bit 257 (3.83.1)

In the Nx25G-EPON and Super-PON PCS, bit 3.83.1 indicates the value to be used for the 257th bit of SP1. See [142.1.3](#) and [144.3.6.7](#) for additional details.

45.2.3.46.6 SP1 balanced (3.83.0)

In the Nx25G-EPON and Super-PON PCS, bit 3.83.0 indicates that repeating SP1 synchronization patterns are to have a balanced number of one and zero bits transmitted. When this bit is set to a zero then SP1 is to remain unbalanced, i.e., SP1 is always transmitted using the values from 3.83.1 and 3.84.0 through 3.99.15. When this bit is set to a one SP1 is to be balanced, i.e., each 257-bit block of SP1 (starting with the second block) is an inversion of the preceding block. See [142.1.3](#) and [144.3.6.7](#) for additional details.

45.2.3.46.7 SP1 pattern (3.84.0 through 3.99.15)

In the Nx25G-EPON and Super-PON PCS, bits 3.84.0 through 3.99.15 indicate the value to be used for the lower 256 bits of the initial SP1 transmitted in a burst. If present, subsequent transmissions of the lower 256 bits of SP1 are determined by bit 3.83.0. See [142.1.3](#) and [144.3.6.7](#) for additional details.

45.2.3.46.8 SP1 length (3.100.15:0)

In the Nx25G-EPON and Super-PON PCS, bits 3.100.15:0 indicate the number of times the 257-bit SP1 is transmitted in a given burst. See [142.1.3](#) and [144.3.6.7](#) for additional details.

45.2.3.46.9 SP2 pattern (3.101.0 through 3.116.15)

In the Nx25G-EPON and Super-PON PCS, bits 3.101.0 through 3.116.15 indicate the value to be used for the lower 256 bits of the initial SP2 transmitted in a burst. If present, subsequent transmissions of the lower 256 bits of SP2 are determined by bit 3.100.15:0. See [142.1.3](#) and [144.3.6.7](#) for additional details.

45.2.3.46.10 SP2 length (3.117.15:0)

In the Nx25G-EPON and Super-PON PCS, bits 3.117.15:0 indicate the number of times the 257-bit SP2 is transmitted in a given burst. See [142.1.3](#) and [144.3.6.7](#) for additional details.

45.2.3.46.11 SP3 pattern (3.118.0 through 3.133.15)

In the Nx25G-EPON and Super-PON PCS, bits 3.118.0 through 3.133.15 indicate the value to be used for the lower 256 bits of the initial SP3 transmitted in a burst. If present, subsequent transmissions of the lower 256 bits of SP3 are determined by bit 3.117.15:0. See [142.1.3](#) and [144.3.6.7](#) for additional details.

45.2.3.46.12 SP3 length (3.134.15:0)

In the Nx25G-EPON and Super-PON PCS, bits 3.134.15:0 indicate the number of times the 257-bit SP3 is transmitted in a given burst. See [142.1.3](#) and [144.3.6.7](#) for additional details.

56. Introduction to Ethernet for subscriber access networks

56.1 Overview

Change the third paragraph in 56.1 as follows:

The relationships between these EFM elements and the ISO/IEC Open System Interconnection (OSI) reference model are shown in Figure 56–1 and Figure 56–2 for point-to-point topologies, Figure 56–3 for 1G-EPON topologies, Figure 56–4 for 10/10G-EPON topologies, Figure 56–5 for 10/1G-EPON topologies, Figure 56–6 for EPoC topologies, ~~and Figure 56–7 for Nx25G-EPON topologies, and Figure 56–7a for Super-PON topologies.~~

Insert a new Figure 56-7a after existing Figure 56-7.

Change the last paragraph in 56.1 as shown:

The EFM architecture is further extended in:

- Clause 75 and Clause 76 by the addition of 10G-EPON.
- Clause 100, Clause 101, and Clause 102 by the addition of EPoC.
- Clause 141, Clause 142, and Clause 143 by the addition of Nx25G-EPON.
- Clause 164 by the addition of Super-PON.

56.1.2 Summary of P2MP sublayers

Change lettered list in 56.1.2 to add new item d) as shown (unchanged list items not shown):

For P2MP optical fiber topologies, EFM supports the following systems:

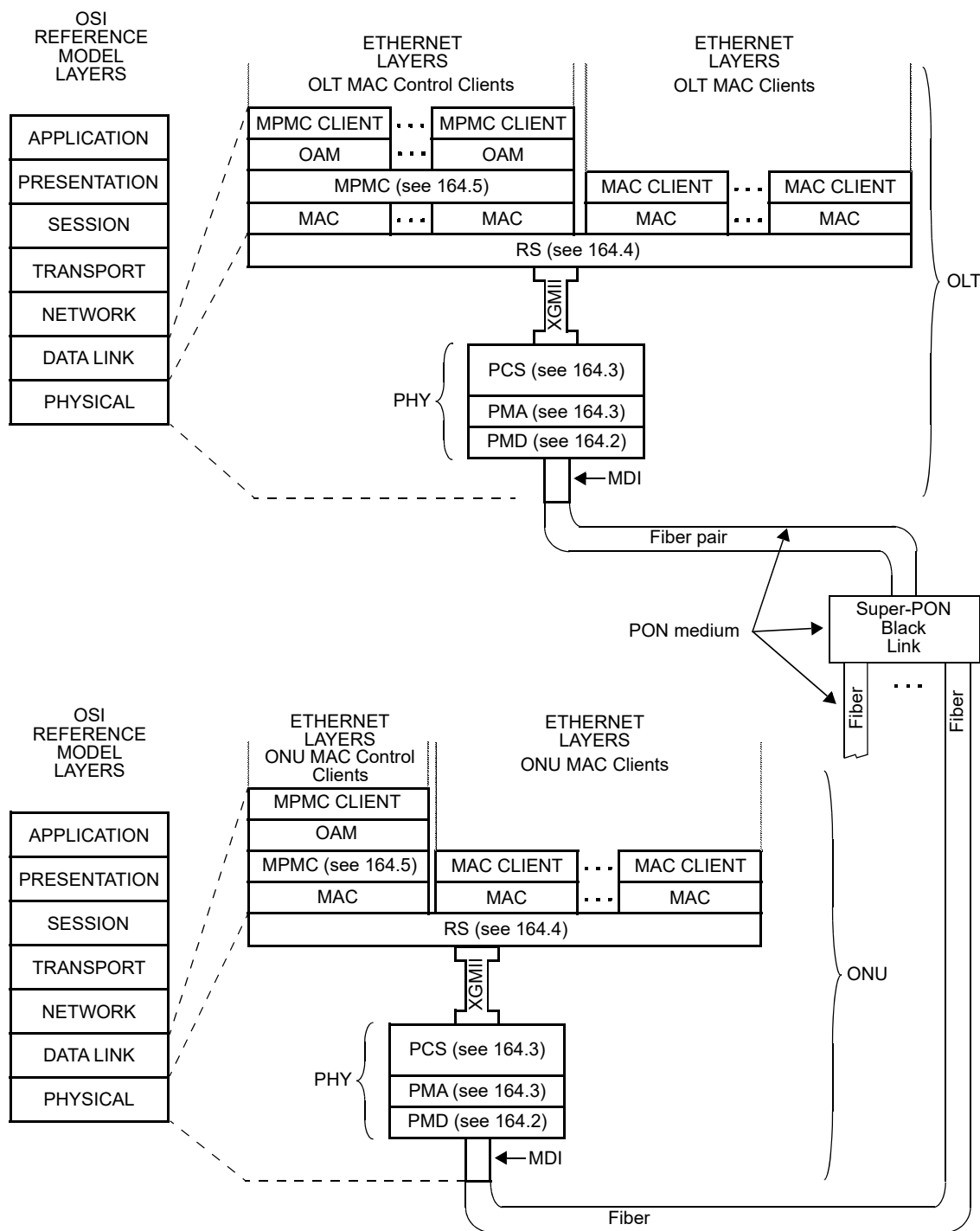
- d) Super-PON (an amplified WDM PON) with a nominal MAC data rate of 10 Gb/s in the downstream direction and 10 Gb/s or 2.5 Gb/s in the upstream direction, with a reach of up to 50 km and up to 1024 ONUs over a point-to-multipoint passive optical distribution network (ODN) using wavelength division multiplexing (WDM). The Super-PON PMA and PCS sublayers are specified in Clause 164.

56.1.2.1 Multipoint MAC Control Protocol (MPCP)

Change paragraph number 1 and 2 in 56.1.2.1 as shown:

The Multipoint MAC Control Protocol (MPCP) for 1G-EPON uses messages, state diagrams, and timers, as defined in Clause 64, to control access to a P2MP topology; Clause 77 defines the messages, state diagrams, and timers required to control access to a P2MP ODN topology in 10G-EPON; ~~and Clause 144 defines the messages, state diagrams, and timers required to control access to a P2MP ODN topology in Nx25G-EPON;~~ and Clause 164 defines the messages, state diagrams, and timers required to control access to a P2MP ODN topology in Super-PON. The issues related to coexistence of 1G-EPON and 10G-EPON on the same fiber plant are described in 77.4.

Every P2MP ODN topology consists of one Optical Line Terminal (OLT) and one or more ONUs, as shown in Figure 56–3, Figure 56–4, Figure 56–5, ~~and Figure 56–7,~~ and Figure 56-7a for 1G-EPON, 10/10G-EPON, 10/1G-EPON, ~~and Nx25G-EPON,~~ and Super-PON, respectively. One of several instances of the MPCP in the OLT communicates with the instance of the MPCP in the ONU. A pair of MPCPs that communicate between the OLT and ONU are a distinct and associated pair.



XGMII= 10 GIGABIT MEDIA INDEPENDENT INTERFACE
MDI = MEDIUM DEPENDENT INTERFACE
OAM = OPERATIONS, ADMINISTRATION & MAINTENANCE
OLT = OPTICAL LINE TERMINAL
RS = RECONCILIATION SUBLAYER
MPMC= MULTIPoint MAC CONTROL

ONU = OPTICAL NETWORK UNIT
PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

**Figure 56-7a—Architectural positioning of EFM:
P2MP Super-PON architecture**

56.1.2.2 Reconciliation Sublayer (RS) and media independent interfaces

Change paragraph number 1 and 4 in 56.1.2.2 as shown:

The **Clause 22** RS and MII, **Clause 35** RS and GMII, and **Clause 46** RS and XGMII, Clause 106 RS and 25GMII, and Clause 132 RS and 50GMII are all employed for the same purpose in EFM, that being the interconnection between the MAC sublayer and the PHY sublayers. Extensions to the **Clause 35** RS for P2MP topologies are described in **Clause 65**, the RS for 10G-EPON P2MP topologies is described in **Clause 76**, the RS for Nx25G-EPON P2MP topologies is described in **Clause 143**, ~~and~~ the RS for EPoC P2MP topologies is described in **Clause 101**, ~~and~~ the RS for Super-PON P2MP topologies is described in **Clause 164**.

This is described in **Clause 65** for EPON, in **Clause 76** for 10G-EPON, in **Clause 143** for Nx25G-EPON, ~~and~~ in **Clause 101** for EPoC, ~~and~~ in **Clause 164** for Super-PON. EFM Copper links use the MII of **Clause 22** operating at 100 Mb/s. This is described in **61.1.4.1.2**.

56.1.3 Physical Layer signaling systems

Insert the following new paragraph in 56.1.3 after the paragraph “All 10G-EPON PMDs are defined in Clause 75.”:

Additionally, EFM introduces a family of Physical Layer signaling systems for Super-PON, as defined in Clause 164.

Insert 10/2.5GBASE-SP1-D and 10/2.5GBASE-SP1-U between 10/1GBASE-PRX-U4 and 10GBASE-PR-D1, and 10GBASE-SP1-D and 10GBASE-SP1-U between 10GBASE-PR-U4 and 10PASS-TS-O in Table 56-1 (unchanged rows not shown):

Table 56–1—Summary of EFM Physical Layer signaling systems

Name	Location	Rate ^a	Nominal reach (km)	Medium	Clause
...					
10/2.5GBASE-SP1-D	OLT	10 Gb/s (tx) 2.5 Gb/s (rx)	50	Two single-mode fibers	164
10/2.5GBASE-SP1-U	ONU	2.5 Gb/s (tx) 10 Gb/s (rx)		One single-mode fiber	
...					
10GBASE-SP1-D	OLT	10 Gb/s (tx) 10 Gb/s (rx)	50	Two single-mode fibers	164
10GBASE-SP1-U	ONU			One single-mode fiber	
...					

^aFor Physical Layer signaling systems the transmit rate is denoted with the abbreviation “(tx)” to the location; whereas the receive rate is denoted with the abbreviation “(rx)”.

Insert 10/2.5GBASE-SP1-D and 10/2.5GBASE-SP1-U between 10/1GBASE-PRX-U4 and 10GBASE-PR-D1, and 10GBASE-SP1-D and 10GBASE-SP1-U between 10GBASE-PR-U4 and 25/10GBASE-PQG-U2 in Table 56–3 and a new column “Super-PON PMD, PMA, PCS, RS, MPCP” between “10G-EPON P2MP MPMC” and “Nx25GBASE-PQ PMD” columns Table 56–3 (unchanged rows and footnotes not shown):

Table 56–3—Nomenclature and clause correlation for optical P2MP systems^a

Nomenclature	Clause																
	57	60				64	65		66	75	76	77	141	142	143	144	164
	OAM	1000BASE-PX10 PMD	1000BASE-PX20 PMD	1000BASE-PX30 PMD	1000BASE-PX40 PMD	P2MP MPMC	P2MP RS, PCS, PMA FEC		1000BASE-X PCS, PMA	10/1GBASE-PRX or 10GBASE-PR PMDs	P2MP RS, PCS, PMA, FEC	10G-EPON P2MP MPMC	Nx25GBASE-PQ PMD	Nx25GBASE-PQ P2MP PCS, PMA, FEC	Nx25GBASE-PQ M2MP MCRS	Nx25GBASE-PQ P2MP MPMC	Super-PON PMD, PMA, PCS, RS, MPCP
...																	
<u>10GBASE-SP1-D</u>	<u>O</u>																<u>M</u>
<u>10GBASE-SP1-U</u>	<u>O</u>																<u>M</u>
...																	
<u>10/2.5GBASE-SP1-D</u>	<u>O</u>																<u>M</u>
<u>10/2.5UBASE-SP1-U</u>	<u>O</u>																<u>M</u>
...																	

^aO = Optional, M = Mandatory

67. System considerations for Ethernet subscriber access networks

67.1 Overview

Insert a new row at end of existing Table 67-1 as follows (unchanged rows not shown):

Table 67-1—Characteristics of the various EFM network media segments

Media type	Rate	Number of PHYs per segment	Nominal reach (km)
Optical Super-PON P2MP segment (10GBASE-SP1, 10/2.5GBASE-SP1)	10 / 10 Gb/s 10 / 2.5 Gb/s	1040 ^a	50

^aAssumes a Super-PON segment including 16 OLT PHYs and 1024 ONU PHYs.

67.2 Discussion and examples of EFM P2MP topologies

Insert the following subclause after 67.2.4:

67.2.5 Super-PON topology and architecture

See 164.1.

Insert Clause 164 and Annex 164A, Annex 164B, and Annex 164C as follows:

164. Physical Layers, Multipoint MAC Control, and management parameters for increased-reach point-to-multipoint Ethernet optical subscriber access (Super-PON)

164.1 Super-PON overview

The name Super-PON indicates a set of specifications for Physical Layers, Multipoint MAC Control sub-layer, and management parameters for optical subscriber access networks supporting point-to-multipoint operations using wavelength division multiplexing over an increased-reach (up to at least 50 km) passive optical network (PON). The defining element of a Super-PON optical distribution network (ODN) is the presence of a passive wavelength router that determines the channels supported by the ODN.

Figure 164–1 shows an exemplary structure of a Super-PON.

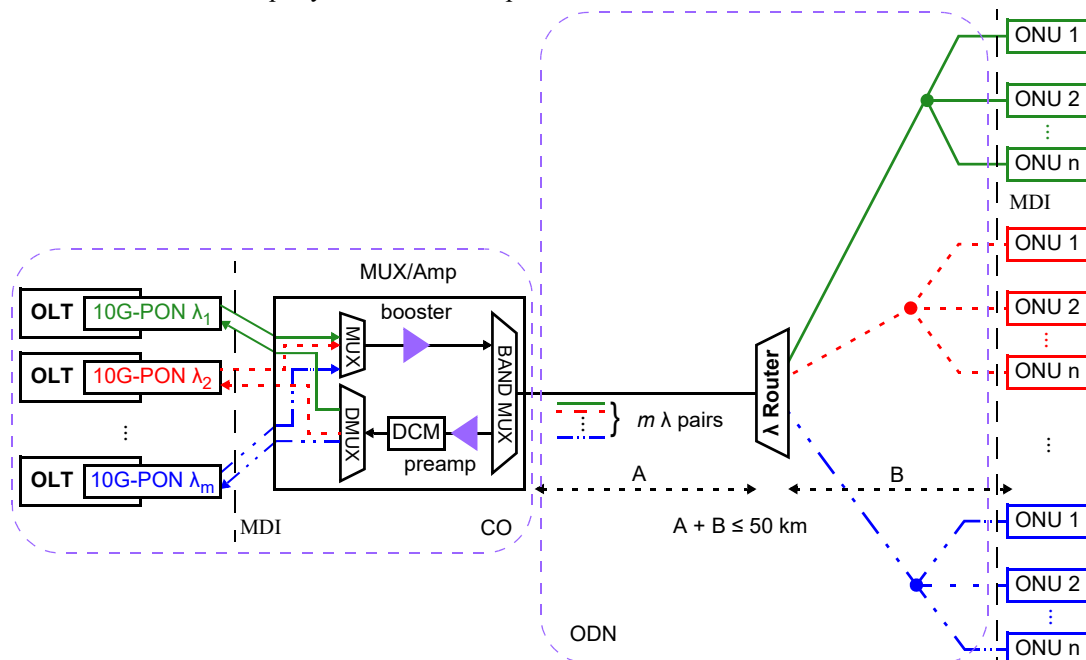


Figure 164–1—Example of Super-PON

As shown in Figure 164–1, the OLT MDI uses a dual-fiber arrangement, while the ONU MDI uses a single-fiber arrangement. The wavelength pairs defining the Super-PON channel plan are shown in Table 164–4.

In the downstream direction, the downstream wavelengths belonging to the channels used by the m OLTs are multiplexed, amplified and finally combined with the upstream wavelengths before getting to the ODN fiber. In the ODN, these wavelengths are routed by the remote wavelength router over its physical ports. Each wavelength reaches a power splitter and then the ONUs served by the origin OLT. The point of attachment to the ODN determines the specific wavelength pair an ONU uses to operate.

In the upstream direction, an ONU transmits using the upstream wavelength belonging to the channel selected by the OLT to which it is connected. The upstream wavelengths are multiplexed by the wavelength router toward the Central Office (CO). In the CO, the upstream wavelengths are separated from the

downstream ones, pre-amplified, processed by a dispersion compensation module (DCM), if needed, and finally demultiplexed for delivery to the appropriate OLT port.

Each OLT port operates over a specified channel determined by the operator in a CO, therefore it is assumed that OLT PMDs support a single wavelength channel. ONUs operate over the channel determined by their point of attachment to the ODN. On the receiving side, Super-PON ONUs use a broadband receiver. On the transmitting side, a variety of implementations are possible for ONUs, including ONUs that support a single channel, fully tunable ONUs able to support all channels, and partially tunable ONUs that support a contiguous subset of the defined channels.

164.2 Super-PON Physical Medium Dependent (PMD) Sublayer and Medium

164.2.1 Overview

164.2 describes the Physical Medium Dependent (PMD) sublayer for Super-PON point-to-multipoint (P2MP) networks (see 164.1) operating at a MAC data rate of 10 Gb/s in the downstream direction and at a MAC data rate of 10 Gb/s or 2.5 Gb/s in the upstream direction. These PMDs are collectively referred to as Super-PON PMDs. Super-PON PMDs supporting the upstream MAC data rate of 10 Gb/s are referred to as symmetric Super-PON PMDs while Super-PON PMDs supporting the upstream MAC data rate of 2.5 Gb/s are referred to as asymmetric Super-PON PMDs.

164.2.1.1 Positioning of the PMD sublayer within the IEEE 802.3 architecture

Figure 164–2 depicts the relationships of the Super-PON PMD sublayer (shown hatched) with other sublayers and the ISO/IEC Open System Interconnection (OSI) reference model.

164.2.2 PMD naming

Super-PON PMD naming conforms to the following convention, with individual elements shown in Table 164–1:

$$r_1/r_2\text{GBASE-SP}g\text{-}dab$$

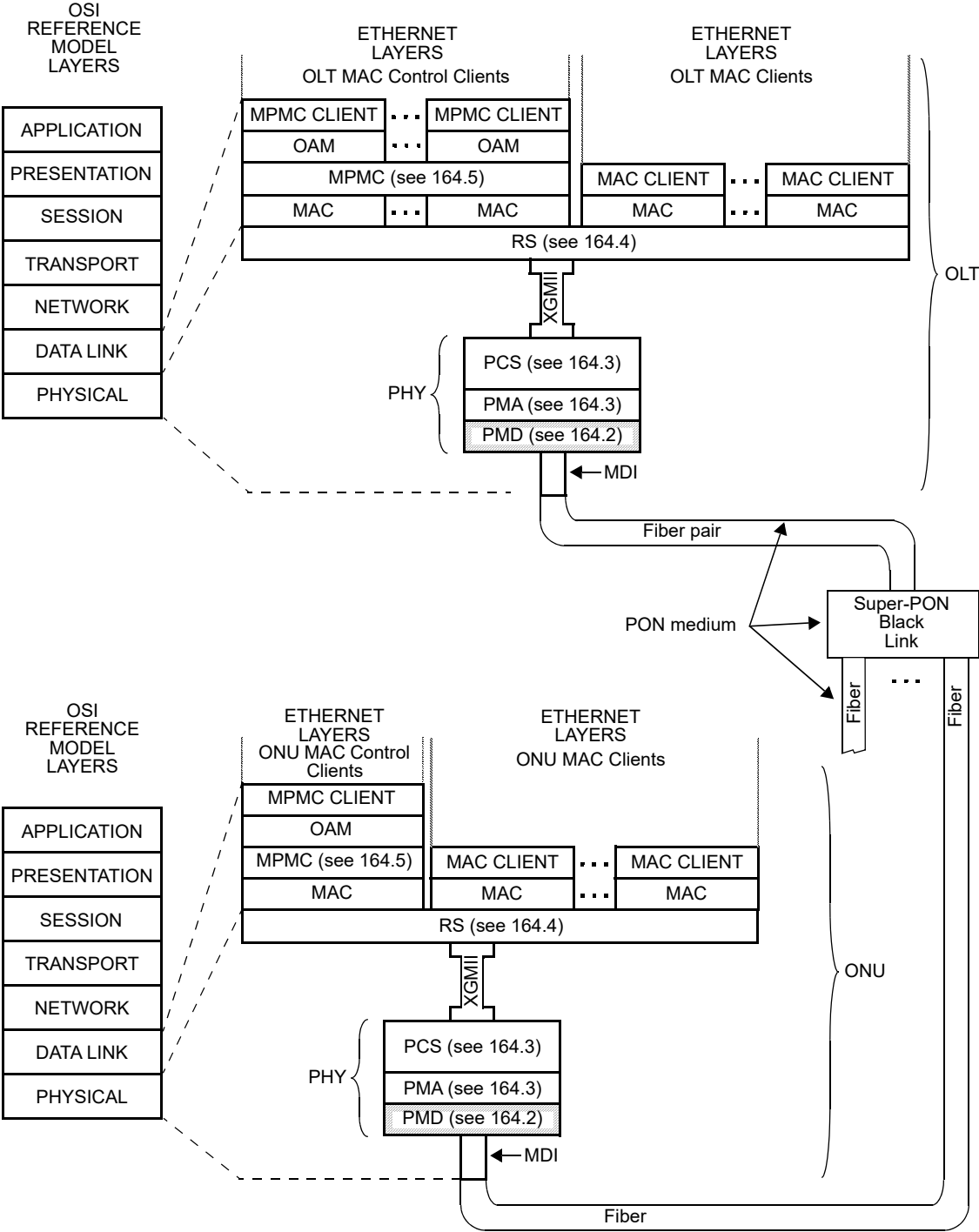
Table 164–1—Super-PON PMD naming convention

Parameter	Explanation	Allowed values
r_1	PMD downstream rate class (in Gb/s)	10
r_2^a	PMD upstream rate class (in Gb/s)	2.5, 10
G	PMD operates at Gigabit rates	—
BASE	—	—
SP	PMD is of Super-PON P2MP type	—
g	PMD FSR set	1, 2
d	PMD direction class type	D, U
a	PMD first transmit channel supported	0 to F^b
b^c	PMD last transmit channel supported	0 to F^c

^aIf r_1 is equal to r_2 (i.e., symmetric-data rate PMDs), r_2 is omitted.

^bSee Table 164–4.

^cIf a is equal to b (i.e., the PMD supports a single channel), then b is omitted. It is assumed that type D PMDs support a single channel.



PMD described in 164.2

XGMII= 10 GIGABIT MEDIA INDEPENDENT INTERFACE

MDI = MEDIUM DEPENDENT INTERFACE

OAM = OPERATIONS, ADMINISTRATION & MAINTENANCE

OLT = OPTICAL LINE TERMINAL

RS = RECONCILIATION SUBLAYER

MPMC= MULTIPPOINT MAC CONTROL

ONU = OPTICAL NETWORK UNIT

PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE

PMA = PHYSICAL MEDIUM ATTACHMENT

PMD = PHYSICAL MEDIUM DEPENDENT

Figure 164-2—Relationship of Super-PON PMD to the ISO/IEC OSI reference model and the IEEE 802.3 Ethernet model

For example, using this convention:

- 10G symmetric OLT PMD supporting only transmit channel 3, using FSR set 1, is identified as 10GBASE-SP1-D3.
- 10G symmetric ONU PMD supporting transmit channels 2 to D, using FSR set 1, is identified as 10GBASE-SP1-U2D.
- 10G/2.5G asymmetric ONU PMD supporting transmit channels 2 to 6, using FSR set 1, is identified as 10/2.5GBASE-SP1-U26.
- 10G/2.5G asymmetric ONU PMD supporting only transmit channel 4, using FSR set 1, is identified as 10/2.5GBASE-SP1-U4.

164.2.3 Supported combinations of OLT and ONU PMDs

Recommended pairings of OLT PMDs with ONU PMDs are shown in Table 164–2.

Table 164–2—OLT and ONU pairings

		ONU PMDs	
		10GBASE-SP1-Uyz	10/2.5GBASE-SP1-Uyz
OLT PMDs	10GBASE-SP1-Dx	10G/10G ^a	N/A
	10/2.5GBASE-SP1-Dx	N/A	10G/2.5G ^a

^aFor the pairing to work, channel x belongs to the channel interval {y, z}.

164.2.4 PMD functional specifications

The Super-PON type PMDs perform the transmit and receive functions that convey data between the PMD service interface and the MDI.

164.2.4.1 PMD service interface

The following specifies the services provided by Super-PON PMDs. These PMD sublayer service interfaces are described in an abstract manner and do not imply any particular implementation.

The PMD Service Interface supports the exchange of a continuous stream of bits between the PMA and PMD entities. The PMD translates the serialized data received from the compatible PMA to and from signals suitable for the specified medium. The following primitives are defined:

- PMD_UNITDATA.request
- PMD_UNITDATA.indication
- PMD_SIGNAL.request
- PMD_SIGNAL.indication

164.2.4.1.1 PMD_UNITDATA.request

This primitive defines the transfer of a serial data stream from the PMA defined in 164.3.4 to the PMD.

The semantics of the service primitive are PMD_UNITDATA.request(tx_bit). The data conveyed by PMD_UNITDATA.request is a continuous stream of bits. The tx_bit parameter can take one of two values: ONE or ZERO.

The PMA defined in 164.3.4 continuously sends the appropriate stream of bits to the PMD for transmission on the medium, at a nominal signaling rate of 10.3125 GBd in the case of Super-PON OLT and symmetric

Super-PON ONU PMDs or at a nominal signaling rate of 2.578125 GBd in the case of asymmetric Super-PON ONU PMDs. Upon receipt of this primitive, the PMD converts the specified stream of bits into the appropriate signals at the MDI.

164.2.4.1.2 PMD_UNITDATA.indication

This primitive defines the transfer of data from the PMD to the PMA defined in 164.3.4.

The semantics of the service primitive are PMD_UNITDATA.indication(rx_bit). The data conveyed by PMD_UNITDATA.indication is a continuous stream of bits. The rx_bit parameter can take one of two values: ONE or ZERO.

The PMD continuously sends a stream of bits to the PMA defined in 164.3.4 corresponding to the signals received from the MDI, at the nominal signaling rate of 10.3125 GBd in the case of Super-PON ONU and symmetric Super-PON OLT PMDs or at the nominal signaling rate of 2.578125 GBd in the case of asymmetric Super-PON OLT PMDs.

164.2.4.1.3 PMD_SIGNAL.request

In the upstream direction, this primitive is generated by the PMA defined in 164.3.4 to turn on and off the transmitter according to the granted time. A signal for laser control is generated as described in [142.2.5.4.3](#) for the PCS defined in 164.3.2.

The semantics of the service primitive are PMD_SIGNAL.request(tx_enable). The tx_enable parameter can take on one of two values: ENABLE or DISABLE, determining whether the PMD transmitter is on (enabled) or off (disabled). The PMA defined in 164.3.4 generates this primitive to indicate a change in the value of tx_enable. Upon the receipt of this primitive, the PMD turns the transmitter on or off as appropriate.

164.2.4.1.4 PMD_SIGNAL.indication

This primitive is generated by the PMD to indicate the status of the signal being received from the MDI. The semantics of the service primitive are PMD_SIGNAL.indication(SIGNAL_DETECT). The SIGNAL_DETECT parameter can take on one of two values: OK or FAIL, indicating whether the PMD is detecting light at the receiver (OK) or not (FAIL). When SIGNAL_DETECT = FAIL, PMD_UNITDATA.indication(rx_bit) is undefined. The PMD generates this primitive to indicate a change in the value of SIGNAL_DETECT. If the MDIO interface is implemented, then PMD_global_signal_detect shall be continuously set to the value of SIGNAL_DETECT.

NOTE—SIGNAL_DETECT = OK does not guarantee that PMD_UNITDATA.indication(rx_bit) is known good. It is possible for a poor quality link to provide sufficient light for a SIGNAL_DETECT = OK indication and still not meet the specified bit error ratio. PMD_SIGNAL.indication(SIGNAL_DETECT) has different characteristics for upstream and downstream links, see 164.2.4.5.

164.2.4.2 PMD block diagram

The PMD sublayer is defined at the reference points shown in Figure 164–3 for Super-PON type PMDs, using the Super-PON Black Link approach.

Test points TP1, TP2, TP3, and TP4 refer to the downstream channel, while test points TP5, TP6, TP7, and TP8 refer to the upstream channel. In the downstream channel, TP2 and TP3 are compliance points, while in the upstream channel TP6 and TP7 are compliance points. TP1, TP4, TP5, and TP8 are reference points for use by implementers. The optical transmit signal is defined at the output end of a patch cord (TP2 for the downstream channel and TP6 for the upstream channel), between 2 m and 5 m in length, of a fiber type consistent with the link type connected to the transmitter. Unless specified otherwise, all transmitter measurements and tests defined in 164.2.9 are made at TP2 or TP6.

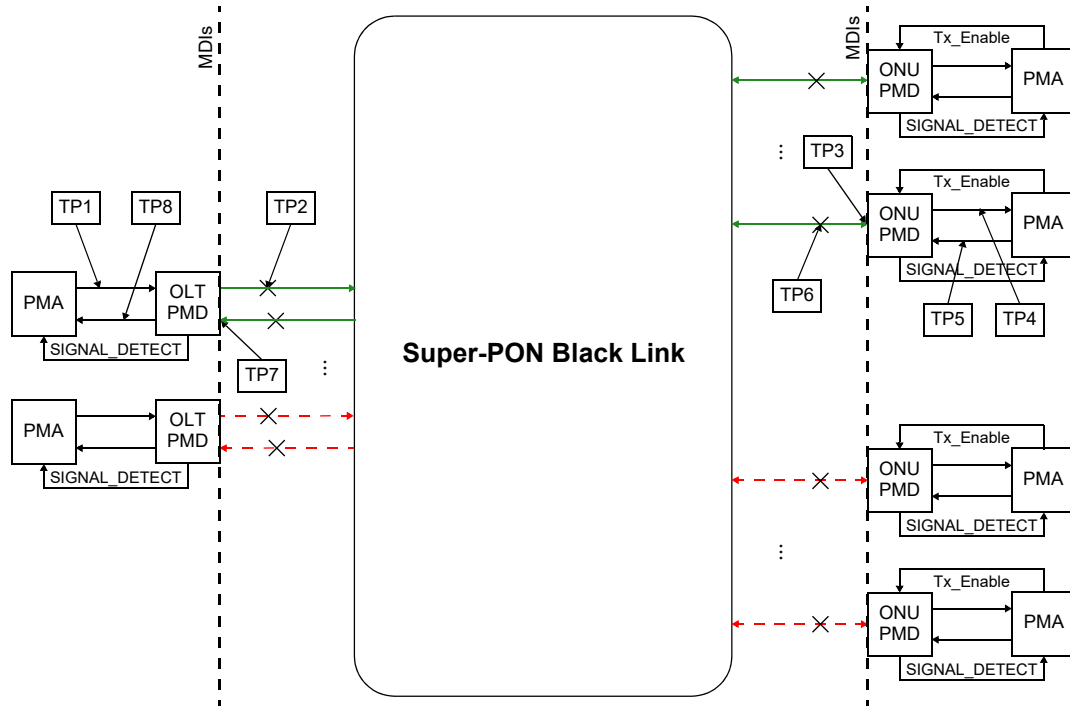


Figure 164-3—Super-PON PMD test points

The optical receive signal is defined at the output of the fiber optic cabling (TP3 for the downstream channel and TP7 for the upstream channel) connected to the receiver. Unless specified otherwise, all receiver measurements and tests defined in 164.2.9 are made at TP3 or TP7.

The electrical specifications of the PMD service interface (TP1 and TP4 for the downstream channel and TP5 and TP8 for the upstream channel) are not system compliance points (these are not readily testable in a system implementation).

An example of a Super-PON Black Link implementation is described in Annex 164A.

164.2.4.3 PMD transmit function

The PMD Transmit function shall convey the bits requested by the PMD service interface primitive `PMD_UNITDATA.request(tx_bit)` to the MDI according to the optical specifications in this clause. In the upstream direction, the flow of bits is interrupted according to `PMD_SIGNAL.request(tx_enable)`. This implies three optical levels, 1, 0, and dark, the latter corresponding to the transmitter being in the OFF state. The higher optical power level shall correspond to `tx_bit = ONE`.

164.2.4.4 PMD receive function

The PMD Receive function shall convey the bits received from the MDI according to the optical specifications in this clause to the PMD service interface using the primitive `PMD_UNITDATA.indication(rx_bit)`. The higher optical power level shall correspond to `rx_bit = ONE`.

164.2.4.5 PMD signal detect function

164.2.4.5.1 ONU PMD signal detect

The PMD Signal Detect function for the continuous mode downstream signal shall report to the PMD service interface, using the primitive `PMD_SIGNAL.indication(SIGNAL_DETECT)`, which is signaled continuously. `PMD_SIGNAL.indication` is intended to be an indicator of the presence of the optical signal. The ONU PMD receiver is not required to verify whether a compliant Super-PON signal is being received.

164.2.4.5.2 OLT PMD signal detect

The response time for the PMD Signal Detect function for the burst mode upstream signal may be longer or shorter than a burst length; thus, it may not fulfill the traditional requirements placed on Signal Detect. `PMD_SIGNAL.indication` is intended to be an indicator of optical signal presence. The signal detect function in the OLT may be realized in the PMD or the PMA defined in 164.3.4. The OLT PMD receiver is not required to verify whether a compliant Super-PON signal is being received.

164.2.4.5.3 Super-PON signal detect functions

The value of the `SIGNAL_DETECT` parameter for Super-PON PMDs shall be generated according to the conditions defined in Table 164–3.

Table 164–3—`SIGNAL_DETECT` value definitions for Super-PON PMDs

PMD type	Receive conditions	<code>SIGNAL_DETECT</code> value
Super-PON type PMD	Average input optical power $\leq$ Signal detect threshold (min) in Table 164–6 or Table 164–8, as applicable	FAIL
	Average input optical power $\geq$ Minimum mean input power in Table 164–6 or Table 164–8 with a compliant signal input, as applicable	OK
	All other conditions	Unspecified

164.2.4.6 PMD transmit enable function for ONU

`PMD_SIGNAL.request(tx_enable)` is defined for all ONU PMDs specified in this clause. `PMD_SIGNAL.request(tx_enable)` is asserted prior to data transmission by the ONU PMDs.

164.2.4.7 PMD delay constraints

The Super-PON PMD delay variation within the PMD shall be less than 0.25 EQT (see 1.4.313).

164.2.5 Wavelength channel allocation

Downstream and upstream channel wavelength assignments shall be as defined in Table 164–4, where frequency values for individual FSR sets (1 and 2) and upstream and downstream bands are normative.

Table 164–4—Super-PON Channel Plan

Channel	FSR set 1				FSR set 2			
	C-band 1 (downstream)		L-band 1 (upstream)		C-band 2 (downstream)		L-band 2 (upstream)	
	Frequency (THz)	Wave length (nm)	Frequency (THz)	Wave length (nm)	Frequency (THz)	Wave length (nm)	Frequency (THz)	Wave length (nm)
0	192.000	1561.42	187.613	1597.93	194.193	1543.78	189.807	1579.46
1	192.100	1560.61	187.711	1597.10	194.294	1542.98	189.906	1578.64
2	192.200	1559.79	187.809	1596.27	194.396	1542.18	190.004	1577.82
3	192.300	1558.98	187.906	1595.44	194.497	1541.38	190.103	1577.00
4	192.400	1558.17	188.004	1594.61	194.598	1540.57	190.202	1576.18
5	192.500	1557.36	188.102	1593.78	194.699	1539.77	190.301	1575.36
6	192.600	1556.56	188.200	1592.95	194.800	1538.97	190.400	1574.54
7	192.700	1555.75	188.297	1592.12	194.901	1538.18	190.499	1573.73
8	192.800	1554.94	188.395	1591.30	195.003	1537.38	190.597	1572.91
9	192.900	1554.13	188.493	1590.47	195.104	1536.58	190.696	1572.09
A	193.000	1553.33	188.590	1589.65	195.205	1535.78	190.795	1571.28
B	193.100	1552.52	188.688	1588.83	195.306	1534.99	190.894	1570.47
C	193.200	1551.72	188.786	1588.00	195.407	1534.19	190.993	1569.65
D	193.300	1550.92	188.883	1587.18	195.508	1533.40	191.092	1568.84
E	193.400	1550.12	188.981	1586.36	195.609	1532.61	191.191	1568.03
F	193.500	1549.32	189.079	1585.54	195.711	1531.82	191.289	1567.22

164.2.6 PMD to MDI optical specifications for OLT PMDs

This subclause details the PMD to MDI optical specifications for OLT PMDs. Specifically, 164.2.6.1 defines the OLT transmit parameters, while 164.2.6.2 defines the OLT receive parameters.

164.2.6.1 OLT transmitter optical specifications

A Super-PON OLT PMD transmitter shall comply with the parameters shown in Table 164–5.

Table 164–5—OLT PMD transmit characteristics

Parameter	10GBASE-SP1-Dx 10/2.5GBASE-SP1-Dx	Unit
Signaling speed (range)	10.3125 ± 100 ppm	GBd
Channel center wavelengths (frequencies)	See Table 164–4, C-band 1 (downstream)	—
Maximum spectral excursion	± 15	GHz
Average launch power, each channel (max)	1.5	dBm
Average launch power, each channel (min)	–2.5	dBm
Minimum side mode suppression ratio (SMSR)	35	dB

Table 164–5—OLT PMD transmit characteristics (*continued*)

Parameter	10GBASE-SP1-Dx 10/2.5GBASE-SP1-Dx	Unit
Minimum channel extinction ratio	8.2	dB
Transmitter eye mask definition {X1, X2, X3, Y1, Y2, Y3}	{0.25, 0.4, 0.45, 0.25, 0.28, 0.4}	UI
Transmitter and dispersion penalty (TDP), for 0 to 910 ps/nm residual chromatic dispersion (CD)	1.0 ^a	dB
RIN _{ISO} (max)	−128	dB/Hz
Average launch power of OFF transmitter (max)	−39	dBm
Optical return loss tolerance (max)	15	dB
Transmitter reflectance (max)	−10	dB

^aA negative chirp transmitter is assumed, which results in a negative dispersion penalty in the positive dispersion region.

164.2.6.2 OLT receiver optical specifications

A Super-PON OLT PMD receiver shall comply with the parameters shown in Table 164–6.

Table 164–6—OLT PMD receive characteristics

Parameter	10GBASE-SP1-Dx	10/2.5GBASE-SP1-Dx	Unit
Signaling speed (range)	10.3125 ± 100 ppm	2.578125 ± 100 ppm	GBd
Channel frequency range	187.600 to 189.092		THz
Bit error ratio (max)	10 ^{−2}		
Maximum mean input power	−6		dBm
Minimum mean input power ^a	−21	−25.5	dBm
Receiver OSNR tolerance ^a	11.9	8.4	dB ^b
Minimum OSNR	14	9.5	dB ^b
Receiver reflectance (max)	−12		dB
Damage threshold	−5		dBm
Signal detect threshold (min)	−45		dBm
T _{receiver_settling} (max)	800		ns

^aReceiver OSNR tolerance is defined at the minimum mean input power with ONU transmitter extinction ratio of 6 dB (see 164.2.9.5).

^bOSNR measured with 0.1 nm spectral resolution.

164.2.7 PMD to MDI optical specifications for ONU PMDs

This subclause details the PMD to MDI optical specifications for ONU PMDs. Specifically, 164.2.7.1 defines the ONU transmit parameters, while 164.2.7.2 defines the ONU receive parameters.

164.2.7.1 ONU transmitter optical specifications

A Super-PON ONU PMD transmitter shall comply with the parameters shown in Table 164–7.

Table 164–7—ONU PMD transmit characteristics

Parameter	10GBASE-SP1-Ux	10/2.5GBASE-SP1-Ux	Unit
Signaling speed (range)	10.3125 ± 100 ppm	2.578125 ± 100 ppm	GBd
Channel center wavelengths (frequencies)	See Table 164–4, L-band 1 (upstream)		—
Maximum spectral excursion (after turn-on time)	± 15		GHz
Average launch power, each channel (max)	8	4.5	dBm
Average launch power, each channel (min)	see Equation (164–1)	–0.5	dBm
Minimum side mode suppression ratio (SMSR) ^a	38		dBm
Minimum channel extinction ratio	see Equation (164–1)	6	dB
Transmitter eye mask definition {X1, X2, X3, Y1, Y2, Y3}	{0.25, 0.4, 0.45, 0.25, 0.28, 0.4}		UI
Transmitter and dispersion penalty (TDP), for –600 to +50 ps/nm residual CD, for –600 to +1020 ps/nm residual CD	2.0	1.0	dB
Average launch power of OFF transmitter (max)	–45		dBm
RIN ₁₅ OMA (max)	–128		dB/Hz
Turn-on time (max)	256		ns
Turn-off time (max)	128		ns
Transmitter reflectance (max)	–10		dB
Optical return loss tolerance	+15		dB

^aIt is assumed the SMSR is measured with only the dc laser bias (no data modulation).

$$P_{\min} = \begin{cases} -1.44 \cdot ER + 12.64 & \text{for } 4.5 \leq ER \leq 6 \\ -ER + 10 & \text{for } 6 \leq ER \leq 7 \\ 3 & \text{for } ER \geq 7 \end{cases} \quad (164-1)$$

where $P_{\min}$ is the minimum power level in dBm and ER is the extinction ratio in dB.

164.2.7.2 ONU receiver optical specifications

A Super-PON ONU PMD receiver shall comply with the parameters shown in Table 164–8.

164.2.8 Super-PON Black Link specification

Super-PON optical path shall comply with the parameters shown in Table 164–9 and Table 164–10.

Table 164–8—ONU PMD receive characteristics

Parameter	10GBASE-SP1-Ux 10/2.5GBASE-SP1-Ux	Unit
Signaling speed (range)	10.3125 ± 100 ppm	GBd
Channel frequency range	191.990 to 193.510	THz
Bit error ratio (max)	10 ⁻²	—
Maximum mean input power	−8	dBm
Minimum mean input power ^a	−31	dBm
Receiver OSNR tolerance ^a	+28	dB ^b
Minimum OSNR	+30	dB ^b
Receiver reflectance (max)	−12	dB
Damage threshold	−2	dBm
Signal detect threshold (min)	−44	dBm

^aReceiver OSNR tolerance is defined at the minimum mean input power with OLT transmitter extinction ratio of 8.2 dB (see 164.2.9.5).

^bOSNR measured with 0.1 nm spectral resolution.

Table 164–9—Optical path OLT to ONU

Parameter	10 Gb/s	Unit
Clear link passband	±15	GHz
Maximum ripple (within the clear link passband)	+2	dB
Maximum (residual) chromatic dispersion	+910	ps/nm
Minimum (residual) chromatic dispersion	0	ps/nm
Minimum optical return loss at transmitter	+42	dB
Maximum discrete reflectance between transmitter and receiver ^a	−26	dB
Maximum differential group delay	+12	ps
Maximum inter-channel crosstalk	−20	dB
Maximum optical path power penalty	+1.1	dB

^a Over all wavelengths specified for FSR set 1 in Table 164–4, including C-band (upstream) and L-band (downstream).

164.2.9 Definitions of optical parameters and measurement methods

The following subclauses describe definitive patterns and test procedures for Super-PON PMDs. Implementers using alternative verification methods need to ensure adequate correlation and allow adequate margin such that specifications are met by reference to the definitive methods. All optical measurements, except TDP and RIN₁₅OMA shall be made through a short patch cable between 2 m and 5 m in length.

Table 164–10—Optical path ONU to OLT

Parameter	10 Gb/s	2.5 Gb/s	Unit
Clear link passband	± 15		GHz
Maximum ripple (within the clear link passband)	+2		dB
Maximum (residual) chromatic dispersion	+50	+1020	ps/nm
Minimum (residual) chromatic dispersion	−600		ps/nm
Minimum optical return loss at transmitter	+32		dB
Maximum discrete reflectance between transmitter and receiver ^a	−26		dB
Maximum differential group delay	+12		ps
Maximum inter-channel crosstalk	−20		dB
Maximum optical path OSNR penalty	2.1	1.1	dB
Maximum loss excursion	10		dB
Burst-mode gain excursion ^b	0.5		dB

^a Over all wavelengths specified for FSR set 1 in Table 164–4, including C-band (upstream) and L-band (downstream).

^b For 4 channels with aligned bursts at the maximum allowed upstream power.

164.2.9.1 Insertion loss

Insertion loss for SMF fiber optic cabling (channel) is defined at the wavelengths specified in 164.2.5, depending on the particular PMD. A suitable test method is described in ITU–T G.650.1.

164.2.9.2 Test patterns

Two types of test patterns are used for testing of 10 Gb/s optical PMDs: square wave (see 52.9.1.2) and patterns 1, 2, or 3 (see 52.9.1.1). 2.5 Gb/s optical PMDs are tested with the same signals with the signaling period multiplied by a factor of 4.

164.2.9.3 Wavelength and side mode suppression ratio (SMSR)

The center wavelength (frequency) and side mode suppression ratio (SMSR) of the OLT shall meet the specifications when measured according to IEC 61280-1-3 under modulated conditions. The center wavelength (frequency) of the ONU shall meet the specification when measured according to IEC 61280-1-3 under modulated conditions. The modulation sequence is Pattern 1 or Pattern 3 (see 52.9.1.1) or a valid Super-PON signal.

The SMSR of the ONU shall meet the specification when measured according to IEC 61280-1-3 under non-modulated conditions.

NOTE—The allowable range of central wavelengths (frequencies) is narrower than the operating wavelength range by the actual RMS spectral width at each extreme.

164.2.9.4 Optical power measurements

Optical power shall meet specifications according to the methods specified in IEC 61280-1-1. A measurement may be made with the port transmitting any valid Super-PON signal.

164.2.9.5 Extinction ratio measurements

The extinction ratio shall meet the specifications when measured according to IEC 61820-2-2 with the port transmitting a valid Super-PON signal, and with minimal back reflections into the transmitter, lower than -20 dB. The test receiver has the frequency response as specified for the transmitter optical waveform measurement.

164.2.9.6 Relative intensity noise optical modulation amplitude (RIN_{15OMA}) measuring procedure

RIN is defined by the measurement methodology of 52.9.6, with an optical return loss value of 15 dB. The measurement filter bandwidth is appropriately equal to the signaling rate (i.e., 10.3 GHz for 10 Gb/s and 2.6 GHz for 2.5 Gb/s).

164.2.9.7 Transmit optical waveform (transmit eye)

The transmitter pulse shape characteristics for Super-PON PMDs at 10G shall meet the specifications of 158.8.7. For 2.5G PMDs, the specifications are similar except for the bandwidth of scope filter response and the jitter corner frequency of the CRU, which are 1/4 of the values for 10G.

164.2.9.8 Transmitter dispersion penalty (TDP)

TDP measurement tests transmitter impairments, including chromatic dispersion effects, due to signal propagation in SMF used in PON. Possible causes of impairment include intersymbol interference, jitter, and RIN. Meeting the separate requirements (e.g., eye mask, spectral characteristics) does not in itself guarantee the TDP. The TDP limit shall be met. TDP is measured as defined in 52.9.10 with an optical channel that meets the requirements listed in 164.2.9.8.2.

164.2.9.8.1 Reference transmitter requirements

See 52.9.10.1. Additionally, the optical signal generation shall include the use of a 50-GHz Gaussian optical filter or similar to emulate the wavelength selective ODN of Super-PON.

164.2.9.8.2 Reference receiver requirements

See 52.9.10.3.

164.2.9.8.3 Test procedure

See 52.9.10, using a reference BER of 10^{-2} and a total optical power for all optical power measurements in lieu of OMA.

164.2.9.9 Receiver OSNR tolerance

See ITU-T G.698.2, section 7.4.3, using a noise bandwidth of 12.5 GHz instead of 0.1 nm.

The jitter in the test transmit signal is defined in 52.8.1 for all 10G signals for the ONU and OLT. For 2.5G receivers in the OLT, the x-axis is divided by exactly 4, to compensate for the 4 times lower signaling rate.

164.2.9.10 Laser on/off timing measurement

T_{on} is defined in 164.2.9.10.1 and shall be less than or equal to 256 ns (defined in Table 164–7).

T_{off} is defined in 164.2.9.10.1 and shall be less than or equal to 128 ns (defined in Table 164–7).

164.2.9.10.1 Definitions

See 141.7.13.1.

164.2.9.10.2 Test method

See 141.7.13.2, replacing “TP4[i]” with “TP4”.

164.2.9.11 Receiver settling timing measurement

$T_{rx_settling}$ is defined in 164.2.9.11.1 and has a value of less than 800 ns. A method for measuring $T_{rx_settling}$ is illustrated in 164.2.9.11.2.

164.2.9.11.1 Definitions

$T_{rx_settling}$ is defined to be the time between when the optical power at TP7 (as in Figure 164–4) reaches the conditions specified in 164.2.9.10.1 for T_{on} and when the electrical modulation (peak-to-peak) at TP8 (as in Figure 164–4) remains within 15% of its steady state amplitude and jitter (see Table 164–6). The $T_{rx_settling}$ time interval is illustrated in Figure 164–4. The data transmitted may be any valid 256B/257B symbol (or a specific power synchronization sequence). The optical signal at TP7 (as in Figure 164–4), at the beginning of the locking, may have any valid 256B/257B pattern, optical power level, jitter, or frequency shift matching the standard specifications.

164.2.9.11.2 Test method

Figure 164–4 illustrates the test setup for measuring the OLT PMD receiver (upstream) $T_{rx_settling}$ time. The optical PMD transmitter has well-known parameters, with a fixed known T_{on} time. After T_{on} time the parameters of the reference transmitter, at TP6 and therefore at TP7, reach within 15% of their steady state values as specified in Table 164–7.

Define $T_{rx_settling}$ time as the time from the Tx_Enable assertion, minus the known T_{on} time, to the time the electrical signal at TP8 reaches within 15% of its steady state conditions.

Conformance needs to be assured for an optical signal at TP7 with any level of its specified parameters before the Tx_Enable assertion. Specifically, the $T_{rx_settling}$ time applies in the following scenarios:

- Switching from a ‘weak’ (minimal received power at TP7) ONU to a ‘strong’ (maximal received power at TP7) ONU, with minimal guard band between.
- Switching from a ‘strong’ ONU to a ‘weak’ ONU, with minimal guard band between.
- Switching from noise level, with maximal duration interval, to ‘strong’ ONU power level.

For a tested PMD receiver with a declared $T_{rx_settling}$ time, measure all PMD receiver electrical parameters at TP8 after $T_{rx_settling}$ from the TX_ENABLE trigger minus the reference transmitter T_{on} , ensuring conformance to within 15% of its specified steady state values.

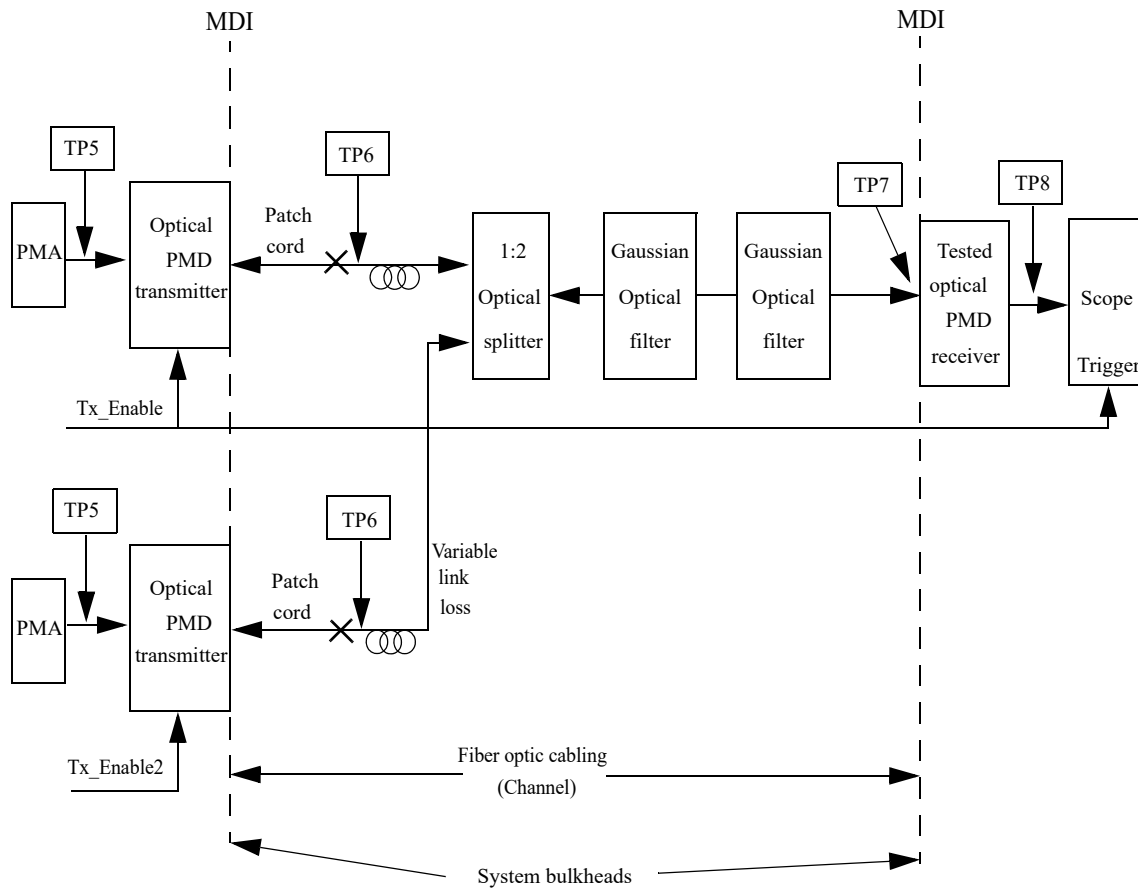


Figure 164-4—Receiver settling time measurement setup

164.2.9.12 Maximum ripple

See ITU-T G.698.2, section 7.3.1. The clear link passband shall be used instead of the maximum spectral excursion.

164.2.9.13 Clear link passband

The clear link passband is the frequency/wavelength range that a wavelength channel is expected to provide for the signal. The maximum ripple parameter applies only for frequencies/wavelengths within the clear link passband.

164.2.9.14 Maximum and minimum residual chromatic dispersion

See ITU-T G.698.2, section 7.3.2.

164.2.9.15 Maximum inter-channel crosstalk

See ITU-T G.698.2, section 7.3.8.

164.2.9.16 Maximum differential group delay

See ITU-T G.698.2, section 7.3.5.

164.2.9.17 Maximum optical path OSNR penalty

See ITU-T G.698.2, section 7.3.10.

164.2.9.18 Minimum optical return loss

See ITU-T G.698.2, section 7.3.3.

164.2.9.19 Maximum spectral excursion

See ITU-T G.989.2, section 11.1.5.2.4.

An ONU transmitter is considered in “stationary wavelength channel state” from “laser on” time after the start of a burst to the end of the burst. This is from the end of the T_{on} period to the start of the T_{off} period in Figure 164–4. For ONU transmitters, contributors to maximum spectral excursion include:

- Any wavelength tuning error from the nominal wavelength value.
- Any wavelength transient effects after the laser on time.

164.2.9.20 Maximum loss excursion

The maximum loss excursion is the maximum difference in optical insertion loss between all channels over upstream wavelengths.

164.2.9.21 Burst-mode gain excursion

The burst-mode gain excursion is the maximum allowed change in gain/loss of a channel in the upstream direction across all upstream traffic loads. The gain excursion specified should not be exceeded if up to four channels on different wavelengths have perfectly aligned bursts.

164.2.10 Environmental, safety, and labeling

164.2.10.1 General safety

All equipment subject to this clause shall conform to the general safety requirements as specified in [J.2](#).

164.2.10.2 Laser safety

Super-PON optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825–1 and IEC 60825–2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product’s laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.

164.2.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

164.2.10.4 Environment

All active components, including OLT and ONU, are expected to operate in a temperature controlled environment. Temperature is expected to be between 0–40 °C; humidity is expected to be less than 80%.

164.2.10.5 PMD labeling

The Super-PON labeling recommendations and requirements are as defined in 52.12.

The list of all supported PMDs is shown in Table 164–2.

Each field-pluggable component shall be clearly labeled with its operating temperature range over which compliance is ensured.

164.2.11 Characteristics of the fiber optic cabling

The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. It also includes a connector plug at each end to connect to the MDI. The fiber optic cabling spans from one MDI to another MDI, as shown in Figure 164–3.

164.2.11.1 Fiber optic cabling model

All fiber types that support single-mode transmission across the wavelength range specified in Table 164–4 may be used. The way to assemble the fiber optical cabling is not defined by this standard. The resulting optical paths shall meet the requirements specified in Table 164–9 and Table 164–10.

164.2.11.2 Medium Dependent Interface (MDI)

The Super-PON type PMD is coupled to the fiber cabling at the MDI. The MDI is the interface between the PMD and the PON medium, as shown in Figure 164–2. Examples of an MDI include the following:

- a) Connectorized fiber pigtail
- b) PMD receptacle

When the MDI is a remateable connection, it shall meet the interface performance specifications of IEC 61753–1. For Super-PON ONU PMD, the MDI is simplex (i.e., it couples to a single fiber). For Super-PON OLT PMD, the MDI is duplex (i.e., it couples to two fibers).

NOTE—Transmitter compliance testing is performed at TP2 or TP6 as defined in 164.2.4.2, not at the MDI.

164.2.12 Protocol implementation conformance statement (PICS) proforma for 164.2, Physical Medium Dependent (PMD) sublayer and medium for Super-PON passive optical networks⁷

164.2.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to 164.2, Physical Medium Dependent (PMD) sublayer and medium for Super-PON passive optical networks, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

164.2.12.2 Identification

164.2.12.2.1 Implementation identification

Supplier ¹	
Contact point for enquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

164.2.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3cs-2022, 164.2, Physical Medium Dependent (PMD) sublayer and medium for Super-PON passive optical networks
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3cs-2022.)	

Date of Statement	
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⁷Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

164.2.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*SPG10D	10GBASE-SP1-D	164.2.6	10G symmetric Super-PON OLT PMD	O.1	Yes [] No []
*SPG10U	10GBASE-SP1-U	164.2.7	10G symmetric Super-PON ONU PMD	O.1	Yes [] No []
*SPG102.5D	10/2.5GBASE-SP1-D	164.2.6	10/2.5G asymmetric Super-PON OLT PMD	O.1	Yes [] No []
*SPG102.5U	10/2.5GBASE-SP1-U	164.2.7	10/2.5G asymmetric Super-PON ONU PMD	O.1	Yes [] No []
*INS	Installation/Cable	164.2.11	Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer	O	Yes [] No []

164.2.12.4 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
FN1	Transmit function	164.2.4.3	Conveys bits from PMD service interface to MDI	M	Yes []
FN2	Transmitter optical signal	164.2.4.3	Highest optical power transmitted is a logic 1	M	Yes []
FN3	Receive function	164.2.4.4	Conveys bits from MDI to PMD service interface	M	Yes []
FN4	Receiver optical signal	164.2.4.4	Higher optical power received is a logic 1	M	Yes []
FN5	ONU signal detect function	164.2.4.5.1	Mapping to PMD service interface	M	Yes []
FN6	ONU signal detect parameter	164.2.4.5.1	Generated according to Table 164-3	M	Yes []
FN7	OLT signal detect function	164.2.4.5.2	Mapping to PMD service interface	O/2	Yes [] No []
FN8	OLT signal detect function	164.2.4.5.2	Provided by higher layer	O/2	Yes [] No []
FN9	OLT signal detect parameter	164.2.4.5.2	Generated according to Table 164-3	O	Yes [] No []
FN10	Delay variation	164.2.4.7	Less than 0.25 EQT	M	Yes []
FN11	Channel plan	164.2.5	Channel to wavelength mapping	M	Yes []

164.2.12.5 PMD to MDI optical specifications for 10GBASE-SP1-D

Item	Feature	Subclause	Value/Comment	Status	Support
SPG10D1	10GBASE-SP1-D transmitter	164.2.6.1	Meets specifications in Table 164–5	SPG10D:M	Yes [] N/A []
SPG10D2	10GBASE-SP1-D receiver	164.2.6.2	Meets specifications in Table 164–6	SPG10D:M	Yes [] N/A []

164.2.12.6 PMD to MDI optical specifications for 10GBASE-SP1-U

Item	Feature	Subclause	Value/Comment	Status	Support
SPG10U1	10GBASE-SP1-U transmitter	164.2.7.1	Meets specifications in Table 164–7	SPG10U:M	Yes [] N/A []
SPG10U2	10GBASE-SP1-U receiver	164.2.7.2	Meets specifications in Table 164–8	SPG10U:M	Yes [] N/A []

164.2.12.7 PMD to MDI optical specifications for 10/2.5GBASE-SP1-D

Item	Feature	Subclause	Value/Comment	Status	Support
SPG102.5D1	10/2.5GBASE-SP1-D transmitter	164.2.6.1	Meets specifications in Table 164–5	SPG102.5D:M	Yes [] N/A []
SPG102.5D2	10/2.5GBASE-SP1-D receiver	164.2.6.2	Meets specifications in Table 164–6	SPG102.5D:M	Yes [] N/A []

164.2.12.8 PMD to MDI optical specifications for 10/2.5GBASE-SP1-U

Item	Feature	Subclause	Value/Comment	Status	Support
SPG102.5U1	10/2.5GBASE-SP1-U transmitter	164.2.7.1	Meets specifications in Table 164–7	SPG102.5U:M	Yes [] N/A []
SPG102.5U2	10/2.5GBASE-SP1-U receiver	164.2.7.2	Meets specifications in Table 164–8	SPG102.5U:M	Yes [] N/A []

164.2.12.9 Definitions of optical parameters and measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	164.2.9.1	2 m to 5 m in length	M	Yes []
OM2T	Center wavelength (frequency) and side mode suppression ratio (SMSR) of the OLT	164.2.9.3	Meets the specifications when measured according to IEC 61280-1-3 under modulated conditions	M	Yes []
OM2U1	Center wavelength (frequency) of the ONU	164.2.9.3	Meets the specifications when measured according to IEC 61280-1-3 under modulated conditions	M	Yes []
OM2U2	Side mode suppression ratio (SMSR) of the ONU	164.2.9.3	Meets the specification when measured according to IEC 61280-1-3 under non-modulated conditions	M	Yes []
OM3	Optical power	164.2.9.4		M	Yes []
OM4	Extinction ratio	164.2.9.5		M	Yes []
OM5	RIN ₁₅ OMA	164.2.9.6		M	Yes []
OM6	Transmitter pulse shape	164.2.9.7	According to 158.8.7 for 10G signal, and according to 158.8.7 with bandwidth of scope filter response and the jitter corner frequency of the CRU, which are 1/4 of the values for 10G	M	Yes []
OM7	Transmitter and dispersion penalty (TDP)	164.2.9.8		M	Yes []
OM8	Receiver OSNR tolerance	164.2.9.10		M	Yes []
OM9	Laser on/off timing measurement	164.2.9.10		M	Yes []
OM10	Receiver settling timing measurement	164.2.9.11		M	Yes []
OM11	Maximum ripple	164.2.9.12		M	Yes []
OM12	Clear link passband	164.2.9.13		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
OM13	Maximum and minimum residual chromatic dispersion	164.2.9.14		M	Yes []
OM14	Maximum inter-channel crosstalk	164.2.9.15		M	Yes []
OM15	Maximum differential group delay	164.2.9.16		M	Yes []
OM16	Maximum optical path OSNR penalty	164.2.9.17		M	Yes []
OM17	Minimum optical return loss	164.2.9.18		M	Yes []
OM18	Maximum spectral excursion	164.2.9.19		M	Yes []
OM19	Maximum loss excursion	164.2.9.20		M	Yes []
OM20	Burst-mode gain excursion	164.2.9.21		M	Yes []
OM21	T _{on} value	164.2.9.10	Less than or equal to 256 ns (defined in Table 164–7)	M	Yes []
OM22	T _{off} value	164.2.9.10	Less than or equal to 128 ns (defined in Table 164–7)	M	Yes []

164.2.12.10 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
FO1	Fiber optic cabling	164.2.11.1	Specified in Table 164–9 and Table 164–10	INS:M	Yes [] N/A []
FO2	MDI requirements	164.2.11.2	Meet the interface performance specifications of IEC 61753–1, if remateable	INS:O	Yes [] No [] N/A []

164.2.12.11 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	164.2.10.1	Conforms to the general safety requirements as specified in J.2	M	Yes []
ES2	Laser safety—IEC Hazard Level 1	164.2.10.2	Conform to Hazard Level 1 laser requirements defined in IEC 60825–1 and IEC 60825–2	M	Yes []
ES3	Documentation	164.2.10.2	Explicitly defines requirements and usage restrictions to meet safety certifications	M	Yes []
ES4	Operating temperature range	164.2.10.4	The operating temperature range is declared	M	Yes []
ES5	Operating temperature range label	164.2.10.5	Provided for field-pluggable components	M	Yes []

164.3 Super-PON Physical Coding Sublayer and Physical Media Attachment

164.3.1 Overview

The Super-PON Physical Coding Sublayer and Physical Media Attachment are based on the Nx25G-EPON Physical Coding Sublayer and Physical Media Attachment (see [Clause 142](#)) with scaled down speeds and support for only one ONU channel.

Subclause 164.3 describes the Physical Coding Sublayer (PCS) with forward error correction (FEC) and Physical Medium Attachment (PMA) used with Super-PON point-to-multipoint (P2MP) networks (see 164.1). This type of network requires that the Multipoint MAC Control sublayer exists above the MACs, as described in 164.5 (see Figure 164–5).

Figure 164–6 illustrates the functional block diagram of the Super-PON PHY with emphasis placed on the PCS. The Super-PON PCS supports Super-PON PMDs, where:

- The receive and transmit paths operate at 10.3125 GBd rate (symmetric Super-PON ONU/OLT), or
- The receive path operates at 10.3125 GBd rate and the transmit path operates at 2.578125 GBd (asymmetric Super-PON ONU), or
- The transmit path operates at 10.3125 GBd rate and the receive path operates at 2.578125 GBd (asymmetric Super-PON OLT).

See 46.1.6 for the definition of TXD, TXC, TX_CLK, RXD, RXC, and RX_CLK.

164.3.1.1 Conventions

See [142.1.1](#).

164.3.1.2 Delay constraints

The combined delay variation through the transmit path of the Super-PON PCS and PMA is expected to be less than 6 EQTs (see 1.4.313) for channels operating at 10.3125 GBd and less than 15 EQTs for channels operating at 2.578125 GBd.

The combined delay variation through the receive path of the Super-PON PCS and PMA is expected to be less than 2 EQTs for channels operating at 10.3125 GBd and less than 5 EQTs for channels operating at 2.578125 GBd.

The aforementioned delay variation limits are applicable only for the data units (either EQ or the corresponding 257-bit block) located at the fixed offset within the FEC codeword.

164.3.1.3 Burst transmission

See [142.1.3](#).

164.3.2 PCS transmit data path

The transmit direction of the Super-PON PCS operates as the transmit direction of the Nx25G-EPON PCS specified in [142.2](#), with the following data rate changes:

- In a Super-PON OLT, the PCS transmit function operates in a continuous mode at the 10.3125 GBd rate.
- In a Super-PON ONU, the PCS transmit function operates in burst mode at the 10.3125 GBd rate (symmetric ONU) or at the 2.578125 GBd rate (asymmetric ONU).

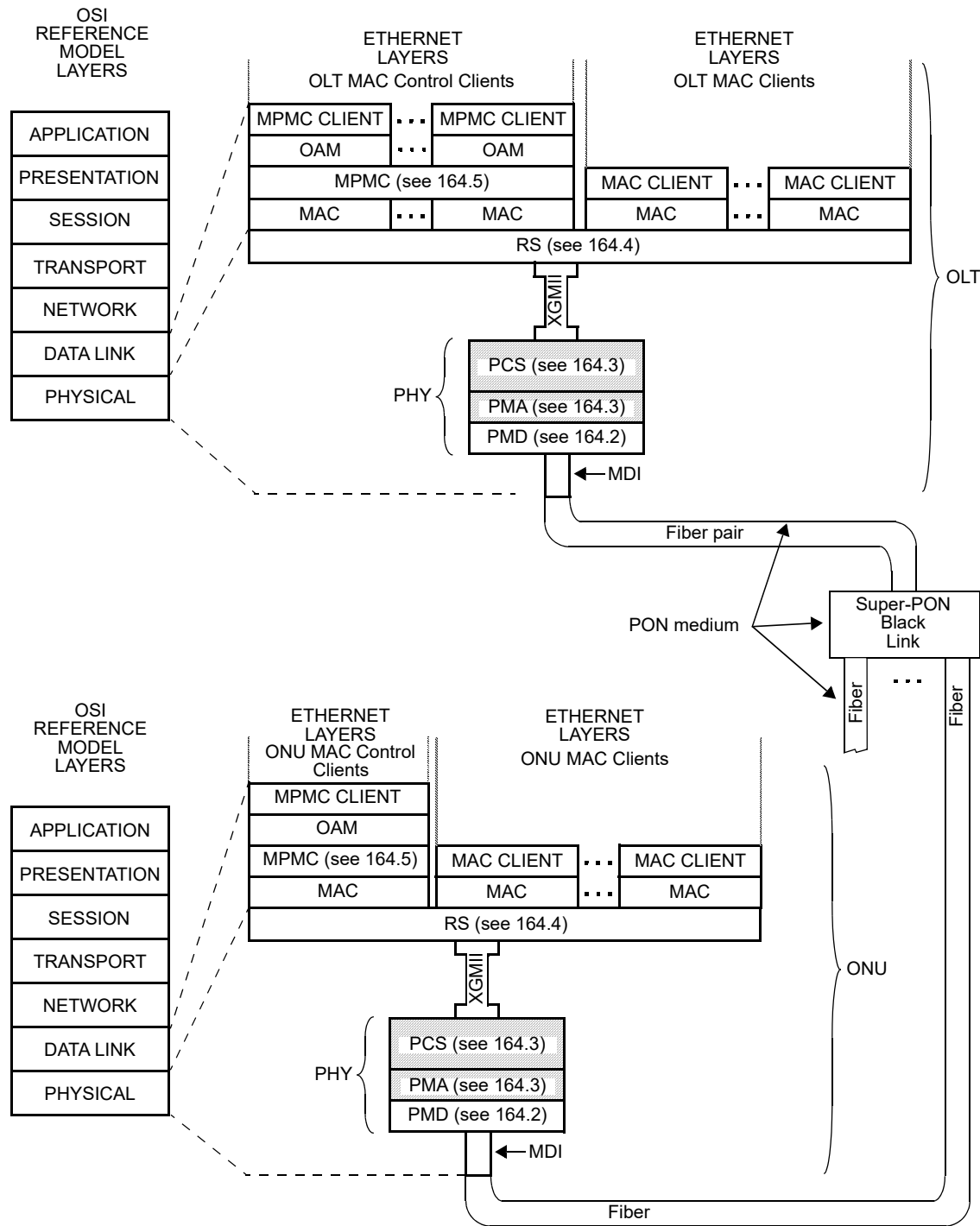


Figure 164-5—Relationship of Super-PON PCS and PMA to the ISO/IEC OSI reference model and the IEEE 802.3 Ethernet model

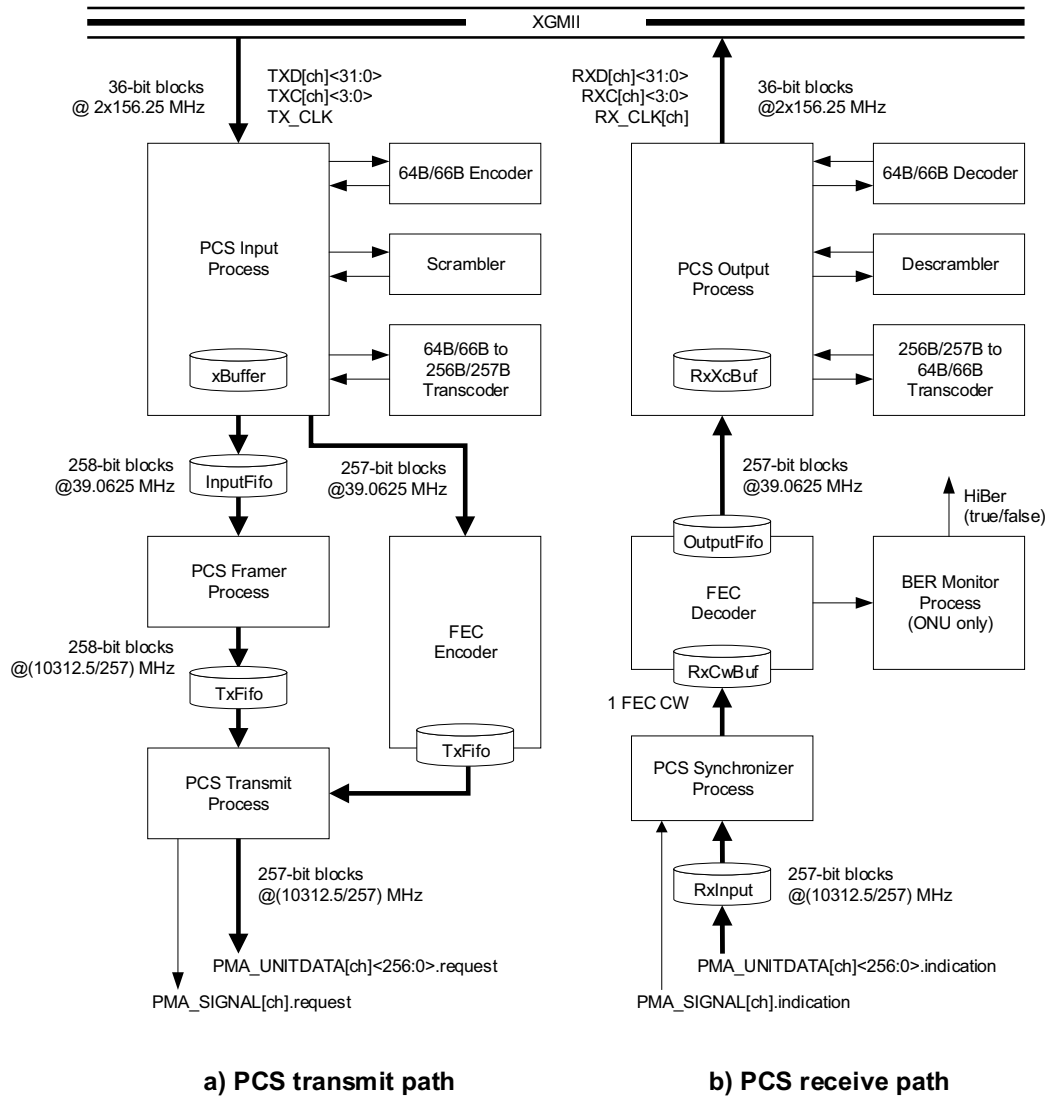


Figure 164-6—PCS Functional Block Diagram

164.3.3 PCS receive data path

The receive direction of the Super-PON PCS operates as the receive direction of the Nx25G-EPON PCS specified in 142.3, with the following data rate changes:

- In a Super-PON ONU, the PCS receive data path operates in a continuous mode at the 10.3125 GBd rate.
- In a Super-PON OLT, the PCS receive data path operates in burst mode at the 10.3125 GBd rate or at the 2.578125 GBd rate.

164.3.4 Super-PON PMA

The PMA adapts the serial PMD service interface (PMD_UNITDATA, see 141.3.3 and 141.3.4) to the 257-bit wide interface of the PCS (PMA_UNITDATA, see 142.4.1). The Super-PON PMA sublayer operates over only one channel, therefore it includes only one instance of the transmit data path and/or the receive data path.

In the downstream direction (from the OLT to the ONUs), the PMA includes a differential encoding option (see 142.4.2 and 142.4.3). This encoding technique facilitates the use of lower bandwidth receivers at the ONUs.

164.3.4.1 Service Interface

The PMA provides a service interface to the PCS. These services are described in an abstract manner and do not imply any particular implementation. The PMA service interface supports the exchange of 257-bit single data-unit vectors between PCS entities. The PMA converts 257-bit single data-unit vectors into bits and passes these to the PMD, and vice versa.

The following primitives are defined:

- PMA_UNITDATA[i].request(tx_code_group<256:0>)
- PMA_UNITDATA[i].indication(rx_code_group<256:0>)
- PMA_SIGNAL[i].request(tx_enable)
- PMA_SIGNAL[i].indication(SIGNAL_OK)

where [i] is always set to zero to indicate the PMA sublayer operates over only one channel.

164.3.4.1.1 PMA_UNITDATA[i].request

This primitive defines the transfer of data (in the form of 257-bit single data-unit vectors) from the PCS to the PMA by the PCS Transmit process, see 142.2.

For semantics of the service primitive, see 142.4.1.1.1.

The PCS continuously sends tx_code_group<256:0> single data-unit vectors to the PMA according to the PMA transmit clock at either (10.3125/257) GHz or (2.578125/257) GHz as defined in 142.4.4.

For effect of receipt, see 142.4.1.1.3.

164.3.4.1.2 PMA_UNITDATA[i]. indication

This primitive defines the transfer of data (in the form of 257-bit single data-unit vectors) from the PMA to the PCS. PMA_UNITDATA[i].indication is used by the PCS receive path processes, see 142.3.5.

For semantics of the service primitive, see 142.4.1.2.1.

The PMA continuously sends rx_code_group<256:0> single data-unit vectors to the PCS according to the PMA transmit clock at either (10.3125/257) GHz or (2.578125/257) GHz as defined in 142.4.4.

For effect of receipt, see 142.4.1.2.3.

164.3.4.1.3 PMA_SIGNAL[i].request

See 142.4.1.3.

164.3.4.1.4 PMA_SIGNAL[i].indication

See 142.4.1.4.

164.3.4.2 Differential encoder

See 142.4.2.

164.3.4.2.1 Differential decoder

See 142.4.3.

164.3.4.3 PMA transmit clock

The data conveyed by PMA_UNITDATA.request() is a 257-bit vector representing a single data-unit which has been prepared for transmission by the PMA client. For the PMA devices transmitting at 10.3125 GBd, the nominal frequency of the PMA transmit clock is 10.3125/257 GHz. For the PMA devices transmitting at 2.578125 GBd, the nominal frequency of the PMA transmit clock is 2.578125 / 257 GHz.

164.3.4.3.1 Loop-timing specifications for ONUs

ONUs shall operate at the same time basis as the OLT, i.e., the ONU PMA transmit clock tracks the ONU PMA receive clock. For the Super-PON ONUs supporting 2.5G transmission (i.e., asymmetric ONUs), the PMA transmit clock is derived from the PMA receive clock with a frequency of exactly 1/4 of the latter.

164.3.4.4 TCDR measurement

The test of the OLT PMA receiver TCDR time assumes that there is an optical PMD transmitter at the ONU with a well-known T_{on} time as defined in 164.2.9.10, and an optical PMD receiver at the OLT with a well-known $T_{rx_settling}$ time as defined in 164.2.9.11. After the $T_{on} + T_{rx_settling}$ time, the parameters at TP8 reach within 15 % of their steady-state values.

Set up the test ONU/OLT test system for 10^{-2} BER. Assuming a 3-zone SP1, SP2, and SP3 upstream ONU burst structure as shown in Figure 142-3, program the ONU SP1 TX pattern length so that the SP1 pattern ends at the precise end of the well-known OLT receiver settling time (within one 257-bit block of SP1, or ~ 10 ns granularity). Starting with the SP2 pattern of zero length (zero 257-bit blocks), test for SP3 detection. If the detection fails, increase the SP2 length by one and repeat the test until SP3 pattern is detected reliably. The number of 257-bit SP2 blocks times the length of each block is the TCDR time, with a margin of error of one 257-bit block time. To reduce hysteresis, increase the number of 257-bit SP2 blocks several hundred nanoseconds beyond this point (20 to 30 additional 257-bit SP2 blocks), and then start decrementing the number of 257-bit SP2 blocks, testing for the SP3 detection at each decrement, until the SP3 SBD is not detected at the OLT. The SP2 block time counting forward and backward shall be less than 400 ns.

164.3.5 Protocol implementation conformance statement (PICS) proforma for 164.3, Super-PON Physical Coding Sublayer and Physical Media Attachment

The supplier of a protocol implementation that is claimed to conform to 164.3, Super-PON Physical Coding Sublayer and Physical Media Attachment, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

164.3.5.1 Identification

164.3.5.1.1 Implementation identification

Supplier ¹	
Contact point for enquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

164.3.5.1.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3cs-2022, 164.3, Super-PON Physical Coding Sublayer and Physical Media Attachment
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3cs-2022.)	

Date of Statement	
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164.3.5.2 PCS capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
PCS1	Transmission bit order	142.2	Per Figure 142–4	M	Yes []
PCS2	Control code values treated as errors	142.2.1	All control code values that do not appear in Table 142–2 are not to be transmitted and are treated as an error if received	M	Yes []
*OLT	OLT functionality		Device supports functionality required for OLT	O/1	Yes [] No []
*ONU	ONU functionality		Device supports functionality required for ONU	O/1	Yes [] No []

164.3.5.3 PCS Processes

Item	Feature	Subclause	Value/Comment	Status	Support
PSD1	FEC Encoder	142.2.4	Encodes the transmitted data stream using a quasi-cyclic QC-LDPC FEC, defined in 142.2.4.1.	M	Yes []
PSD1a	FEC codeword shortening	142.2.4.2	Supports FEC shortening	M	Yes []
PSD1b	FEC encoding process	142.2.4.2	Per 142.2.4.2	M	Yes []
PSD2	Input Process	142.2.5.4.1	As depicted in Figure 142–9	M	Yes []
PSD3	Framer Process	142.2.5.4.2	As depicted in Figure 142–10	M	Yes []
PSD4	Transmit Process	142.2.5.4.3	As depicted in Figure 142–11	M	Yes []
PSD5	64B/66B Decoder	142.3.4	As depicted in Figure 49-17	M	Yes []
PSD6a	Synchronizer Process in OLT	142.3.5.4	As depicted in Figure 142–14, for every enabled receive channel	OLT:M	Yes [] N/A []
PSD6b	Synchronizer Process in ONU	142.3.5.5	As depicted in Figure 142–15, for every enabled receive channel	ONU:M	Yes [] N/A []
PSD7	Output Process	142.3.5.7	As depicted in Figure 142–17, for every enabled receive channel	M	Yes []
PSD8	PCS ONU BER monitor Process	142.3.5.7	As depicted in Figure 142–16 for every enabled receive channel	ONU:M	Yes [] N/A []

164.3.5.4 PMA Processes

Item	Feature	Subclause	Value/Comment	Status	Support
PMA1	Differential Encoder in OLT	142.4.2	As depicted in Figure 142–18	OLT:M	Yes [] N/A []
PMA2a	Differential Decoder in ONU	142.4.3	As depicted in Figure 142–19	ONU:M	Yes [] N/A []
PMA2b	Automatic detection of differential encoding	142.4.3	ONU implements automatic detection of RX path differential encoding and enables decoder as appropriate	ONU:M	Yes [] N/A []
PMA3	ONU loop-timing	164.3.4.3.1	ONU PMA transmit clock tracks the ONU PMA receive clock	ONU:M	Yes [] N/A []
PMA4	TCDR measurement	164.3.4.4	The SP2 block time counting forward and backward is less than 400 ns	OLT:M	Yes [] N/A []
PMA5	<i>PMA_SIGNAL[i].request</i> , value in OLT	142.4.1.3	In OLT, this primitive always takes on the value of ON	OLT:M	Yes [] N/A []

164.4 Super-PON Reconciliation Sublayer

164.4.1 Overview

The Super-PON Reconciliation Sublayer is based on the Nx25G-EPON Reconciliation Sublayer (see [Clause 143](#)) with scaled down speeds and support for only one ONU channel.

The Super-PON Reconciliation Sublayer enables multiple MACs to interface with one XGMII. The Super-PON Reconciliation Sublayer is the same as the Nx25G-EPON Multi-Channel Reconciliation Sublayer (MCRS) using only one xMII interface. Figure 164–7 shows the relationship between this RS and the ISO/IEC OSI reference model.

The RS adapts the bit-serial protocols of the MAC to the parallel format of the Physical Coding Sublayer (PCS) service interface. Subclause 164.4 defines an RS as an interface between the MAC sublayer and one xMII, where xMII is used as a generic term for the XGMII interface operating at the 10 Gb/s or 2.5 Gb/s speeds.

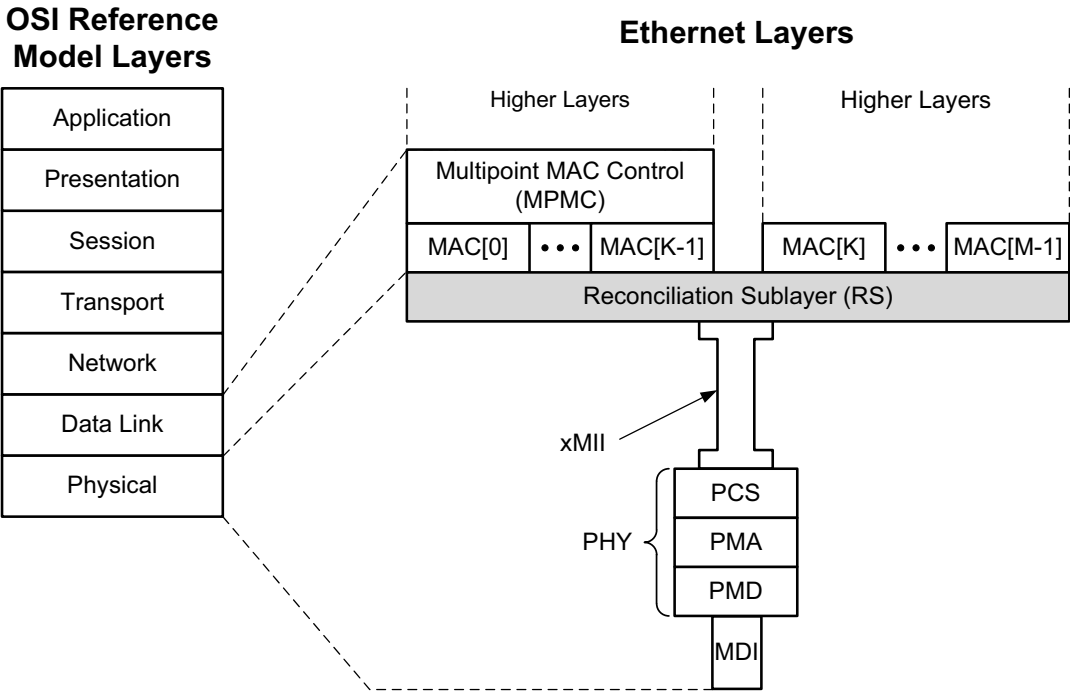


Figure 164–7—Relationship of RS to the OSI Reference Model

164.4.2 Summary of major concepts

See [143.2](#).

164.4.3 RS functional specifications

See [143.3](#).

164.4.4 Super-PON RS requirements

164.4.4.1 Super-PON architecture

This subclause describes the RS requirements for Super-PON networks. The RS is used with Super-PON networks in order to interface one MAC instance with one XGMII channel in each direction. Figure 164–8 illustrates the relationship of the RS and the OSI protocol stack for Super-PON.

164.4.4.2 RS channels

An RS channel that carries information from the OLT to the ONU is referred to as the downstream channel, and the channel that carries information from an ONU to the OLT is referred to as the upstream channel.

The Super-PON architecture shall implement a single RS channel in each direction. Each RS channel is bound to a separate PCS instance via a separate XGMII instance.

164.4.4.3 Symmetric and asymmetric data rates

The Super-PON architecture supports symmetric (10G/10G) and asymmetric data rates (10G/2.5G).

In asymmetric systems, the asymmetric data rate is achieved via the RS channel rate asymmetry, where a single downstream RS channel operates at 10 Gb/s and a single upstream RS channel operates at 2.5 Gb/s. Additional details for RS implementations supporting the channel rate asymmetry are provided in 164.4.4.7.

164.4.4.4 Super-PON application-specific parameters

For definitions of constants, variables, and functions, see [143.3.3](#) (transmit direction) and [143.3.4](#) (receive direction).

164.4.4.4.1 Constants

ADJ_BLOCK_SIZE
Value: 257

NUM_CH
Value: 1

RATE_ADJ_SIZE
Value: 3

164.4.4.4.2 Transmit variables

EnvTx

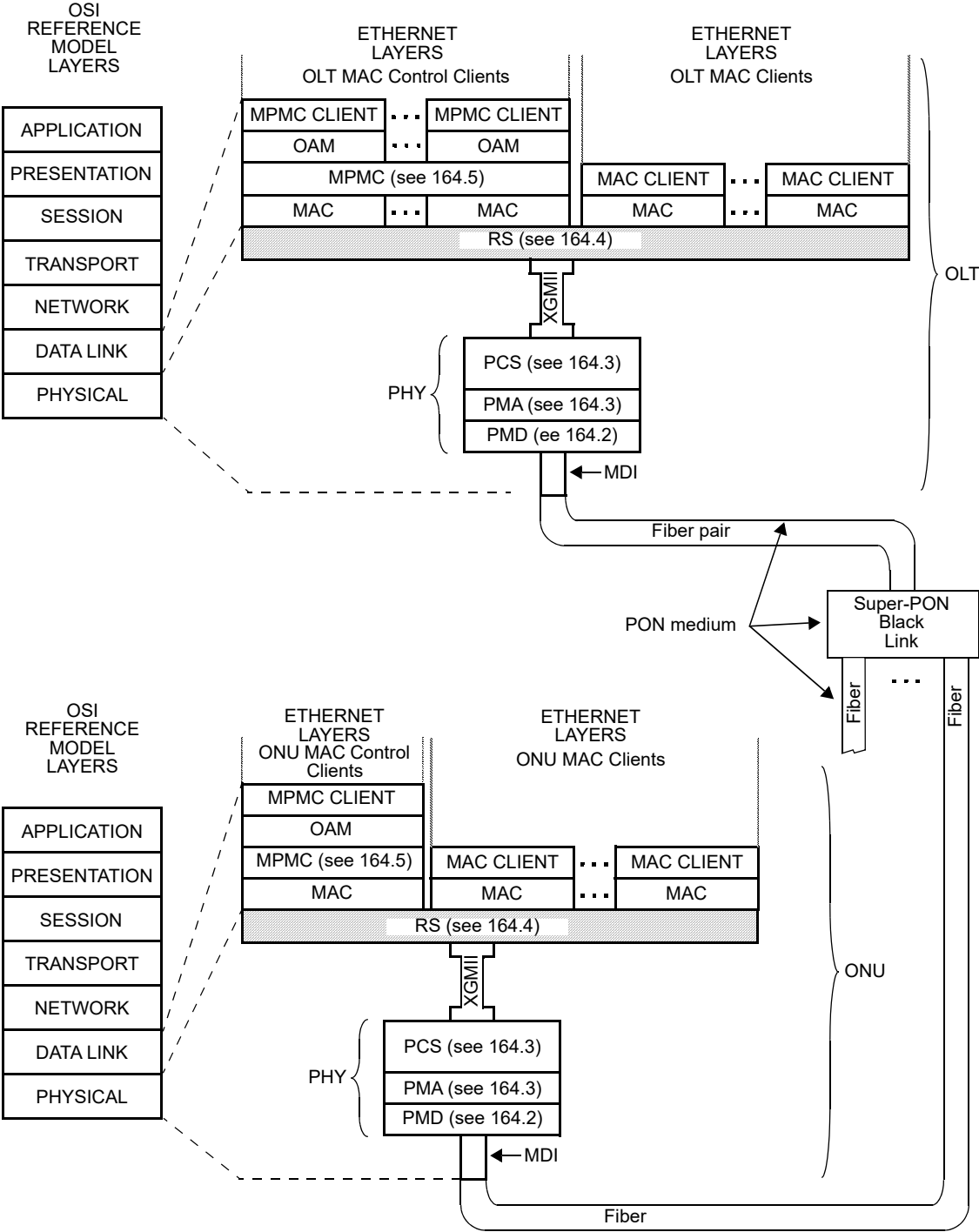
Description: Since there is no timing jitter or channel skew to be removed at the transmitting device, the size of EnvTx buffer may be reduced to only two rows. If this optimization is implemented, the variables rRow and wRow are represented by 1-bit unsigned integers.

164.4.4.5 RS time synchronization

See [143.4.2](#).

164.4.4.6 Delay variability constraints

See [143.4.3](#).



RS described in 164.4

XGMII= 10 GIGABIT MEDIA INDEPENDENT INTERFACE

MDI = MEDIUM DEPENDENT INTERFACE

OAM = OPERATIONS, ADMINISTRATION & MAINTENANCE

OLT = OPTICAL LINE TERMINAL

RS = RECONCILIATION SUBLAYER

MPMC= MULTIPPOINT MAC CONTROL

ONU = OPTICAL NETWORK UNIT

PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE

PMA = PHYSICAL MEDIUM ATTACHMENT

PMD = PHYSICAL MEDIUM DEPENDENT

Figure 164-8—Relationship of Super-PON RS to the ISO/IEC OSI reference model and the IEEE 802.3 Ethernet model

164.4.4.7 Asymmetric rate operation

In asymmetric Super-PON systems, downstream transmission uses one channel operating at 10 Gb/s, while the upstream transmission uses one channel operating at 2.5 Gb/s. Figure 164–9 illustrates the layering diagram of asymmetric Super-PON OLT and ONU. In the OLT, the RS sublayer serves MAC entities supporting the transmit data rate of 10 Gb/s and the receive data rate of 2.5 Gb/s. In the ONU, the RS sublayer serves MAC entities supporting the transmit data rate of 2.5 Gb/s and the receive data rate of 10 Gb/s.

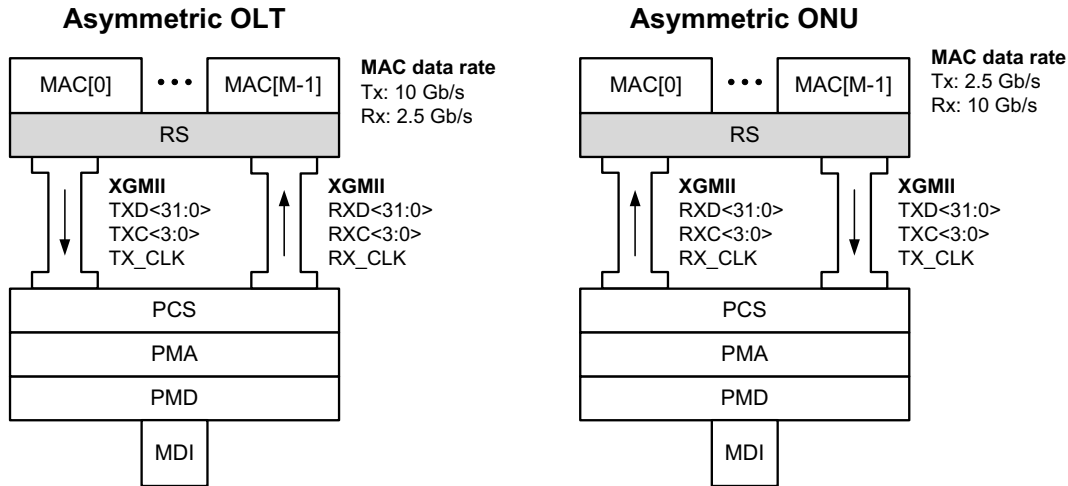


Figure 164–9—Asymmetric OLT and ONU layering diagram

Because of the required close coupling between the RS clock (InClk, see 143.3.3.4 and OutClk, see 143.3.4.3) and MPCP clock (LocalTime, see 144.2.1.2), the RS buffer read pointers advance by one every EQT, i.e., downstream and upstream channels within RS operate at a nominal data rate of 10 Gb/s. To adapt the RS channel rate to the MAC data rate of 2.5 Gb/s, the RS channel is throttled by inserting a padding EQ at the rate of 3 padding EQs per every 4 EQTs. The transfer of information through the 10 Gb/s RS channel is illustrated in Figure 164–10.

The padding EQs are interleaved with information EQs using the following pattern:

<information EQ> <padding EQ> <padding EQ> <padding EQ>

The usage of the padding EQs is entirely confined to the RS sublayer and does not affect the definition of interfaces to either of the adjacent sublayers. Therefore the definition of the padding EQ format and values are left to implementations.

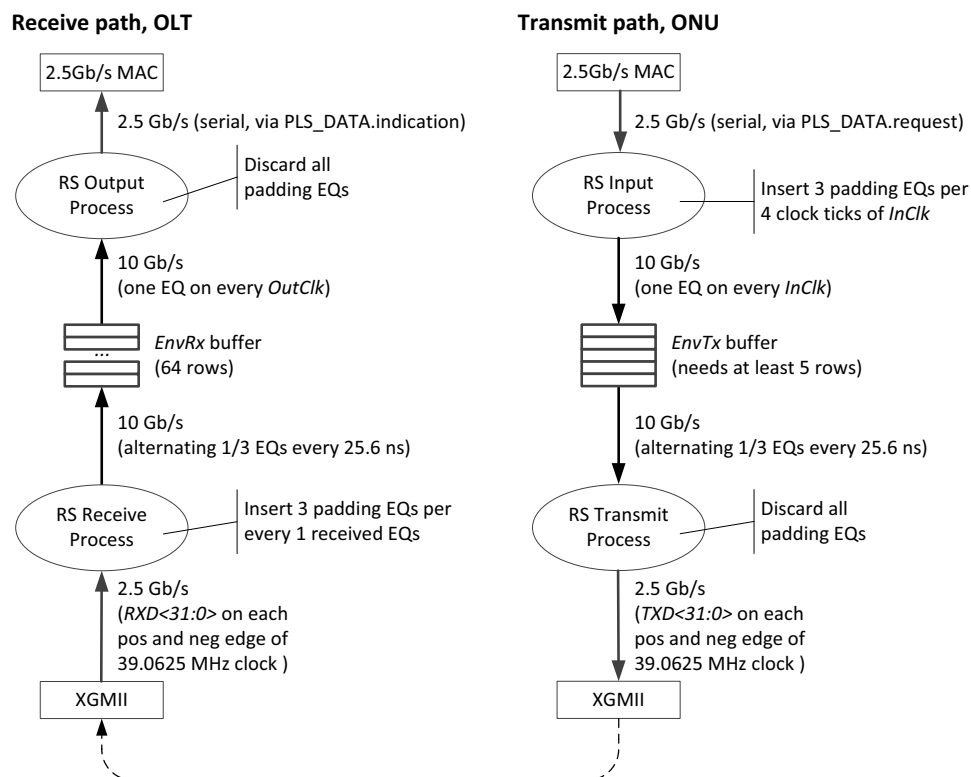


Figure 164–10—Upstream channel operating at 2.5 Gb/s

164.4.5 Protocol implementation conformance statement (PICS) proforma for 164.4, Super-PON Reconciliation Sublayer

The supplier of a protocol implementation that is claimed to conform to 164.4, Super-PON Reconciliation Sublayer, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

164.4.5.1 Identification

164.4.5.1.1 Implementation identification

Supplier ¹	
Contact point for enquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

164.4.5.1.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3cs-2022, 164.4, Super-PON Reconciliation Sublayer
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3cs-2022.)	

Date of Statement	
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164.4.5.2 Generic MCRS

Item	Feature	Subclause	Value/Comment	Status	Support
MC1	Envelope Header structure	143.3.2	Uses the format shown in Figure 143–10	M	Yes []
MC2	Input Process	143.3.3.6.1	Implements the state diagram as depicted in Figure 143–12	M	Yes []
MC3	Transmit Process	143.3.3.6.2	Implements the state diagram as depicted in Figure 143–13	M	Yes []
MC4	Receive Process	143.3.4.5.1	Implements the state diagram as depicted in Figure 143–15	M	Yes []
MC5	Output Process	143.3.4.5.2	Implements the state diagram as depicted in Figure 143–16	M	Yes []

164.4.5.3 RS in Super-PON

164.4.5.3.1 Major capabilities/option

Item	Feature	Subclause	Value/Comment	Status	Support
*102.5G	10/2.5G-EPON functionality	164.4.4.3	Device supports functionality required for 10/2.5G-EPON	O.1	Yes [] No []
*1010G	10/10G-EPON functionality	164.4.4.3	Device supports functionality required for 10/10G-EPON	O.1	Yes [] No []

164.4.5.3.2 RS implementation in Super-PON

Item	Feature	Subclause	Value/Comment	Status	Support
EPON1	Number of RS channels	164.4.4.2	Implement a single channel in downstream direction and a single channel in upstream direction	102.5G:M or 1010G:M	Yes [] N/A []

164.5 Super-PON Multipoint MAC Control Sublayer

164.5.1 Overview

The Super-PON Multipoint MAC Control Sublayer is based on the Nx25G-EPON Multipoint MAC Control Sublayer (see [Clause 144](#)) with scaled down speeds and support for only one ONU channel.

Subclause 164.5 updates the management parameters in [Clause 144](#) to define the mechanisms and control protocols required in order to reconcile Super-PON point-to-multipoint (P2MP) networks (see 164.1) into the Ethernet framework.

The Multipoint MAC Control (MPMC) sublayer defined in 164.5 includes the Multipoint Control Protocol (MPCP) responsible for arbitration of TDM-based access to the point-to-multipoint (P2MP) medium. The MPMC functionality shall be implemented for subscriber access devices containing point-to-multipoint (P2MP) Physical Layer devices defined in 164.5.

Figure 164–11 illustrates the functional block diagram of Super-PON OLT and ONUs with emphasis placed on the MPMC.

164.5.1.1 Principles of point-to-multipoint operation

See [144.1.1](#).

164.5.1.2 Position of Multipoint MAC Control (MPMC) within the IEEE 802.3 hierarchy

Figure 164–11 depicts the architectural positioning of the MPMC sublayer with respect to the MAC and the MPMC client. The MPMC sublayer extends the MAC Control sublayer to support multiple clients and additional MAC control functionality.

164.5.1.3 Functional block diagram

See [144.1.3](#).

NOTE—Super-PON does not use the Channel Control Protocol (CCP) defined for Nx25G-EPON.

164.5.1.4 Service interfaces

See [144.1.4](#).

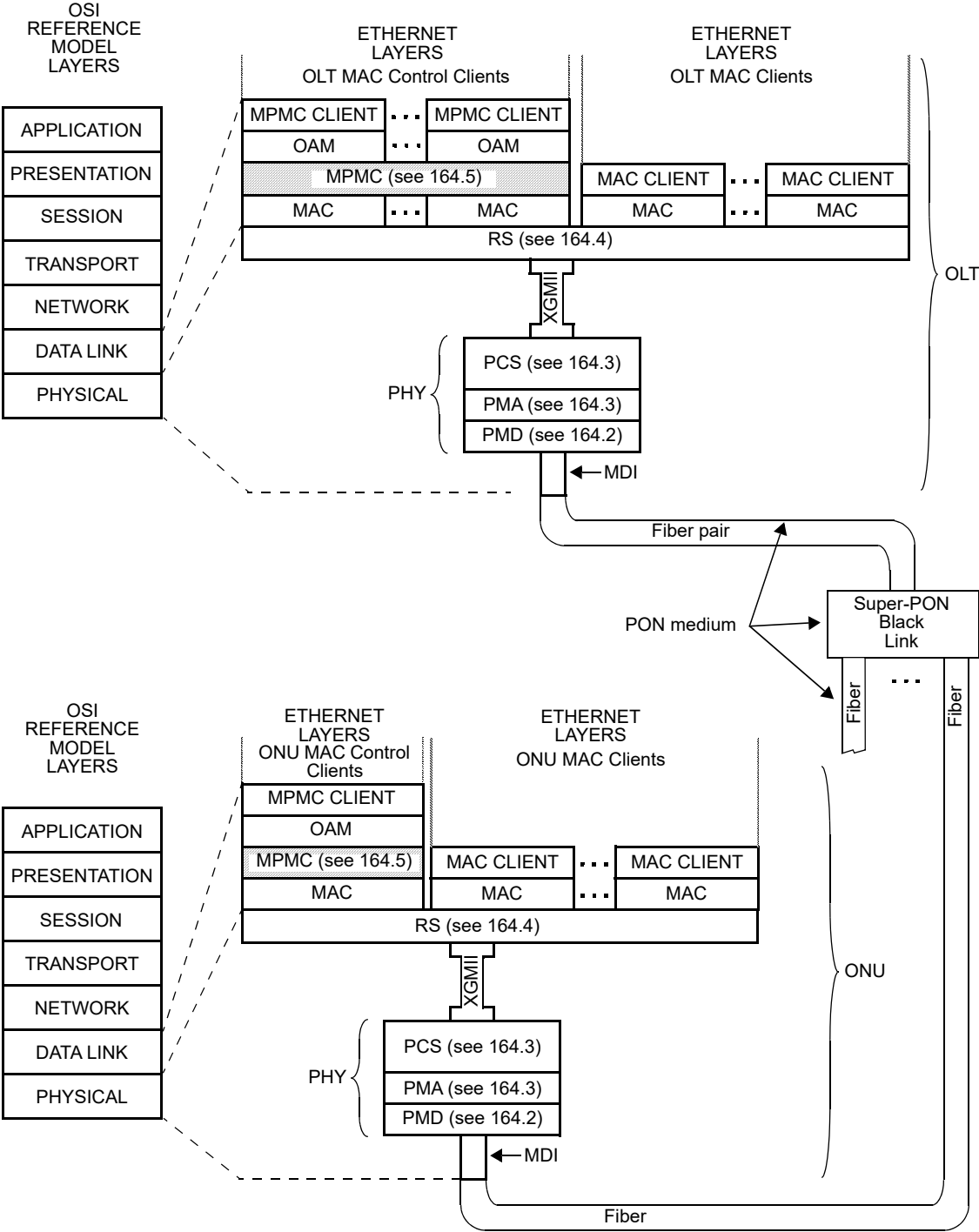
164.5.1.5 Conventions

See [142.1.1](#).

164.5.2 Protocol-independent operation

164.5.2.1 Control Parser and Control Multiplexer

The Control Parser (see [Figure 144–5](#)) is responsible for opcode-independent parsing of MAC frames and forwarding these frames to other processes for opcode-specific operations. The Control Parser also extracts the value of the Timestamp field from all MPCPDUs that contain this field and checks whether the timestamp drift value is within the acceptable range. There are no interfaces connecting the Control Parser to MAC Clients.



MPMC described in 164.5

XGMII= 10 GIGABIT MEDIA INDEPENDENT INTERFACE

MDI = MEDIUM DEPENDENT INTERFACE

OAM = OPERATIONS, ADMINISTRATION & MAINTENANCE

OLT = OPTICAL LINE TERMINAL

RS = RECONCILIATION SUBLAYER

MPMC= MULTIPPOINT MAC CONTROL

ONU = OPTICAL NETWORK UNIT

PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE

PMA = PHYSICAL MEDIUM ATTACHMENT

PMD = PHYSICAL MEDIUM DEPENDENT

Figure 164-11—Relationship of Super-PON MPMC to the ISO/IEC OSI reference model and the IEEE 802.3 Ethernet model

The Control Multiplexer (see [Figure 144–6](#)) is responsible for forwarding frames received from multiple opcode-specific processes to the underlying MAC sublayer. The Control Multiplexer inserts the timestamp value into all MPCPDUs that carry the Timestamp field. There are no interfaces connecting the Control Multiplexer to MAC Clients.

164.5.2.1.1 Constants

DRIFT_THOLD

Type: Integer

Description: This constant holds the maximum amount of drift allowed before a timestamp drift error is declared. Exceeding this drift causes ONU deregistration (either self-deregistration or deregistration by the OLT).

Value: 2 (for the receive channels operating at 10 Gb/s) or 3 (for the receive channels operating at 2.5 Gb/s)

Unit: EQT

164.5.2.1.2 Counters

LocalTime

Type: 32-bit unsigned

Description: This variable holds the value of the local timer used to control MPCP operation. This variable is advanced once per EQT. At the OLT the counter shall track the XGMII transmit clock, while at the ONU the counter shall track the XGMII receive clock. For accuracy of the receive clock, see 164.3.4.3.1. In the ONU, this variable is updated with the received timestamp value by the Control Parser Process (see [144.2.1.5](#)).

164.5.2.1.3 Variables

See [144.2.1.3](#).

164.5.2.1.4 Functions

See [144.2.1.4](#).

164.5.2.1.5 Control Parser state diagram

See [144.2.1.5](#).

164.5.2.1.6 Control Multiplexer state diagram

See [144.2.1.6](#).

164.5.3 Multipoint Control Protocol (MPCP)

164.5.3.1 Principles of Multipoint Control Protocol (MPCP)

See [144.3.1](#).

164.5.3.1.1 Ranging measurement and time synchronization

Ranging measurement and time synchronization are specified in [144.3.1.1](#). For Super-PON:

- In the OLT, the LocalTime counter is synchronized with the OLT XGMII transmit clock and increments synchronously with InClk (see [143.3.3.4](#)).

- In the ONU, the LocalTime counter is synchronized with the XGMII receive clock and increments synchronously with OutClk (see 143.3.3.4).

164.5.3.1.2 Granting access to the PON media by the OLT

See 144.3.1.2.

164.5.3.2 MPCP block diagram

See 144.3.2.

164.5.3.3 Delay variability requirements

The MPCP protocol relies on strict timing based on distribution of timestamps. A compliant implementation needs to guarantee a constant delay through the MAC and PHY in order to maintain the correctness of the timestamping mechanism. The actual delay is implementation dependent; however, a complying implementation shall maintain the combined delay variation through the MAC and PHY of less than one EQT for channels operating at 10.3125 GBd and less than two EQTs for channels operating at 2.578125 GBd.

164.5.3.4 Logical link identifier (LLID) types

See 144.3.4.

164.5.3.5 Allocation of LLID values

See 144.3.5.

164.5.3.6 MPCPDU structure and encoding

See 144.3.6.

164.5.3.6.1 GATE description

The GATE message is specified in 144.3.6.1. Super-PON uses only upstream channel 0 in the ChannelMap bits (see Table 144-2).

164.5.3.6.2 REPORT description

See 144.3.6.2.

164.5.3.6.3 REGISTER_REQ description

The REGISTER_REQ message is specified in 144.3.6.3. For Super-PON the RegisterRequestInfo field has the structure shown in Table 164–11.

Table 164–11—RegisterRequestInfo field

Bit	Flag field	Values
0	Reserved	Ignored on Reception
1	ONU is 10G upstream capable	0 – ONU transmitter is not capable of 10 Gb/s 1 – ONU transmitter is capable of 10 Gb/s
2	Reserved	Ignored on Reception

Table 164–11—RegisterRequestInfo field (continued)

Bit	Flag field	Values
3	ONU is 2.5G upstream capable	0 – ONU transmitter is not capable of 2.5 Gb/s 1 – ONU transmitter is capable of 2.5 Gb/s
4	Reserved	Ignored on Reception
5	10G registration attempt	0 – 10 Gb/s registration is not attempted 1 – 10 Gb/s registration is attempted
6	Reserved	Ignored on Reception
7	2.5G registration attempt	0 – 2.5 Gb/s registration is not attempted 1 – 2.5 Gb/s registration is attempted
15:8	Reserved	Ignored on Reception

164.5.3.6.4 REGISTER description

See 144.3.6.4.

164.5.3.6.5 REGISTER_ACK description

See 144.3.6.5.

164.5.3.6.6 DISCOVERY description

The DISCOVERY message is specified in 144.3.6.6. For Super-PON the DiscoveryInfo field has the structure shown in Table 164–12.

The flags in the DiscoveryInfo field allow the OLT to exercise discovery admission control over the unregistered ONUs. Specifically, the following ONU behavior is defined:

- The ONU shall not generate/transmit a REGISTER_REQ MPCPDU using the 10 Gb/s upstream channel if the OLT did not open the 10 Gb/s discovery window, i.e., if bit 5 was set to 0.
- The ONU shall not generate/transmit a REGISTER_REQ MPCPDU using the 2.5 Gb/s upstream channel if the OLT did not open the 2.5 Gb/s discovery window, i.e., if bit 7 was set to 0.

The values of the DiscoveryInfo field flags are set by the OLT MPMC client and may change from one discovery attempt to the next. The OLT MPMC client may allow a concurrent registration of ONUs with different rates by setting bits 5 and 7 to 1. The processing of DiscoveryInfo flags by the ONU and the ONU behavior in dual-rate systems is further specified in 164.5.3.9.

Table 164–12—DiscoveryInfo field

Bit	Flag field	Values
0	Reserved	Ignored on Reception
1	OLT is 10G upstream capable	0 – OLT does not support 10 Gb/s reception 1 – OLT supports 10 Gb/s reception
2	Reserved	Ignored on Reception
3	OLT is 2.5G upstream capable	0 – OLT does not support 2.5 Gb/s reception 1 – OLT supports 2.5 Gb/s reception
4	Reserved	Ignored on Reception
5	OLT is opening 10G discovery window	0 – OLT is not capable of receiving 10 Gb/s data in this window 1 – OLT is capable of receiving 10 Gb/s data in this window

Table 164–12—DiscoveryInfo field

Bit	Flag field	Values
6	Reserved	Ignored on Reception
7	OLT is opening 2.5G discovery window	0 – OLT is not capable of receiving 2.5 Gb/s data in this window 1 – OLT is capable of receiving 2.5 Gb/s data in this window
9:8	Reserved	Ignored on Reception
13:10	Super-PON Channel information	Encodes the channel number (see Table 164–4) the OLT is operating on
15:14	Reserved	Ignored on Reception

164.5.3.6.7 SYNC_PATTERN description

See 144.3.6.7.

164.5.3.7 Discovery process

See 144.3.7.

164.5.3.7.1 Constants

ACK_GATE_LIMIT

As defined in 144.3.7.2.

DISCOVERY_MARGIN

Type: Integer

Description: This constant holds the extra margin reserved at the end of a discovery grant to accommodate the largest possible round-trip time on a given ODN. The round-trip time also includes any internal delays in the OLT and ONU, such as FEC encoding and decoding delays.

Value: 78,906 (505 μ s for ODN with 50 km reach)

Unit: EQT

MISSED_REPORT_LIMIT

As defined in 144.3.7.2.

REQ_LENGTH

As defined in 144.3.7.2.

SP_COUNT

As defined in 144.3.7.2.

164.5.3.7.2 Counters

See 144.3.7.2.

164.5.3.7.3 Variables

BEGIN

As defined in 144.3.7.3

ChState

Type: Array of eight Boolean values

Description: The value of this variable represents a binary-encoded status of upstream channels at the ONU. Super-PON uses only one channel at the ONU, therefore bit 0 shall be always set to 1 (i.e., channel enabled) and bits 1 through 7 shall be set to 0. The value of each bit has the following meaning:

1 = channel is enabled

0 = channel is disabled

DeregistrationTrigger

As defined in 144.3.7.3.

GrantEndTime

As defined in 144.3.7.3.

GrantMargin

As defined in 144.3.7.3.

MaxDelay

As defined in 144.3.7.3.

MissedReportCount

As defined in 144.3.7.3.

Onu10GCapable

Type: Boolean

Description: This variable is set to true if the ONU is capable at transmitting at a line rate of 10.3125 GBd. Otherwise, it is set to false.

Onu2_5GCapable

Type: Boolean

Description: This variable is set to true if the ONU is capable at transmitting at a line rate of 2.578125 GBd. Otherwise, it is set to false.

OnuRssiLocal

As defined in 144.3.7.3.

MissedReportCount

As defined in 144.3.7.3.

RegAllowed

Type: Boolean

Description: This variable is set to true if upon the verification of the various fields of the DISCOVERY MPCPDU, the ONU has determined that it is allowed to transmit a REGISTER_REQ in the current discovery window. The RegAllowed is an alias for the following code:

RegAllowed =

```
// 1) Upstream channel is available
((MsgDiscovery.ChannelMap AND ChState) != 0) AND

// 2) RSSI is within the allowed limits
OnuRssiLocal ≥ MsgDiscovery.OnuRssiMin AND
OnuRssiLocal ≤ MsgDiscovery.OnuRssiMax AND

// 3) 10G discovery is open and ONU is 10G-capable,
// OR (2.5G discovery is open and ONU is 2.5G-capable
// AND the OLT and/or the ONU are not 10G-capable)
// (see Discovery in dual-rate systems, 164.5.3.9)
((MsgDiscovery.DiscoveryInfo[6] == 1 AND Onu10GCapable) OR
(MsgDiscovery.DiscoveryInfo[5] == 1 AND Onu2_5GCapable) AND
(MsgDiscovery.DiscoveryInfo[2] == 0 OR !Onu10GCapable));
```

Registered

As defined in 144.3.7.3.

RegStart

As defined in 144.3.7.3.

SpSeq

As defined in 144.3.7.3.

164.5.3.7.4 Functions

See 144.3.7.4.

164.5.3.7.5 Messages

See 144.3.7.5.

164.5.3.7.6 Discovery Initiation state diagram

See 144.3.7.6.

164.5.3.7.7 Registration Completion state diagram

See 144.3.7.7.

164.5.3.7.8 ONU Registration state diagram

See 144.3.7.8.

164.5.3.8 Granting process

See 144.3.8.

164.5.3.9 Discovery process in dual-rate systems

The MPCP Discovery Process (see 164.5.3.7) facilitates the coexistence of different types of Super-PON ONUs on the same PON. The coexistence mode allows symmetric and asymmetric ONUs to be deployed on the same ODN and to be connected to the same Super-PON OLT.

164.5.3.9.1 OLT rate-specific discovery

The DISCOVERY MPCPDU (see 164.5.3.6.6) includes the DiscoveryInfo field, which gives the Super-PON OLT control over the types of ONUs allowed to participate in the given discovery window. Using the DiscoveryInfo field, the OLT indicates its receive line rate capabilities (10 Gb/s and/or 2.5 Gb/s) as well as the specific line rate(s) allowed in the given discovery window. The OLT may open separate (non-overlapping) discovery windows for 10 Gb/s and 2.5 Gb/s transmission using two separate DISCOVERY MPCPDUs or it may open a single discovery window for 10 Gb/s and 2.5 Gb/s line rates using a single DISCOVERY MPCPDU.

Table 164–13 illustrates the different types of ONUs that may respond to a DISCOVERY MPCPDU with the given settings of the DiscoveryInfo sub-fields.

Table 164–13—DISCOVERY MPCPDUs for Super-PON ONU types

ONU types targeted by DISCOVERY MPCPDU		DiscoveryInfo field value			
Symmetric	Asymmetric	Upstream capable		Discovery window	
		10G	2.5G	10G	2.5G
X		1	0/1	1	0
	X	0/1	1	0	1
X	X	1	1	1	1

164.5.3.9.2 ONU rate-specific registration

An unregistered Super-PON ONU is capable of receiving a DISCOVERY MPCPDU transmitted by the OLT on the DISC_PLID. When received by a symmetric Super-PON ONU, the DISCOVERY MPCPDU is parsed, and if a 10 Gb/s discovery window is opened, the ONU may attempt to register, if other conditions are also satisfied (see definition of the RegAllowed variable in 164.5.3.7.3). When received by an asymmetric Super-PON ONU, the DISCOVERY MPCPDU is parsed, and if a 2.5 Gb/s discovery window is opened, the ONU may attempt to register.

In general, the ONUs attempt to register using the highest upstream transmission rate supported by the OLT and the ONU. If the OLT advertised itself as 10G-capable, but in the current DISCOVERY MPCPDU it has not enabled the 10 Gb/s discovery window, the 10G-capable ONU skips such discovery attempts and waits for a future discovery window in which 10 Gb/s transmission is enabled.

Table 164–14 shows the action the ONU shall take based on the ONU transmit capabilities and the received discovery information.

The ONU transmits the REGISTER_REQ MPCPDU in envelopes with the discovery PLID (DISC_PLID, see 144.3.5).

Table 164–14—ONU action during discovery window

<i>DiscoveryInfo</i> field				ONU upstream capability		ONU action
Upstream capability		Discovery window				
2.5G	10G	2.5G	10G	2.5G	10G	
1	0	1	0	1	0/1	Attempt 2.5G registration
1	0/1	1	0/1	1	0	Attempt 2.5G registration
0/1	1	0/1	1	0/1	1	Attempt 10G registration
1	1	0	1	1	0	Wait for 2.5G discovery window
1	1	1	0	0/1	1	Wait for 10G discovery window

164.5.4 Protocol implementation conformance statement (PICS) proforma for 164.5, Super-PON Multipoint MAC Control Sublayer

The supplier of a protocol implementation that is claimed to conform to 164.5, Super-PON Multipoint MAC Control Sublayer, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

164.5.4.1 Identification

164.5.4.1.1 Implementation identification

Supplier ¹	
Contact point for enquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

164.5.4.1.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3cs-2022, 164.5, Super-PON Multipoint MAC Control
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3cs-2022.)	
Date of Statement	

164.5.4.2 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*OLT	OLT functionality	144.1	Device supports functionality required for OLT	O/1	Yes [] No []
*ONU	ONU functionality	144.1	Device supports functionality required for ONU	O/1	Yes [] No []

164.5.4.3 PICS proforma tables for Multipoint MAC Control

164.5.4.3.1 Clock tracking

Item	Feature	Subclause	Value/Comment	Status	Support
CLK1	Clock tracking at OLT	164.5.2.1.2	<i>LocalTime</i> tracks the XGMII transmit clock	OLT:M	Yes [] N/A []
CLK2	Clock tracking at ONU	164.5.2.1.2	<i>LocalTime</i> tracks the XGMII receive clock	ONU:M	Yes [] N/A []

164.5.4.3.2 LLID

Item	Feature	Subclause	Value/Comment	Status	Support
LL1a	Reserved LLID values, transmit	144.3.5	Do not transmit envelopes with <i>ESC_LLID</i> or a reserved value in the LLID field	M	Yes []
LL1b	Reserved LLID values, receive	144.3.5	Ignore envelopes with <i>ESC_LLID</i> or a reserved value in the LLID field	M	Yes []
LL2a	Unregistered ONU, accept envelopes	144.3.5	Accept only the envelopes containing the <i>DISC_PLID</i> value in the LLID field	ONU:M	Yes [] N/A []
LL2b	Unregistered ONU, ignore envelopes	144.3.5	Ignore envelopes containing values other than <i>DISC_PLID</i> in the LLID field	ONU:M	Yes [] N/A []
LL3a	Registered ONU, accept envelopes	144.3.5	Accept envelopes containing the following values in the LLID field: — The specific PLID value assigned to this ONU during registration; — The specific MLID value assigned to this ONU during registration — Broadcast PLID (<i>BCAST_PLID</i>) — Broadcast MLID (<i>BCAST_MLID</i>) — Any ULID or GLID assigned to this ONU by management	ONU:M	Yes [] N/A []
LL3b	Registered ONU, ignore envelopes	144.3.5	Ignore envelopes containing the <i>DISC_PLID</i> value in the LLID field	ONU:M	Yes [] N/A []

164.5.4.3.3 Protocol-independent state diagrams

Item	Feature	Subclause	Value/Comment	Status	Support
SM1	Control Parser	144.2.1.5	Meets the requirements of Figure 144–5	M	Yes []
SM2	Control Multiplexer	144.2.1.6	Meets the requirements of Figure 144–6	M	Yes []

164.5.4.3.4 MPCP

Item	Feature	Subclause	Value/Comment	Status	Support
MP1	GATE structure	144.3.6.1	As in Figure 144–12	M	Yes []
MP1a	Grant start time and size	144.3.6.1	Transmission on each channel starts at the ONU's local time equal to the <i>StartTime</i> value and has the length as necessary to transmit all allocated envelopes (the sum of all <i>EnvLength</i> fields) together with the associated optical and FEC overhead	ONU:M	Yes [] N/A []
MP1b	Fragmentation	144.3.6.1	When flag <i>F</i> is set to 0, do not fragment new frames	ONU:M	Yes [] N/A []
MP1c	Forced report	144.3.6.1	When flag <i>FR</i> is set to 1, report the total length of the frames (including IPG and preamble), queued for transmission on this specific LLID	ONU:M	Yes [] N/A []
MP1d	MPCPDU fragmentation	144.3.6.1	ONU does not fragment MPCPDU frames, regardless of the value of the <i>Fragmentation</i> flag in the <i>EnvAlloc</i> structure that allocates a PLID envelope	ONU:M	Yes [] N/A []
MP2	REPORT structure	144.3.6.2	As in Figure 144–13	M	Yes []
MP3	REGISTER_REQ structure	144.3.6.3	As in Figure 144–14	M	Yes []
MP4	REGISTER structure	144.3.6.4	As in Figure 144–15	M	Yes []
MP5	REGISTER_ACK structure	144.3.6.5	As in Figure 144–16	M	Yes []
MP6	DISCOVERY structure	144.3.6.6	As in Figure 144–17	M	Yes []
MP6a	Discovery attempt	144.3.6.6	Attempt to register on a single channel only	ONU:M	Yes [] N/A []
MP6b	<i>OnuRssiMin</i> trigger	144.3.6.6	Generate a REGISTER_REQ message in the given discovery window only when measured RSSI is greater or equal to <i>OnuRssiMin</i>	ONU:M	Yes [] N/A []
MP6c	<i>OnuRssiMax</i> trigger	144.3.6.6	Generate a REGISTER_REQ message in the given discovery window only when measured RSSI is smaller than or equal to <i>OnuRssiMax</i>	ONU:M	Yes [] N/A []
MP6d	Discovery attempt at 10 Gb/s	164.5.3.6.6	ONU does not generate REGISTER_REQ MPCPDU using the 10 Gb/s upstream channel if the OLT did not open the 10 Gb/s discovery window, i.e., if the bit 5 was set to 0	ONU:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
MP6e	Discovery attempt at 2.5 Gb/s	164.5.3.6.6	ONU does not generate REGISTER_REQ MPCPDU using the 2.5 Gb/s upstream channel if the OLT did not open the 2.5 Gb/s discovery window, i.e., if the bit 7 was set to 0	ONU:M	Yes [] N/A []
MP7	SYNC_PATTERN structure	144.3.6.7	As in Figure 144–18	M	Yes []
MP8	MPCPDU timing on transmit	144.3.1.1	When multiple MPCPDUs are transmitted within a single envelope, all these MPCPDUs have the same <i>Timestamp</i> field value, referencing the transmission time of the ESH.	M	Yes []
MP9a	Discovery Initiation in OLT	144.3.7.6	Meets the requirements of Figure 144–20, single instance associated with DISC_PLID	OLT:M	Yes [] N/A []
MP9b	Discovery Completion in OLT	144.3.7.7	Meets the requirements of Figure 144–21, one instance for each unicast PLID being registered	OLT:M	Yes [] N/A []
MP9c	Discovery in ONU	144.3.7.8	Meets the requirements of Figure 144–22, single instance	ONU:M	Yes [] N/A []
MP10a	GATE Generation	144.3.8.7	Meets the requirements of Figure 144–23, one instance for each registered unicast PLID	OLT:M	Yes [] N/A []
MP10b	GATE Reception	144.3.8.8	Meets the requirements of Figure 144–24, single instance	ONU:M	Yes [] N/A []
MP11a	Envelope Commitment in OLT	144.3.8.9	Meets the requirements of Figure 144–25, single instance	OLT:M	Yes [] N/A []
MP11b	Envelope Commitment in ONU	144.3.8.10	Meets the requirements of Figure 144–26, single instance	ONU:M	Yes [] N/A []
MP11c	Envelope Activation	144.3.8.11	Meets the requirements of Figure 144–27, single instance	M	Yes []
MP12	ONU multi-rate discovery	164.5.3.9.2	ONU takes action based on on the ONU transmit capabilities and the received discovery information, as shown in Table 164–14	ONU:M	Yes [] N/A []
MP13	MPCP delay variability	164.5.3.3	Maintain the combined delay variation through the MAC and PHY of less than one EQT for channels operating at 10.3125 GBd and less than two EQTs for channels operating at 2.578125 GBd	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
MP14a	Granting overlapping envelopes to the PLID	144.3.1.2	OLT does not allocate overlapping envelopes to the PLID, except the fully overlapping envelopes (see Figure 143–5)	OLT:M	Yes [] N/A []
MP14b	Processing partially overlapping PLID envelope allocations	144.3.1.2	If ONU receives partially overlapping PLID envelope allocations, it chooses only one of these envelopes for MPCPDU transmission, and only if the envelope length is enough for at least one complete MPCPDU	ONU:M	Yes [] N/A []

Table 164A–1—Expected loss of a 50 km Super-PON access link

Component	Count	Typical loss [dB]	Worst case loss [dB]	Total loss [dB]
Fiber [50 km]	—	0.24 / km	0.24 / km	12
Connectors ^a	6	0.2	0.5	1.5
Splices ^b	17	0.05	0.2	1
CAWG ^c	1	4	4	4
1:64 splitter ^d	1	20.5	20.5	20.5
Margin	1	2	2	2
Total				41

^a Assumes 1 bad connector and 5 good ones per link.

^b Assumes 10 inline splices and 7 for components/cables changes.

^c Assumes low-loss design.

^d Assumes premium grade optical splitters.

164A.2 MUX/Amp

This subclause describes an example implementation of the MUX/Amp, with components arranged as shown in Figure 164A–1.

164A.2.1 MUX/DEMUX characteristics

The MUX and DEMUX may be implemented using flat-top arrayed waveguide gratings (AWG). The MUX and DEMUX functions in the MUX/Amp do not require cyclical properties, so they may be designed to have a much wider free spectral range (FSR). Each AWG operates over one 16-channel wavelength band: the MUX operates over C-band 1; the DEMUX operates over L-band 1 (see Table 164–4). The recommendations for the flat-top AWG based MUX/DEMUX are shown in Table 164A–2.

Table 164A–2—Recommendations for the flat-top AWG based MUX/DEMUX

Parameter	Condition	Min	Max	Unit
Number of channels		16		—
Number of bands		1		—
Channel spacing (MUX) ^a		98		GHz
Operational wavelength (MUX)		1547.715	1563.047	nm
Channel spacing (DEMUX)		100.0		GHz
Operational wavelength (DEMUX)		1583.907	1599.594	nm
Clear pass band (CPB)	Relative to nominal	–15	+15	GHz
In-channel ripple	Over CPB for all SOP		1.0	—
Insertion loss (IL)	Over Channel passband (CBP) for all SOP		5.5	dB
Adjacent channel crosstalk	Over CPB of adjacent channels	–23		dB

Table 164A–2—Recommendations for the flat-top AWG based MUX/DEMUX (*continued*)

Parameter	Condition	Min	Max	Unit
Non-adjacent channel crosstalk	Over CPB of non-adjacent channels	–28		dB
Total integrated crosstalk	Sum of all crosstalk	–20		dB
Operating temperature		0	40	°C

^a Channel spacing for the four wavelength bands are different because of cyclical AWG implementation constraints. The channel plan specified in Table 164–4 allows the use of a single cyclical AWG for the 4-band wavelength router.

164A.2.2 Booster optical amplifier characteristics

The downstream booster optical amplifier may be implemented using a constant gain high-power erbium-doped fiber amplifier (EDFA). EDFAs are chosen to reduce inter-channel interactions between the WDM channels. Gain flattening is needed over the 16-channel operational range to reduce the imbalance between wavelength channels after amplification. The recommended specifications for the constant gain EDFA based downstream booster amplifier are shown in Table 164A–3.

Table 164A–3—Recommendations for the constant gain EDFA-based downstream booster amplifier

Parameter	Condition	Min	Max	
Operating wavelength range ^a		1547.715	1563.047	nm
Single channel input power		–8	–5	dBm
Number of channels		1	16	—
Small signal gain	Over all SOP	21		dB
Gain ripple	16-channel input, over all input powers		2	dB
Saturation power		26.5		dBm
Noise figure			12	dB
Operating temperature		0	40	°C

^a C-band 1 (see Table 164–4).

The output of the booster optical amplifier may be either hazard level 2M or 3, depending on the configuration chosen by the operator. For deployments using class 3 lasers, fault detection with automatic power reduction is recommended.

164A.2.3 Band MUX characteristics

The band MUX is responsible for combining the multiplexed downstream and upstream signals into a bi-directional signal for transmission over a single fiber ODN. Since the wavelengths for a next-generation system are already defined (i.e., L-band 2 and C-band 2), the band-mux also has expansion ports to allow the next generation system to be inserted without the need to interrupt service. The band MUX may be implemented using three thin film filters (TFF) in two stages. Additional stages can be inserted for monitoring, if desired. The recommended specifications for the TFF based band MUX are shown in Table 164A–4.

Table 164A–4—Recommendations for the TFF-based band MUX

Parameter	Condition	Min	Max	Unit
Number of ports	(not including common)	4		—
Channel wavelengths	L-band 1	1583.907	1599.594	—
	L-band 2	1565.602	1581.109	
	C-band 1	1547.715	1563.047	
	C-band 2	1530.233	1545.393	
Insertion loss			1	dB
In-band ripple			1	dB
Channel uniformity			1	dB
Adjacent channel crosstalk		−40		dB
Non-adjacent channel crosstalk		−40		dB
Operating temperature		0	40	C

164A.2.4 Preamp characteristics

The upstream optical preamplifier may be implemented using a gain-clamped EDFA. Similar to the downstream power amplifier, EDFAs are preferred to reduce the interaction between the WDM channels. The time period of upstream bursts are long enough to cause gain fluctuations in a standard EDFA. Gain clamping is therefore used to avoid gain variations for the burst-mode upstream signals. There are multiple methods for optical gain clamping; the exact implementation is not prescribed. The recommended specifications for the gain-clamped EDFA based upstream preamplifier are shown in Table 164A–5.

Table 164A–5—Recommendations for the gain-clamped EDFA-based upstream preamplifier

Parameter	Condition	Min	Max	
Operating wavelength range ^a		1583.907	1599.594	nm
Single channel input power	Continuous power	−38	−18	dBm
Number of channels		1	16	—
Small signal gain	Over all SOP	25		dB
Gain ripple	16-channel input, over all input powers		2	dB
Saturation power	For all 16 channels	10		dBm
Noise figure			5.5	dB
Burst length ^b		0.32	500	μs
Burst-mode gain excursion ^c			0.5	dB
Operating temperature		0	40	°C

^a L-band 1 (see Table 164–4).

^b Excluding preambles for transmitter/receiver stabilization and CDR.

^c The burst-mode gain excursion is the power ratio at the start of a burst to instantaneous power at the end of the longest defined burst for a pulse of 6 dB higher in power than the defined max single channel input power.

The input power into the upstream preamplifier is expected to be very low. Therefore, the noise figure (NF) and gain are critical parameters. The output power is low, thus simplifying the gain clamping implementation.

164A.2.5 DCM characteristics

The dispersion compensation module (DCM) may be implemented using any technology. Known solutions include the following:

- Fiber Bragg Grating (FBG) DCMs
- Negative Dispersion Fiber DCMs

FBG DCMs are likely to be channelized and able to cover the full C- and L-bands. They have a fixed insertion loss and a fixed latency. The latency of FBG DCMs is very small (i.e., $<0.025 \mu\text{s}$). This means the optical line terminal's (OLT) calculation of the length from the ranging will be unaffected by the DCM. As FBG DCMs are likely to be channelized, it is important to make sure the amplitude and phase ripple are minimal between $\pm 15 \text{ GHz}$ of the nominal channel frequency.

Fiber-based DCM are not channelized and their insertion loss and latency is dependent on the length of the fiber used to build them. To compensate for 50 km of standard single-mode fiber, around 10 km of dispersion compensating fiber (DCF) is used. This results in around $25 \mu\text{s}$ of latency. Because a DCM is only used for the upstream (US) direction, there will be imbalance in the latency, with greater latency in the upstream direction. PON protocols are tolerant of latency asymmetry so this is expected to not be a problem, even if this will lead to an overestimation of the link length by the OLT of $\sim 5 \text{ km}$. Other time sensitive protocols (e.g., time transport protocols) may break when using this type of DCM.

164A.3 Wavelength Router

The cyclical wavelength router is best implemented using a Cyclical Arrayed Waveguide Grating (CAWG). AWGs are naturally cyclical. For the Super-PON wavelength router, the FSR for the CAWG should be approximately 2200 GHz wide, spanning the width of each wavelength band including all the unused channels. The response is defined only for the channels in the four wavelength-bands used in Super-PON. However, the cyclical nature of the AWG will actually repeat for many more cycles. It is possible to use those additional cycles for applications not defined in this standard, such as OTDR.

In addition, this device is intended to be deployed in the outside plant as part of the ODN. Therefore, the device should be able to operate over an extended temperature range. Multiple athermalization methods exist for AWGs and a specific athermalization method is not prescribed.

The recommended specifications for the cyclical AWG based wavelength router are shown in Table 164A–6.

Table 164A–6—Recommendations for the CAWG-based wavelength router

Parameter	Condition	Min	Max	Unit
Number of channels		16		—
Number of bands		4		—
Channel spacing (MUX)	L-band 1 L-band 2 C-band 1 C-band 2	97.7 98.8 100.0 101.2		GHz
Operational wavelength (MUX) ^a	L-band 1 L-band 2 C-band 1 C-band 2	1583.907 1565.602 1547.715 1530.233	1599.594 1581.109 1563.047 1545.393	nm
Clear pass band (CPB)	Relative to nominal	–15	+15	GHz
In-channel ripple	Over CPB for all SOP		1.0	—
Insertion loss (IL)	Over CPB for all SOP		4.0 ^b	dB
Adjacent channel crosstalk	Over CPB of adjacent channels	–23		dB
Non-adjacent channel crosstalk	Over CPB of non-adjacent channels	–28		dB
Total integrated crosstalk	Sum of all crosstalk	–20		dB
Temperature range		–40	65	°C

^a The full channel plan is shown in Table 164–4.

^b The low-loss implementation of the CAWG is only used for deployments using 64-port splitters up to 50 km. In this implementation a Mach-Zehnder interferometer precedes the AWG, which reduces the loss of the AWG device. A standard AWG design can be used for lower loss ODNs, such as ODNs with less than 40 km of fiber plant or ODNs using 32-port or lower count splitters.

164A.4 Power splitter

The power splitters for Super-PON are as those used in other PON standards. In order to reduce the link loss, premium low-loss splitters are used for links approaching 50-km using a 64-port splitter to keep the total link loss below 41 dB.

The insertion loss of 20.5 dB for a 64-port splitter is not exotic and may be achieved by most splitter manufacturers by selecting the lower loss units from a batch.

For links shorter than 40 km, the loss through the fiber is 2.8 dB lower. This allows the use of a standard 21 dB loss 64-port splitter.

164A.5 Link budget analysis

164A.5.1 Downstream for a 50-km link

Figure 164A–2 shows the downstream link components and expected losses/gains for each component. The expected minimum power for different locations in the ODN is shown at the bottom of the figure. For simplicity, all connector, splice, and link aging losses are lumped together right before the receiver (ONU).

The low power and high extinction ratio of the downstream OLT transmitter favor a deeply biased electron absorption modulator or a Mach Zehnder modulator. These solutions are able to create zero or even negative chirp, thereby producing a negligible dispersion penalty for up to 50 km of transmission. This is reflected by the defined residual CD range in Table 164–5 and the absence of a DCM in Figure 164A–2.

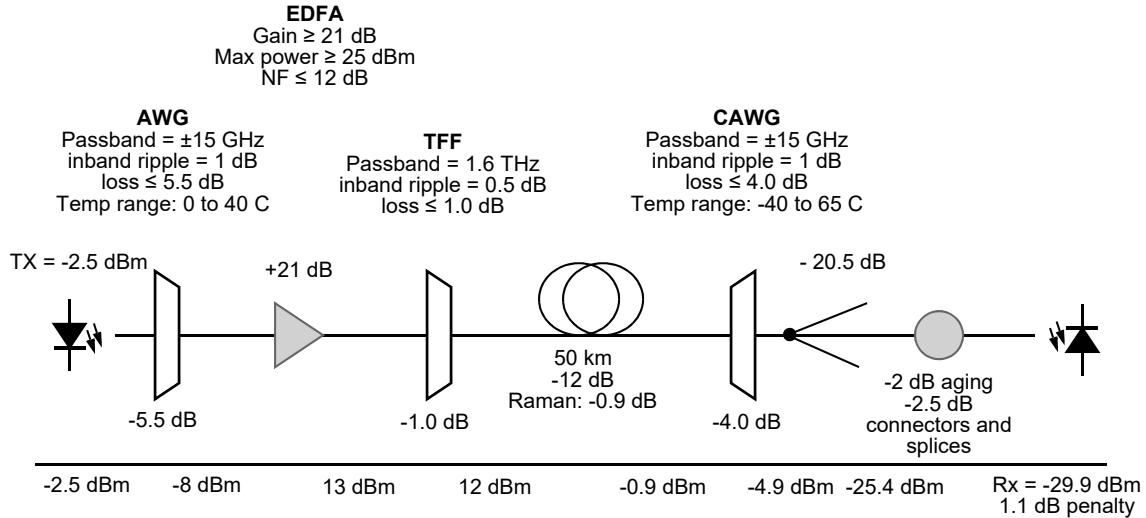


Figure 164A–2—Downstream link budget

164A.5.2 Upstream for a 50-km link

Figure 164A–3 shows the upstream link components and expected losses/gains for each component. The expected minimum power for different locations in the ODN is shown at the bottom of the figure. For simplicity, all connector, splice, and link aging losses are lumped together right before the transmitter (ONU).

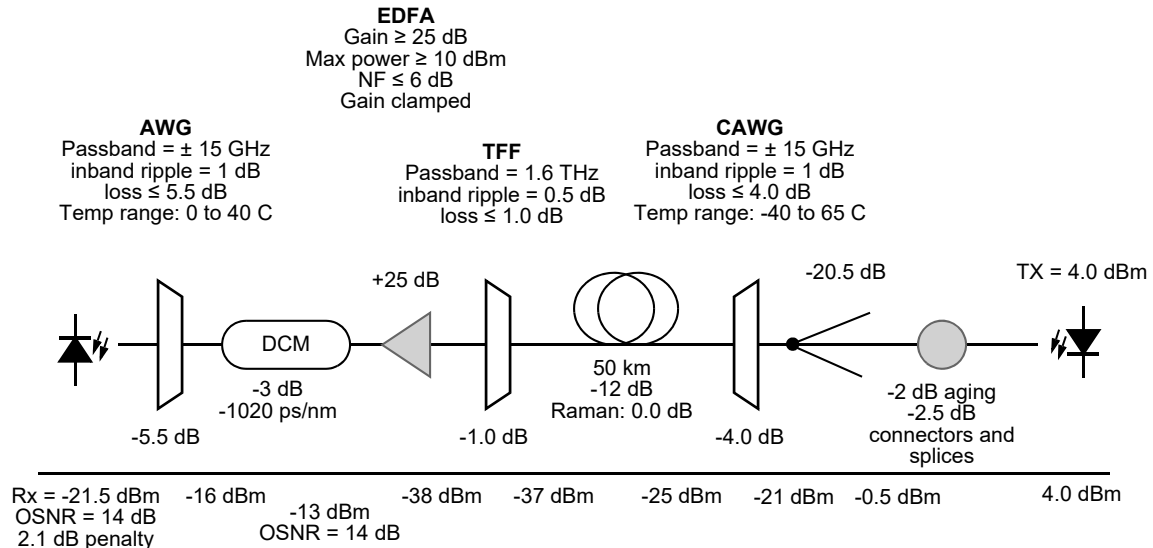


Figure 164A–3—Upstream link budget

164A.5.3 Raman Effect Penalty

The Stimulated Raman Scattering (SRS) transfers power from shorter wavelength channels to the longer wavelength channels. The total power transferred is proportional to the product of all the powers at all the wavelengths.

In the Super-PON system, the downstream signal power is substantially higher than the upstream signal power in the AWG feeder fiber, the only location where combined powers are sufficient for substantial Raman interaction. The percentage of power transferred from the upstream signals is determined by the power in the downstream signals.

Using L-band wavelengths for upstream, the Raman effect transfers power to the upstream signals. This is illustrated by the negative Raman penalty in Figure 164A–4. This calculation assumed the use of 16 channels for US and DS. The channel with the highest Raman penalty is shown. A negative penalty means that power is transferred into the channel via SRS, thus receiving gain.

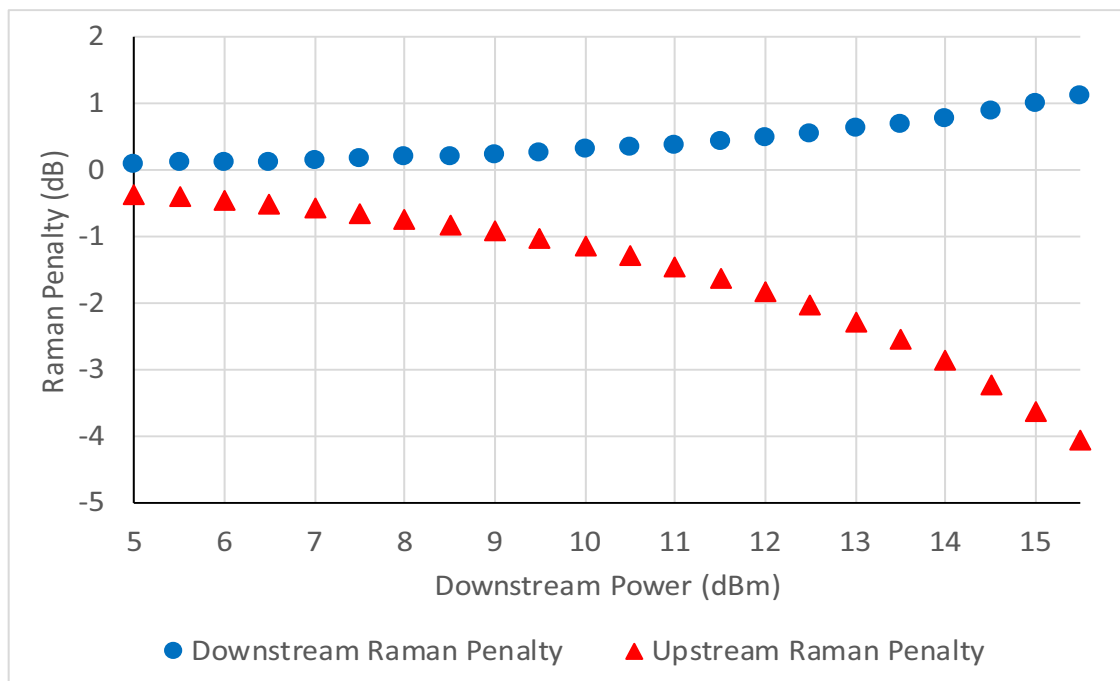


Figure 164A–4—Raman penalty

This Raman gain cannot be relied upon for the following two reasons:

- The gain in power will only be a fraction if fewer wavelengths are used.
- The gain from a modulated pump will also result in additional noise.

Therefore, the Raman gain/penalty is assumed to be zero when calculating the upstream powers for upstream operation.

The Raman effect between the 16 downstream channels causes the longer wavelength channels to steal power from the shorter wavelength channels. However, this effect is weaker due to the close pump-Stokes separation. Again, because the upstream powers are so much lower in power, the power transferred to the upstream wavelengths is very small compared to the power transferred among the downstream channels. The predicted downstream penalty may be handled by increasing the gain and maximum power of the downstream EDFA in the MUX/Amp module. In the downstream link budget, the expected Raman power penalty is added to the loss of the fiber.

Figure 164A–5 shows the power levels as a function of the link budget, assuming a 2.1 dB transmission penalty for the upstream and 1.1 dB transmission penalty for the downstream. A 41-dB link budget requires an upstream launch power level of 4.0 dBm at 10 Gb/s.

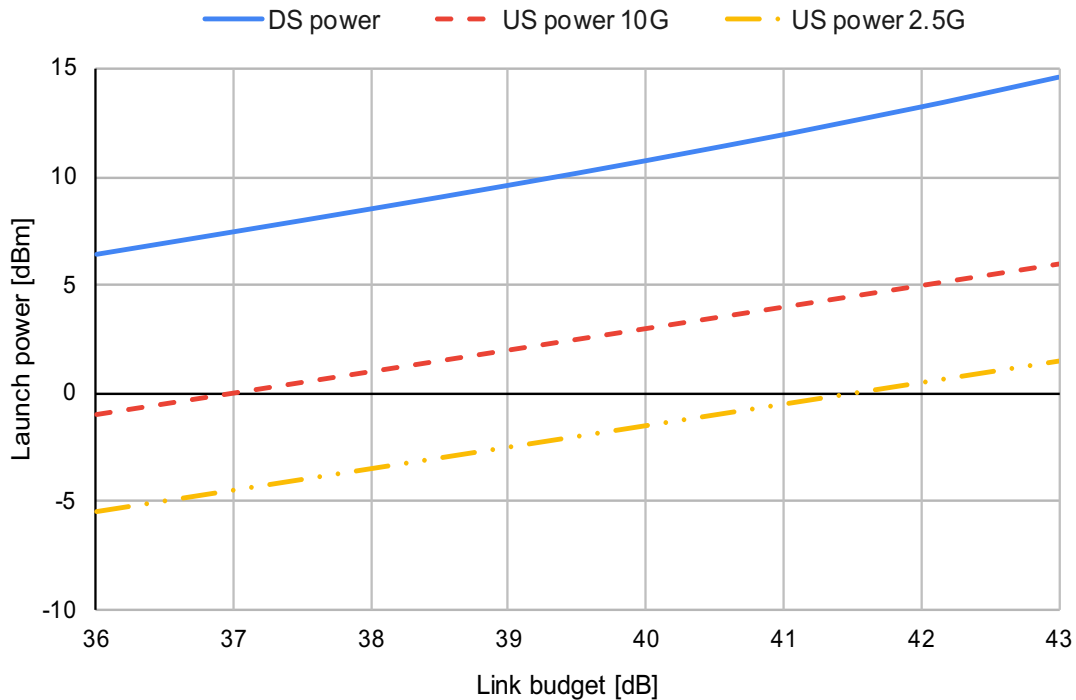


Figure 164A–5—ONU power levels

All Raman penalties are calculated assuming the power on all downstream channels is 2-dB higher than the minimum downstream power. This is to account for some channel imbalance in the downstream channels without causing an excessively large Raman penalty on the lowest power channel(s). Note that the typical allowed variation in OLT output power is 4 dB. Allowing for 2-dB higher power for the Raman calculation assumes (and requires) an operator to perform some closed-loop balancing of channel powers for each ODN unless a tighter than usual spread in OLT transmitter power can be guaranteed by the OLT optical module.

164A.5.4 Dispersion compensation

The operational dispersion range selected for Super-PON was based on the following needs:

- Desire to enable Directly Modulated Laser (DML) and Electroabsorption Modulated Laser (EML) solutions.
- Need to reduce the total penalty from transmission for all transmitter types.

DMLs typically have a high positive chirp, which is very difficult to avoid. This causes a large transmission penalty if there is substantial positive residual chromatic dispersion (CD) in the link. DML transmitters need CD compensation to operate in a 50-km link.

For EML transmitters, the chirp is typically determined by the bias point of the modulator. A low bias results in higher transmission powers, positive chirp, and lower extinction ratio (ER). A high bias results in lower transmission powers, low or even negative chirp, and higher ER. The requirement of 4-dBm output power is relatively high for EML transmitters, so it likely requires a moderately low modulator bias point, resulting in a positive chirp.

The performance of typical DML and EML solutions for different residual chromatic dispersion is shown in Figure 164A–6. Both solutions are positively chirped so perform better with negative residual dispersion. The higher chirp factor of the DML transmitter results in a lower tolerance in the positive and negative residual dispersion bounds for a 1-dB penalty. These results suggest that a residual dispersion between -600 ps/nm and $+50$ ps/nm allows the EML and DML transmitters to operate within 1 dB penalty.

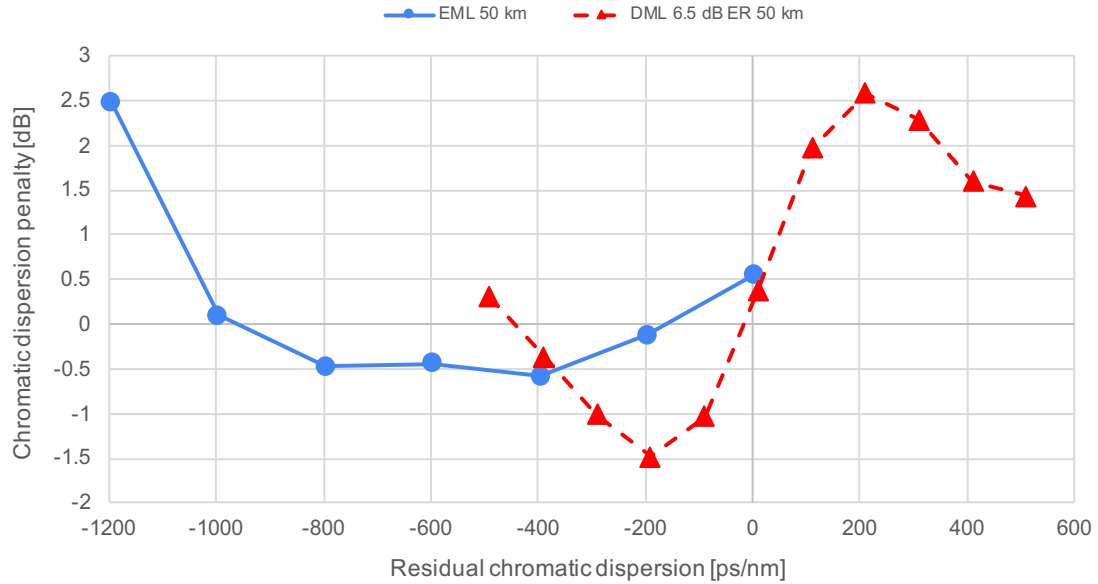


Figure 164A–6—Chromatic dispersion penalty for DML and EML transmitters

Annex 164B

(informative)

Cabled fiber link attributes used for system design

164B.1 Overview

A concatenated link usually includes a number of spliced factory lengths of optical fiber cable. The transmission parameters for concatenated links need to take into account not only the performance of the individual cable lengths but also the statistics of concatenation.

The transmission characteristics of factory length optical fiber cables has a certain probability distribution which can be taken into account if the most economic designs are to be obtained. This annex needs to be interpreted, therefore, with the statistical nature of the various parameters in mind.

Link attributes such as end-to-end attenuation, chromatic dispersion, or PMD are affected by factors other than optical fiber cables, by such things as splices, passive components, and installation. These factors are not specified in this annex. The estimation methods of link parameters needed for system design are based on measurements, modeling or other considerations. The following provides additional description, methodology, and examples of how cabling link attenuation attributes specifically can be statistically estimated.

164B.2 Attenuation

The mean attenuation, A , of a link is given by Equation (164B-1):

$$A = \alpha \times L + \alpha_s \times x + \alpha_c \times y \quad (164B-1)$$

where

- α represents the mean attenuation coefficient of the fiber cables in a link
- α_s represents the mean splice loss
- x represents the number of splices in a link
- α_c represents the mean loss of line connectors
- y represents the number of line connectors in a link (if provided)
- L represents the link length

A suitable margin needs to be allocated for future modifications of cable configurations (additional splices, extra cable lengths, aging effects, temperature variations, etc.). The previous equation does not include the loss of equipment connectors. The attenuation budget used in designing an actual system should account for the statistical variations in these parameters.

The cabled fiber attenuation link design value (LDV) provides a statistical upper limit for the α parameter in Equation (164B-1).

The attenuation LDV calculation starts with establishing a statistical distribution. Let x_i and L_i be the attenuation coefficient (dB/km) and length, respectively, of a fiber in the i^{th} cable in a concatenated link of N cables. The attenuation coefficient, x_N (dB/km), of this link is given by Equation (164B-2):

$$x_N = \frac{\sum_{i=1}^N L_i \times x_i}{\sum_{i=1}^N L_i} = \frac{1}{L_{link}} \times \sum_{i=1}^N L_i \times x_i \quad (164B-2)$$

If it is assumed that all cable section lengths are less than some common value, L_{Cab} , and simultaneously reducing the number of assumed cable sections to $M = L_{link}/L_{Cab}$, then, for a link comprised of equal-length cables, $L_i = L_{Cab}$, equation above becomes:

$$x_N \leq x_M = \frac{L_{Cab}}{L_{link}} \times \sum_{i=1}^M x_i = \frac{1}{M} \times \sum_{i=1}^M x_i \quad (164B-3)$$

The variation in the concatenated link attenuation coefficient, x_M , is less than the variation in the individual cable sections, x_i , because of the averaging of the concatenated fibers.

Once the attenuation distribution has been established, the Monte Carlo method is used to determine the probability density, f_{link} , of the concatenated link attenuation coefficients without making any assumption about its form. This method simulates the process of building links by sampling the measured attenuation population repeatedly.

Attenuation coefficients are measured on a sufficiently large number of segments so as to characterize the underlying distribution. This data is then used to compute the attenuation coefficient for a single path in a concatenated link.

Computation of the attenuation LDV is made by randomly selecting M values from the measured attenuation coefficients, and adding them according to Equation (164B-3). The computed attenuation LDV is placed in a table or a histogram of values derived from other random samplings. The process is repeated until a sufficient number of attenuation LDV values has been computed to produce a high density (0.01 dB/km) histogram of the concatenated attenuation LDV distribution. If the histogram is used directly, without any additional characterization such as Gaussian fitting, the number of resamples is expected to be at least 10 000.

Because of the central limit theorem, the histogram of cabled fiber attenuation LDV tends to converge to distributions that can be described with a minimum of two parameters. Hence, the histogram is fit to a parametric distribution that enables extrapolation to probability levels that are smaller than what would be implied by the sample size. The two parameters invariably represent two aspects of the distributions: the central value and the variability about the central value.

To obtain probability levels of $Q = 10^{-3}$ using a pure numeric approach requires Monte Carlo simulations of at least 10^4 samples. Once this is complete, attenuation LDV is interpolated from the associated cumulative probability density function.

Considerations of multiple distributions, multiple fiber types, and cable constructions led to the values shown in Table 164B–1.

Table 164B–1—Attenuation and chromatic dispersion LDVs

Attribute	Detail	Value
Link Design Values (LDV)	M	8 cables ^a
	Q	0.1%
Cabled Fiber Attenuation LDV	Maximum from 1490 nm to 1625 nm	0.24 dB/km
Chromatic Dispersion LDV	Minimum at 1550nm	16.7 ps/nm x km ^b
	Maximum at 1550nm	17.4 ps/nm x km ^b
	Minimum at 1625 nm	21.2 ps/nm x km ^b
	Maximum at 1625 nm	21.8 ps/nm x km ^b

^aThis could be representative of a 40 km link using 5 km cable sections. The LDVs are conservative for links with more segments.

^bUsing linear interpolation, minimum and maximum chromatic dispersion LDV values can be calculated for system design purposes across the relevant C and L Band wavelengths in Table 164–4.

164B.3 Chromatic dispersion

The chromatic dispersion in ps/nm can be calculated from the chromatic dispersion coefficients of the factory lengths, assuming a linear dependence on length, and with due regard for the signs of the coefficients.

A similar process to the attenuation LDV can be performed with chromatic dispersion for a particular wavelength, thus, establishing a chromatic dispersion LDV.

Annex 164C

(informative)

Super-PON spectral excursion measurement

164C.1 Methodology

Measuring the spectral excursion for upstream ONU transmitters requires the summation of the continuous mode spectral excursion and the transient effects associated with laser turn-on. Wavelength measuring devices, such as optical spectral analyzers, are not typically designed to measure dynamic effects. A potential method to quantify the transient spectral excursion is described next:

- Position between the transmitter and a linear PIN photodiode a 50 GHz tunable Gaussian optical filter with the center wavelength detuned from the ONU's continuous-mode wavelength by 25 GHz, as shown in Figure 164C–1. This results in 3 dB less power into the photodiode relative to ideal alignment between the filter's center frequency to the transmitter.
- Use a burst-mode signal to drive the ONU transmitter with bursts of >100 us.
- Capture the power envelope of the time domain waveform output from the photodiode. The offset optical filter translates frequency shifts into a power envelope change. An example measurement is show in Figure 164C–2.
- Measure the difference in the power envelope between laser on time after the burst start and after the laser has reached steady state.
- Convert the change in the measured power envelope into a transient frequency shift by using the Gaussian relationship of the filter.

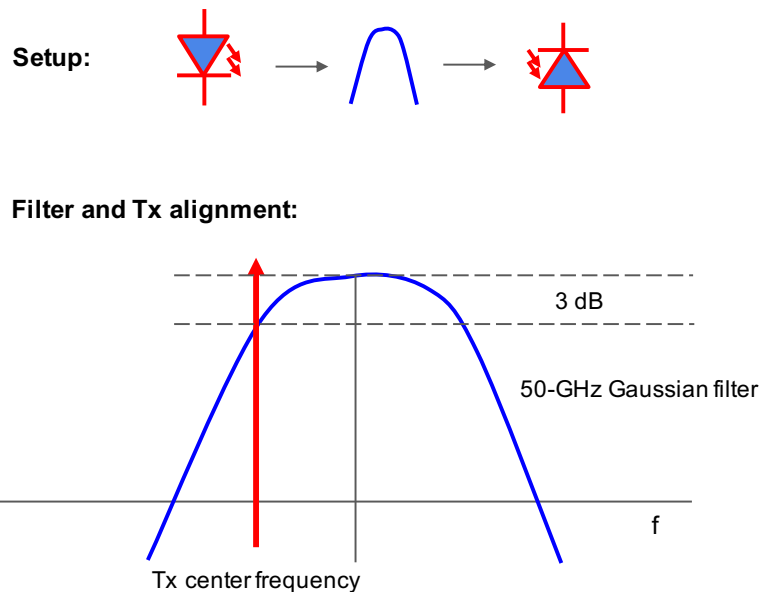


Figure 164C–1—Experimental setup for testing for transient wavelength drift

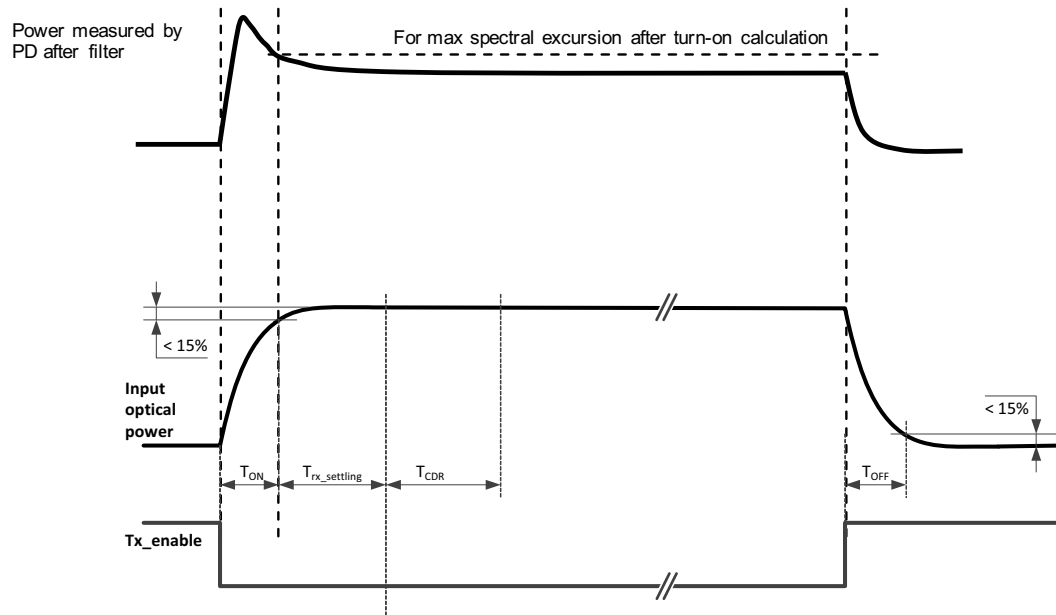


Figure 164C-2—A potential power envelope output after filtering

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